



Honeywell

Fail Safe Control Safety Manual

Version 500

Revision 01 (01/98)

FS40-520

PM.MAN.8047

Total Plant



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Printed in the Netherlands

Version 500
Revision 01 (01/98)

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Abbreviations

AC		Alternating current
AI		Analog input
AK		Anforderungsklasse (requirement class)
AO		Analog output
BI		Multiple input
BO		Multiple output
CE		Conformité Européenne
CP		Central part
CPU		Central processing unit
CSA		Canadian Standards Association
DBM		Diagnostic and battery module
DC		Direct current
DI		Digital input
DIN		Deutscher Industrienorm (German industrial standard)
DO		Digital output
DCS		Distributed control system
E/E/PES		Electrical/Electronic/Programmable electronic system
EEA		European Economic Area
EEC		European Economic Community
EMC		Electromagnetic compatibility
EPROM		Erasable programmable read-only memory
ESD		Emergency shutdown
EU		European Union
EUC		Equipment under control
F&G		Fire & Gas
FAT		Factory acceptance test
FB		Function block
FLD		Functional logic diagram
FMEA		Failure mode effect analysis
FS		Fail-safe
FSC		Fail Safe Control
FSC-DS		Fail Safe Control Development System
H&B		Hartmann & Braun
H-bus		Horizontal bus
HBD		Horizontal bus driver
HSMS		Honeywell Safety Management Systems
I		Input
I/O		Input/output
IC		Input channel
ICS		Integrated control system
IEC		International Electrotechnical Commission
IM		Input module
NFS		Non fail-safe
O		Output
OC		Output channel
OLM		On-line modification
OM		Output module



Abbreviations (continued)

PC	Personal computer
PES	Programmable electronic system
PST	Process safety time
PSU	Power supply unit
RAM	Random-access memory
SER	Sequence-of-event recording
SIL	Safety integrity level
SMOD	Secondary means of de-energization
SOE	Sequence of events
TPS	TotalPlant Solution
TÜV	Technischer Überwachungsverein
UL	Underwriters Laboratories
V-bus	Vertical bus
VBD	Vertical bus driver
WD	Watchdog



References

For FSC documentation:

Publication Title	Publication Number
FSC Safety Manual	PM.MAN.8047
FSC Hardware Manual	PM.MAN.8048
FSC Software Manual	PM.MAN.8025

For FSC-SM documentation:

Publication Title	Publication Number	Binder Title	Binder Number
FSC Safety Manager Installation Guide	FS20-500	Implementation FSC Safety Manager	TPS 3076
FSC Safety Manager Implementation Guidelines	FS11-500	Implementation FSC Safety Manager	TPS 3076
FSC Safety Manager Control Functions	FS09-500	Implementation FSC Safety Manager	TPS 3076
FSC Safety Manager Parameter Reference Dictionary	FS09-550	Implementation FSC Safety Manager	TPS 3076
FSC Safety Manager Configuration Forms	FS88-500	Implementation FSC Safety Manager	TPS 3076
FSC Safety Manager Service Manual	FS13-500	Implementation FSC Safety Manager	TPS 3076





Section 1 – Introduction

1.1 System overview

Section

This section provides general information on the FSC system and its compliance to standards, as well as a glossary of terms. It covers the following topics:

Subsection	Topic	See page
1.1	System overview	1
1.2	Standards compliance.....	2
1.3	Definitions	7

System overview

The Fail Safe Control (FSC) system is a microprocessor-based control system for safety applications. The system can be configured in a number of different basic arrangements depending on the requirement class of the process and the availability required

The safety of the FSC system is obtained through its specific design for these applications. This design includes facilities for self-testing of all FSC modules through software and specialized hardware based on a failure mode effect analysis (FMEA) for each module. Additional software routines are included to guarantee proper execution of the software. This approach can be classified as software diversity. These features maintain fail-safe operation of the FSC system even in the single-channel configurations. By placing these single-channel versions in parallel, one gets not only safety but also availability: *proven availability*.

The FSC system and the FSC user station (with the FSC Navigator software) from Honeywell Safety Management Systems B.V. provide the means to guarantee optimum safety and availability. To achieve these goals, it is essential that the system is operated and maintained by authorized and qualified staff. If it is operated by unauthorized or unqualified persons, severe injuries or loss of production may result. This Safety Manual covers the applications of the FSC system for requirement classes (German: Anforderungsklassen) AK1 to AK6 in accordance with DIN V 19250 of May 1994. This Safety Manual also covers the applications which must comply with IEC 61508.

1.2 Standards compliance

Standards

This subsection lists the standards that FSC complies with, and also provides some background information on CE marking (EMC directive and Low Voltage directive).

Table 1-1 FSC compliance to standards

Standard	Title	Remarks
DIN V 19250 (1/89, 5/94)	Measurement and control. Fundamental safety aspects to be considered for measurement and control equipment. (German title: Leittechnik. Grundlegende Sicherheitsbetrachtungen für MRS-Schutzeinrichtungen)	Safety applications up to safety class AK 8
DIN V 0801 (1/90) and Amendment A (10/94)	Principles for computers in safety-related systems. (German title: Grundsätze für Rechner in Systemen mit Sicherheitsaufgaben)	Microprocessor-based safety systems
VDE 116 (10/89)	Electrical equipment of furnaces. (German title: Elektrische Ausrüstung von Feuerungsanlagen)	
EN 54 part 2 (01/90)	Components of automatic fire detection systems, Introduction (German title: Bestandteile automatischer Brandmeldeanlagen)	
EN 50081-2-1993	Electromagnetic compatibility – Generic emission standard, Part 2: Industrial environment	
EN 50082-2-1993	Electromagnetic compatibility – Generic immunity standard, Part 2: Industrial environment	
EN 61131-2-1994	Programmable controllers. Part 2: Equipment requirements and tests	
UL 1998	Safety-related software, first edition	Underwriters Laboratories
UL 508	Industrial control equipment, sixteenth edition	Underwriters Laboratories
UL 991	Test for safety-related controls employing solid-state devices, second edition	Underwriters Laboratories
CSA C22.2	Process control equipment. Industrial products.	Canadian Standards Association No. 142 (R1993)



Table 1-1 FSC compliance to standards (continued)

Standard	Title	Remarks
DIN IEC 60068	Basic environmental testing procedures	
DIN IEC 60068 Part 2-1	Cold test	0°C (32°F); 16 hours; system in operation; reduced power supply voltage (-15%) U=20.4 Vdc or (-10%); U=198 Vac
DIN IEC 60068 Part 2-1	Cold test	-5°C (23°F); 16 hours; system in operation
DIN IEC 60068 Part 2-2	Dry heat test	up to 60°C (140°F); 16 hours; system in operation; increased power supply voltage (+15%): U=27.6 Vdc or (+10%): U=242 Vac
DIN IEC 60068 Part 2-3	Test Ca: damp heat, steady state	21 days at +40°C (104°F), 95% relative humidity; function test after cooling
DIN IEC 60068 Part 2-3	Test Ca: damp heat, steady state	96 hours at +40°C (104°F), 95% relative humidity; system in operation
DIN IEC 60068 Part 2-6	Environmental testing – Part 2: Tests – Test Fc: vibration (sinusoidal)	Excitation: sine-shaped with sliding frequency; Frequency range: 10-150 Hz Loads: 10-57 Hz; 0.075 mm 57-150 Hz; 1 G Duration: 10 cycles (20 sweeps) per axis No. of axes: 3 (x, y, z) Traverse rate: 1 oct/min System in operation
DIN IEC 60068 Part 2-27	Environmental testing – Part 2: Tests – Test Ea: shock	Half sinus shock 1 shock per direction (6 in total) Maximum acceleration: 15 G Shock duration: 11 ms System in operation

CE marking

The CE mark (see Figure 1-1) is a compliance symbol which indicates that a product meets the requirements of the EU directives that apply to that product. CE (Conformité Européenne) marking is a prerequisite to marketing FSC systems in the European Union.

EU directives are documents issued on the authority of the Council of the European Union. They set out requirements and regulations for certain categories of products or problem areas. The directives apply not only to the member countries of the European Union but to the whole European Economic Area (EEA), which is made up of Austria, Belgium, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Italy, Liechtenstein, Luxembourg, the Netherlands, Norway, Portugal, Spain, Sweden and the United Kingdom.

The directives have the following key objectives:

- free movement of goods within the EU/EEA geographical regions through harmonization of standards and elimination of trade barriers,

- safety of persons, their property and of animals, and

- protection of the environment.



Figure 1-1 CE mark

For control products like FSC, a number of EU directives apply. The FSC product is compliant with two of these: the Electromagnetic Compatibility (EMC) Directive (89/336/EEC) and the Low Voltage Directive (73/23/EEC). Each is discussed in more detail below.



EMC directive (89/336/EEC)

One of the EU directives that FSC complies with is the EMC directive, or *Council Directive 89/336/EEC of 3 May 1989 on the approximation of the laws of the Member States relating to electromagnetic compatibility* as it is officially called. It "applies to apparatus liable to cause electromagnetic disturbance or the performance of which is liable to be affected by such disturbance" (Article 2).

The EMC directive defines protection requirements and inspection procedures relating to electromagnetic compatibility for a wide range of electric and electronic items.

Within the context of the EMC directive, 'apparatus' means all electrical and electronic appliances together with equipment and installations containing electrical and/or electronic components.

'Electromagnetic disturbance' means any electromagnetic phenomenon which may degrade the performance of a device, unit of equipment or system. An electromagnetic disturbance may be electromagnetic noise, an unwanted signal or a change in the propagation medium itself.

'Electromagnetic compatibility' is the ability of a device, unit of equipment or system to function satisfactorily in its electromagnetic environment without introducing intolerable electromagnetic disturbances to anything in that environment.

There are two sides to electromagnetic compatibility: emission and immunity. These two essential requirements are set forth in Article 4, which states that an apparatus must be constructed so that:

- (a) the electromagnetic disturbance it generates does not exceed a level allowing radio and telecommunications equipment and other apparatus to operate as intended;
- (b) the apparatus has an adequate level of intrinsic immunity of electromagnetic disturbance to enable it to operate as intended.

The EMC directive was originally published in the Official Journal of the European Communities on May 23, 1989. The directive became effective on January 1, 1992, with a four-year transitional period. During the transitional period, a manufacturer can choose to meet existing national laws (of the country of installation) or comply with the EMC directive (demonstrated by the CE marking and Declaration of Conformity). The transitional period ended on December 31, 1995, which meant that as of January 1, 1996 compliance with the EMC directive became **mandatory** (a legal requirement). All electronic products may now only be marketed in the European Union if they meet the requirements laid down in the EMC directive. This also applies to FSC system cabinets.



Low voltage directive (73/23/EEC)

The FSC product also complies with the low voltage directive, or *Council Directive 73/23/EEC of 19 February 1973 on the harmonization of the laws of the Member States relating to electrical equipment designed for use within certain voltage limits* as it is officially called. It states that "electrical equipment may be placed on the market only if, having been constructed in accordance with good engineering practice in safety matters in force in the Community, it does not endanger the safety of persons, domestic animals or property when properly installed and maintained and used in applications for which it was made" (Article 2).

The low voltage directive defines a number of principal safety objectives that electrical equipment must meet in order to be considered "safe".

Within the context of the low voltage directive, 'electrical equipment' means any equipment designed for use with a voltage rating of between 50 and 1,000 V for alternating current (AC) and between 75 and 1,500 V for direct current (DC).

The low voltage directive was originally published in the Official Journal of the European Communities on March 26, 1973. It was amended by Council Directive 93/68/EEC, which became effective on January 1, 1995, with a two-year transitional period. During the transitional period, a manufacturer can choose to meet existing national laws (of the country of installation) or comply with the low voltage directive (demonstrated by the CE marking and Declaration of Conformity). The transitional period ended on December 31, 1996, which meant that as of January 1, 1997 compliance with the low voltage directive became **mandatory** (a legal requirement). All electronic products may now only be marketed in the European Union if they meet the requirements laid down in the low voltage directive. This also applies to FSC system cabinets.



1.3 Definitions

Definitions

This section provides a list of essential safety terms that apply to the FSC system. All definitions have been taken from IEC 61508-4 (FDIS version, February '98).

Dangerous failure

Failure which has the potential to put the safety-related system in a hazardous or fail-to-function state.

NOTE: Whether or not the potential is realized may depend on the channel architecture of the system; in systems with multiple channels to improve safety, a dangerous hardware failure is less likely to lead to the overall dangerous or fail-to-function state.

Error

Discrepancy between a computed, observed or measured value or condition and the true, specified or theoretically correct value or condition.

EUC risk

Risk arising from the EUC or its interaction with the EUC control system.

Failure

The termination of the ability of a functional unit to perform a required function.

NOTE 1: The definition in IEC 61508-4 is the same, with additional notes.

NOTE 2: See Figure 1-2 for the relationship between faults and failures, both in IEC 61508 and IEC 61509.

NOTE 3: Performance of required functions necessarily excludes certain behaviour, and some functions may be specified in terms of behaviour to be avoided. The occurrence of such behaviour is a failure.

NOTE 4: Failures are either random (in hardware) or systematic (in hardware or software).

Fault

Abnormal condition that may cause a reduction in, or loss of, the capability of a functional unit to perform a required function

NOTE: IEC 61508-4 defines "fault" as a state characterized by the inability to perform a required function, excluding the inability during preventative maintenance or other planned actions, or due to lack of external resources.

Functional safety

Part of the overall safety relating to the EUC and the EUC control system which depends on the correct functioning of the E/E/PE safety-related systems, other technology safety-related systems and external risk reduction facilities.

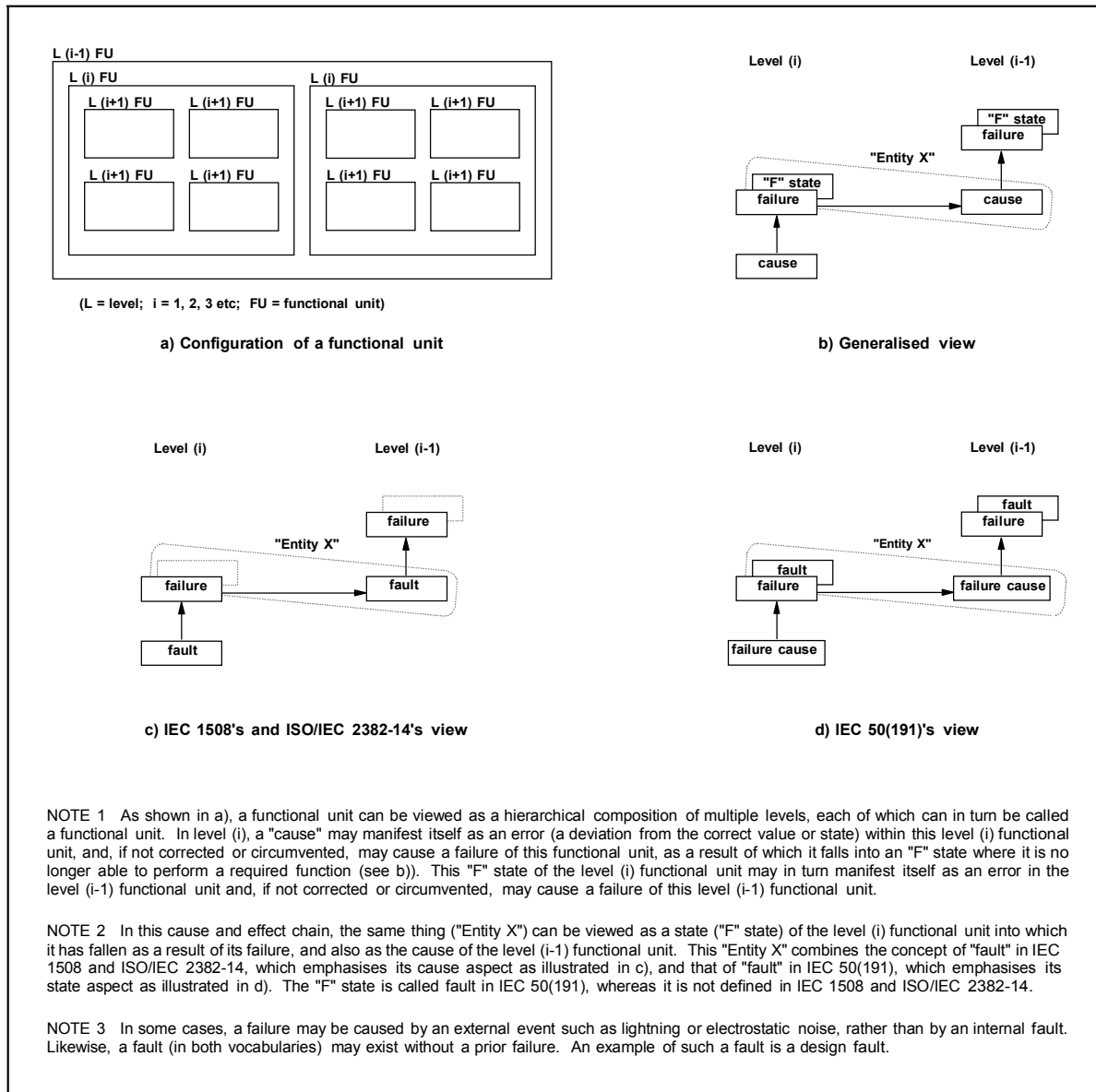


Figure 1-2 Failure model

Functional safety assessment

Investigation, based on evidence, to judge the functional safety achieved by one or more E/E/PE safety-related systems, other technology safety-related systems or external risk reduction facilities.

Human error

Mistake.
Human action or inaction that produces an unintended result.

**Hardware safety integrity**

Part of the safety integrity of the safety related systems relating to random hardware failures in a dangerous mode of failure

NOTE: The term relates to failures in a dangerous mode. That is, those failures of a safety-related system that would impair its safety integrity. The two parameters that are relevant in this context are the overall dangerous failure rate and the probability of failure to operate on demand. The former reliability parameter is used when it is necessary to maintain continuous control in order to maintain safety, the latter reliability parameter is used in the context of safety-related protection systems.

Mode of operation

Way in which a safety-related system is intended to be used, with respect to the frequency of demands made upon it in relation to the proof check frequency, which may be either:

low demand mode - where the frequency of demands for operation made on a safety-related system is not significantly greater than the proof check frequency; or

high demand or continuous mode - where the frequency of demands for operation made on a safety-related system is significantly greater than the proof check frequency

NOTE: Typically for low demand mode, the frequency of demands on the safety-related system is the same order of magnitude as the proof test frequency (i.e. months to years where the proof test interval is a year). While typically for high demand or continuous mode, the frequency of demands on the safety-related system is hundreds of times the proof test frequency (i.e. minutes to hours where the proof test interval is a month).

Programmable electronic system (PES)

System for control, protection or monitoring based on one or more programmable electronic devices, including all elements of the system such as power supplies, sensors and other input devices, data highways and other communication paths, and actuators and other output devices (see Figure 1-3).

NOTE: The structure of a PES is shown in Figure 1-3 a). Figure 1-3 b) illustrates the way in which a PES is represented in IEC 61508, with the programmable electronics shown as a unit distinct from sensors and actuators on the EUC and their interfaces, but the programmable electronics could exist at several places in the PES. Figure 1-3 c) illustrates a PES with two discrete units of programmable electronics. Figure 1-3 d) illustrates a PES with dual programmable electronics (i.e. two channel), but with a single sensor and a single actuator.

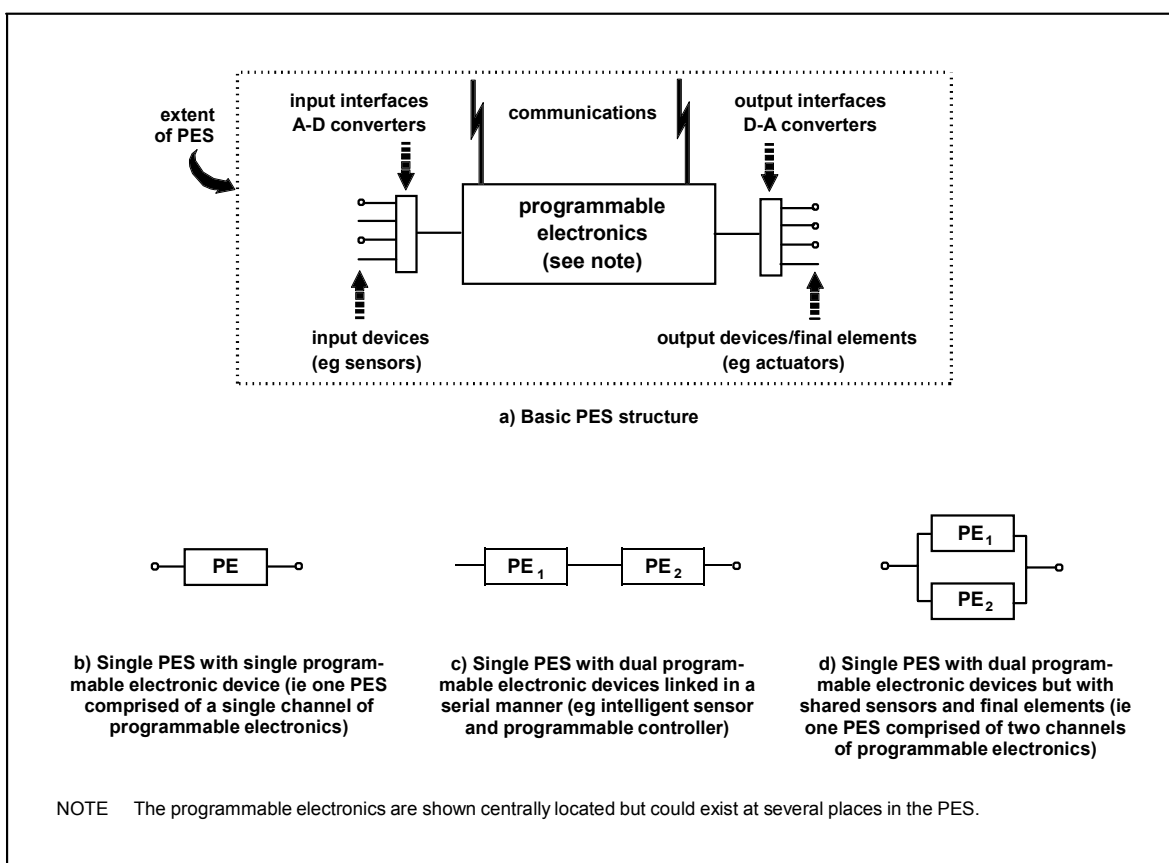


Figure 1-3 Programmable electronic system (PES): structure and terminology

Risk Combination of the probability of occurrence of harm and the severity of that harm.

Safe failure Failure which does not have the potential to put the safety-related system in a hazardous or fail-to-function state.

NOTE: Whether or not the potential is realized may depend on the channel architecture of the system; in systems with multiple channels to improve safety, a safe hardware failure is less likely to result in an erroneous shutdown.

Safety Freedom from unacceptable risk.

Safety integrity level (SIL) Discrete level (one out of a possible four) for specifying the safety integrity requirements of the safety functions to be allocated to the E/E/PE safety-related systems, where safety integrity level 4 has the highest level of safety integrity and safety integrity level 1 has the lowest.

NOTE 1: The target failure measures for the safety integrity levels are specified in Table 1-2 and Table 1-3.



Table 1-2 Safety integrity levels: target failure measures for a safety function, allocated to an E/E/PE safety-related system operating in low demand mode of operation

Safety integrity level	Low demand mode of operation (average probability of failure to perform its design function on demand)
4	10^{-5} to 10^{-4}
3	10^{-4} to 10^{-3}
2	10^{-3} to 10^{-2}
1	10^{-2} to 10^{-1}
NOTE: See notes 3 to 7 below for details on interpreting this table.	

Table 1-3 Safety integrity levels: target failure measures for a safety function, allocated to an E/E/PE safety-related system operating in high demand or continuous mode of operation

Safety integrity level	High demand or continuous mode of operation (probability of a dangerous failure per hour)
4	10^{-9} to 10^{-8}
3	10^{-8} to 10^{-7}
2	10^{-7} to 10^{-6}
1	10^{-6} to 10^{-5}
NOTE: See notes 3 to 7 below for details on interpreting this table.	

NOTE 3: The parameter in Table 1-3 for high demand or continuous mode of operation, probability of a dangerous failure per hour, is sometimes referred to as the frequency of dangerous failures, or dangerous failure rate, in units of dangerous failures per hour.

NOTE 4: This document sets a lower limit on the target failure measures, in a dangerous mode of failure, that can be claimed. These are specified as the lower limits for safety integrity level 4 (i.e. an average probability of failure of 10^{-5} to perform its design function on demand, or a probability of a dangerous failure of 10^{-9} per hour). It may be possible to achieve designs of safety-related systems with lower values for the target failure measures for non-complex systems, but it is considered that the figures in the table represent the limit of what can be achieved for relatively complex systems (for example programmable electronic safety-related systems) at the present time.

NOTE 5: The target failure measures that can be claimed when two or more E/E/PE safety-related systems are used may be better than those indicated in Table 1-2 and Table 1-3 providing that adequate levels of independence are achieved.

NOTE 6: It is important to note that the failure measures for safety integrity levels 1, 2, 3 and 4 are target failure measures. It is accepted that only with respect to the hardware safety integrity will it be possible to quantify and apply reliability prediction techniques in assessing whether the target failure measures have been met. Qualitative techniques and judgements have to be made with respect to the precautions necessary to meet the target failure measures with respect to the systematic safety integrity.

NOTE 7: The safety integrity requirements for each safety function shall be qualified to indicate whether each target safety integrity parameter is either:

the average probability of failure to perform its design function on demand (for a low demand mode of operation); or

the probability of a dangerous failure per hour (for a high demand or continuous mode of operation).

Safety lifecycle Necessary activities involved in the implementation of safety-related systems, occurring during a period of time that starts at the concept phase of a project and finishes when all of the E/E/PE safety-related systems, other technology safety-related systems and external risk reduction facilities are no longer available for use.

Safety-related system Designated system that both:

- implements the required safety functions necessary to achieve or maintain a safe state for the EUC, and
- is intended to achieve, on its own or with other E/E/PE safety-related systems, other technology safety-related systems or external risk reduction facilities, the necessary safety integrity for the required safety functions

NOTE 1: The term refers to those systems, designated as safety-related systems, that are intended to achieve, together with the external risk reduction facilities, the necessary risk reduction in order to meet the required tolerable risk.

NOTE 2: The safety-related systems are designed to prevent the EUC from going into a dangerous state by taking appropriate action on receipt of commands. The failure of a safety-related system would be included in the events leading to the identified hazard or hazards. Although there may be other systems having safety functions, it is the safety-related systems that have been designated to achieve, in their own right, the required tolerable risk. Safety-related systems can broadly be divided into safety-related control systems and safety-related protection systems, and have two modes of operation.

NOTE 3: Safety-related systems may be an integral part of the EUC control system or may interface with the EUC by sensors and/or actuators. That is, the required safety integrity level may be achieved by implementing the safety functions in the EUC control system (and possibly by additional separate and independent systems as well) or the safety functions may be implemented by separate and independent systems dedicated to safety.



NOTE 4: A safety-related system may:

- a) be designed to prevent the hazardous event (i.e. if the safety-related systems perform their safety functions then no hazard arises). The key factor here is the ensuring that the safety-related systems perform their functions with the degree of certainty required (for example, for the specified functions, that the average probability of failure should not be greater than 10^{-4} to perform its design function on demand).
- b) be designed to mitigate the effects of the hazardous event, thereby reducing the risk by reducing the consequences. As for a), the probability of failure on demand for the specified functions (or other appropriate statistical measure) should be met.
- c) be designed to achieve a combination of a) and b).

NOTE 5: A person can be part of a safety-related system. For example, a person could receive information from a programmable electronic device and perform a safety task based on this information, or perform a safety task through a programmable electronic device.

NOTE 6: The term includes all the hardware, software and supporting services (e.g. power supplies) necessary to carry out the specified safety function (sensors, other input devices, final elements (actuators) and other output devices are therefore included in the safety-related system).

NOTE 7: A safety-related system may be based on a wide range of technologies including electrical, electronic, programmable electronic, hydraulic and pneumatic.

Systematic safety integrity

Part of the safety integrity of safety-related systems relating to systematic failures in a dangerous mode of failure

NOTE: Systematic safety integrity cannot usually be quantified (as distinct from hardware safety integrity which usually can).

Validation

Confirmation by examination and provision of objective evidence that the particular requirements for a specific intended use are fulfilled.



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Section 2 – FSC configurations

2.1 Configuration overview

Section

This section provides information on the various FSC configurations. It covers the following topics:

Subsection	Topic	See page
2.1	Configuration overview	15
2.2	Single processor and single I/O	16
2.3	Redundant processor and single I/O	17
2.4	Redundant processor and redundant I/O	18
2.5	Redundant processor with redundant and single I/O	20

Basic configurations

The Fail Safe Controller supports two basic setups (single and redundant) which can be combined into four different configurations (see Table 2-1).

Table 2-1 FSC configurations

Processor	I/O	I/O racks	See subsection	Remark
Single	Single	Multiple	2.2	
Redundant	Single	Multiple	2.3	processors with inter processor communication.
Redundant	Redundant	Multiple	2.4	processors with inter processor communication, and redundancy for safety and/or availability.
Redundant	Redundant and Single	Multiple	2.5	processors with inter processor communication, redundancy for safety and/or availability, and single I/O for non-critical functions (e.g. lamps).

All FSC configurations can be used for safety applications. The preferred configuration depends on the safety requirements. The FSC configurations defined in Table 2-1 are discussed in more detail in subsections 2.2 to 2.5.

2.2 Single processor and single I/O

This FSC configuration has a single processor (Central Part) and single input and output (I/O) modules (see Figure 2-1). The I/O modules are controlled via the Vertical Bus Driver (VBD), which is located in the Central Part, and the Vertical bus (V-Bus), which controls up to 10 I/O racks. Each I/O rack is controlled via the Horizontal Bus Driver (HBD). No redundancy is present except as built into those modules where redundancy is required for safety (e.g. memory and watchdog).

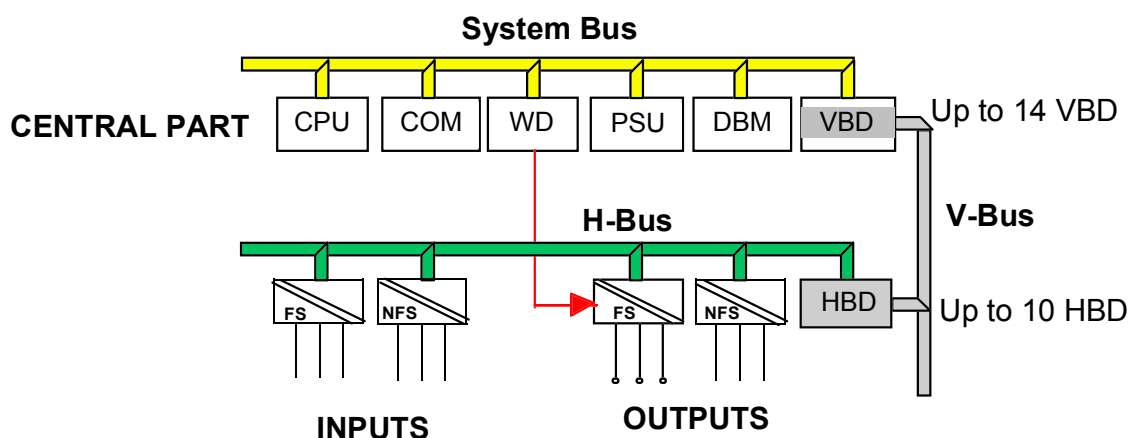


Figure 2-1 Single processor, single I/O configuration

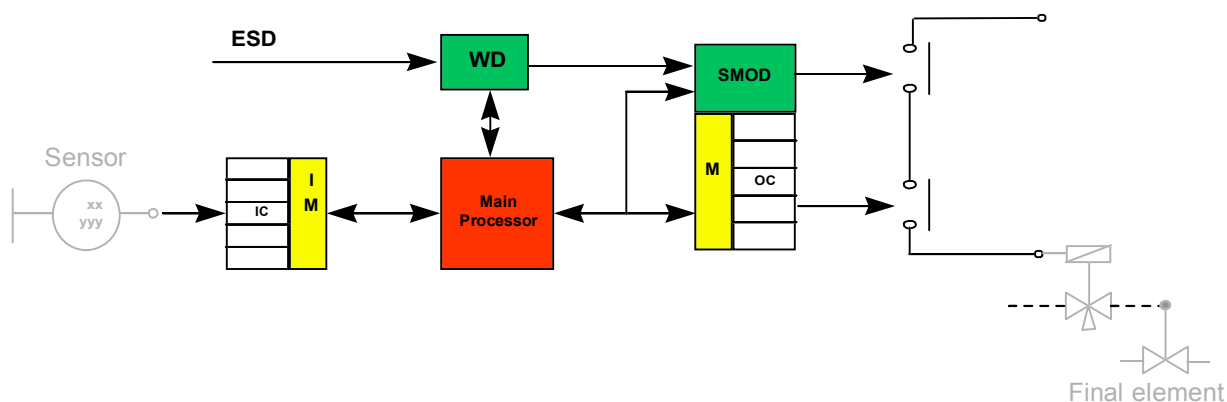


Figure 2-2 Functional diagram: single processor, single I/O



2.3 Redundant processor and single I/O

This FSC configuration has a redundant processor (Central Parts) and single input and output (I/O) modules (see Figure 2-3 and Figure 2-4). The I/O modules are controlled via the VBDs, which are located in each Central Part, and the V-Bus, which controls up to 10 I/O racks. Each I/O rack is controlled via the HBD. The processor is fully redundant, which allows continuous operation and bumpless (zero-delay) transfer in case of a processor failure.

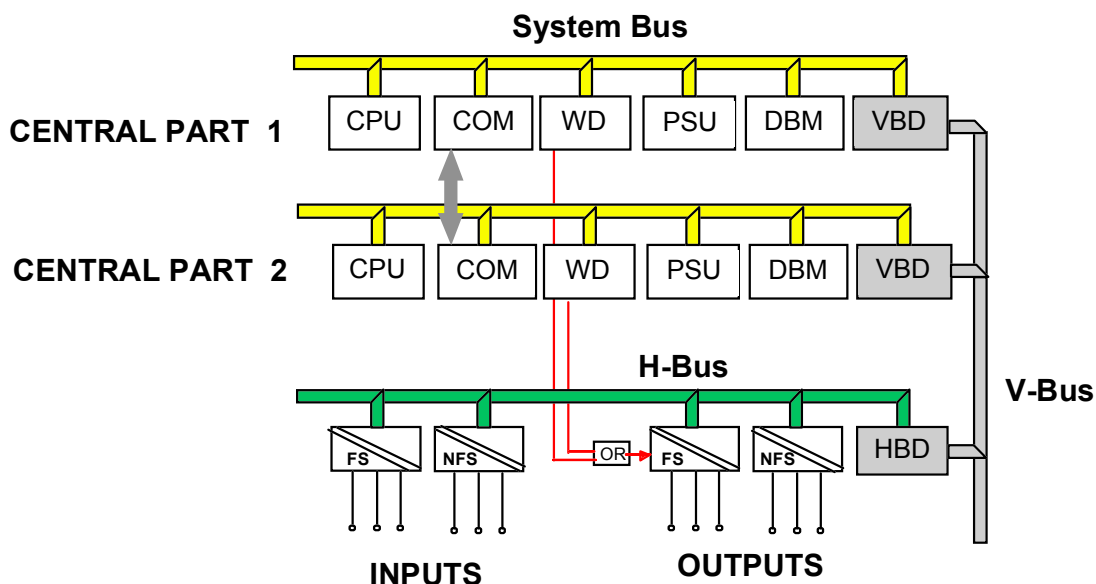


Figure 2-3 Redundant processor, single I/O configuration

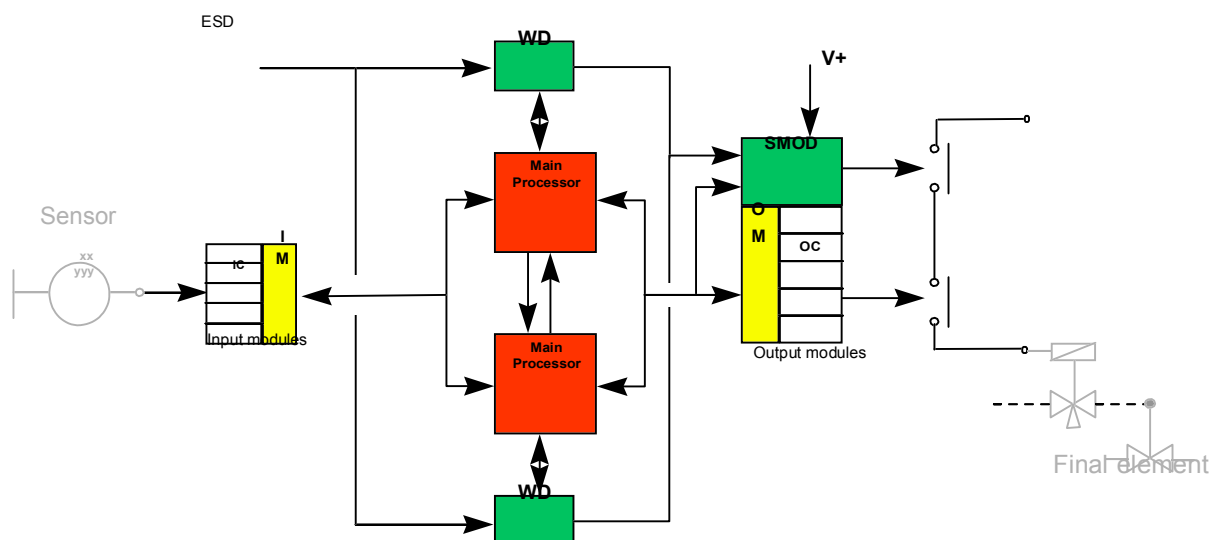


Figure 2-4 Functional diagram: redundant processor, single I/O

2.4 Redundant processor and redundant I/O

This FSC configuration has a redundant processor (Central Parts) and redundant input and output (I/O) modules (outputs connected in parallel with optional cross-monitoring) (see Figure 2-5 and Figure 2-6).

The I/O modules are controlled via the VBDs, which are located in each Central Part and the V-Bus, which controls up to 10 I/O racks. Each I/O rack is controlled via the HBD. The processor and I/O are fully redundant, which allows continuous operation and bumpless (zero-delay) transfer in case of a processor or I/O failure.

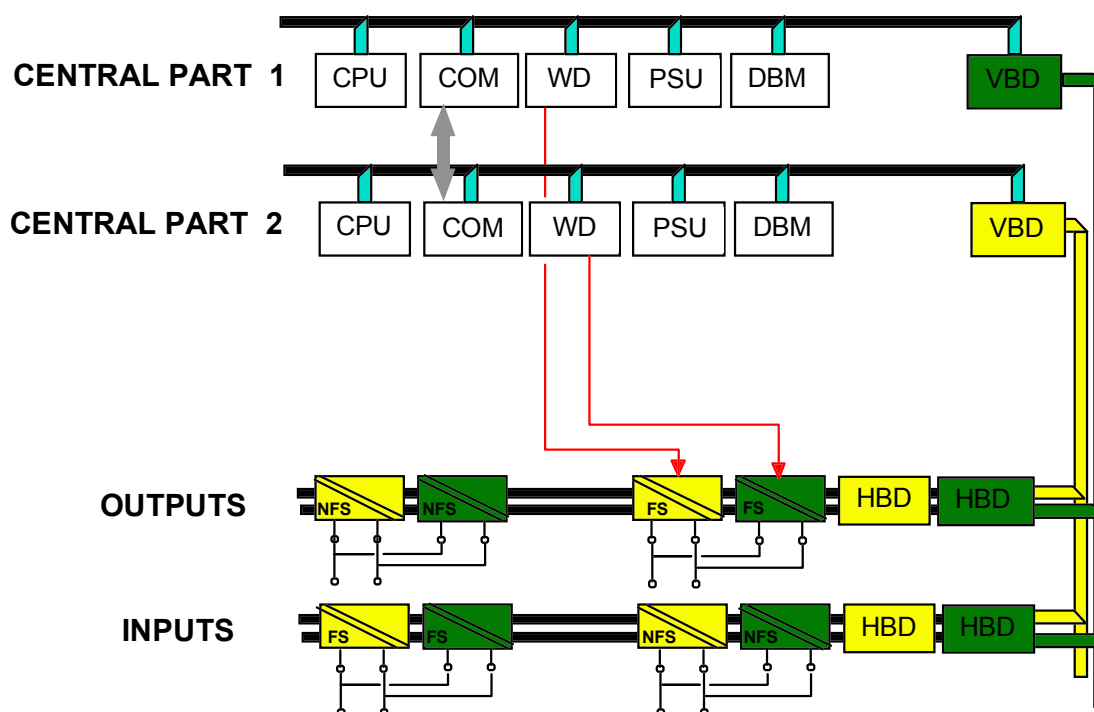


Figure 2-5 Redundant processor, redundant I/O configuration

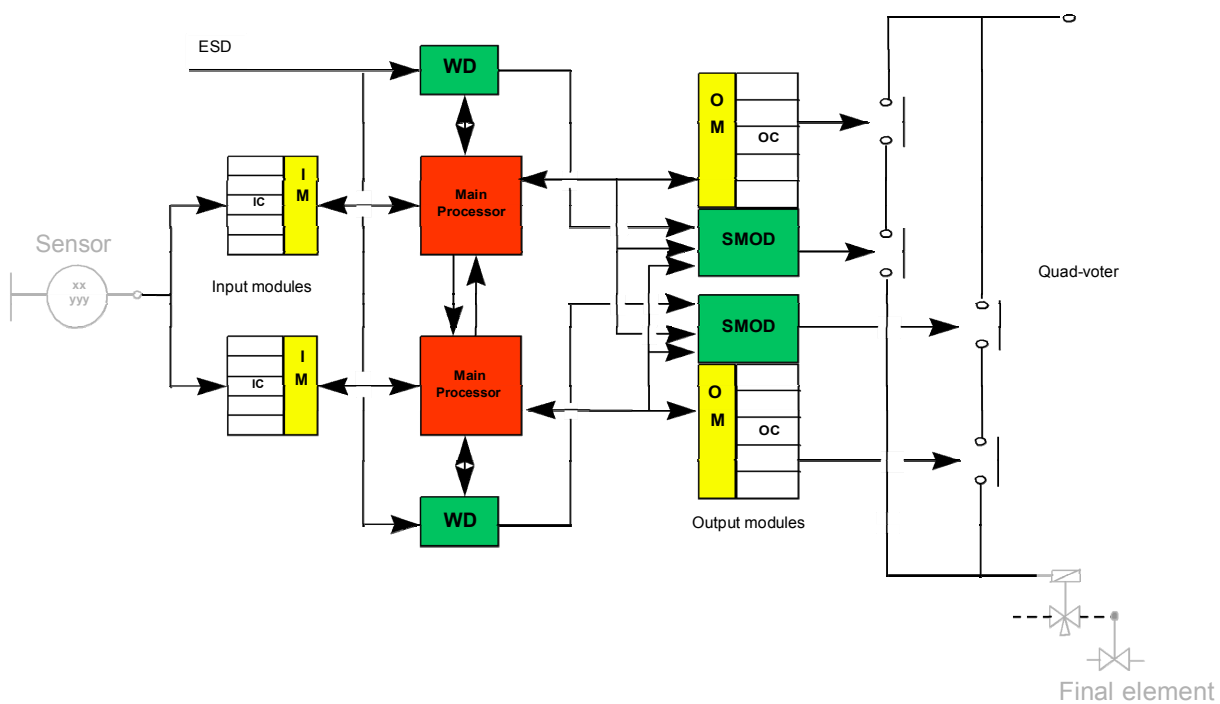


Figure 2-6 Functional diagram: redundant processor, redundant I/O

2.5 Redundant processor with redundant and single I/O

This FSC configuration has a redundant processor (Central Part) and redundant input and output (I/O) modules (outputs connected in parallel with optional cross-monitoring) combined with single input and output modules (see Figure 2-7 and Figure 2-8).

The I/O modules are controlled via the VBDs, which are located in each Central Part, and the V-Bus, which controls up to 10 I/O racks. Each I/O rack is controlled via the HBD. The processor and I/O are fully redundant, which allows continuous operation and bumpless (zero-delay) transfer in case of a processor or I/O failure of the redundant I/O modules.

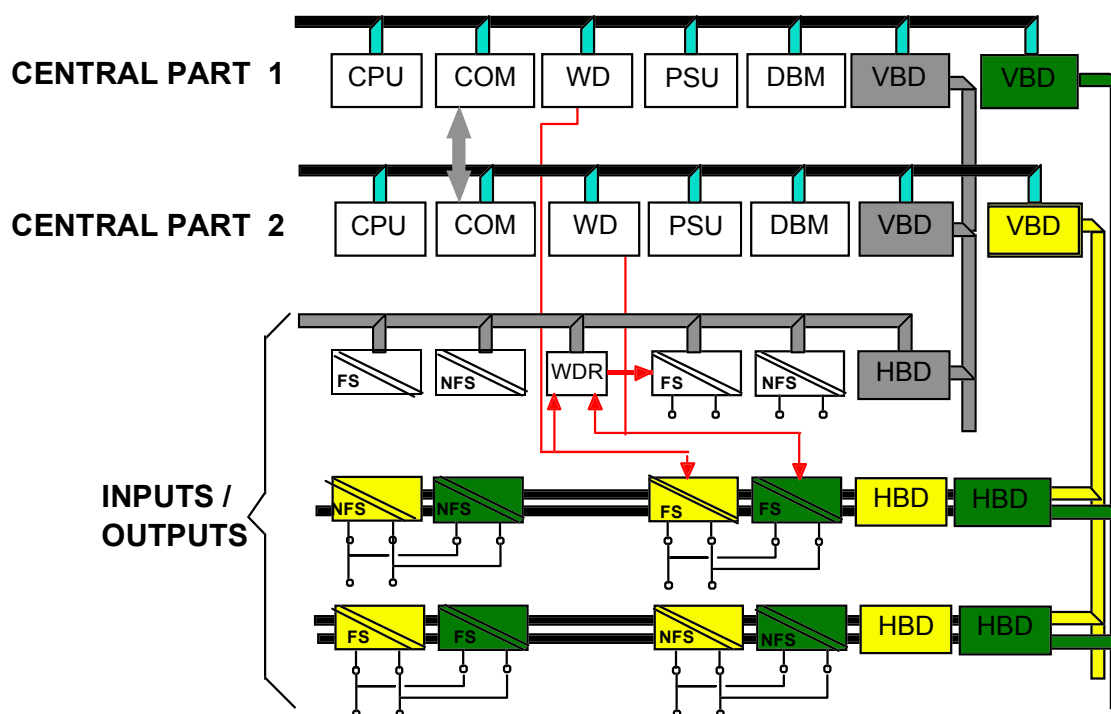


Figure 2-7 Redundant processor with redundant and single I/O configuration

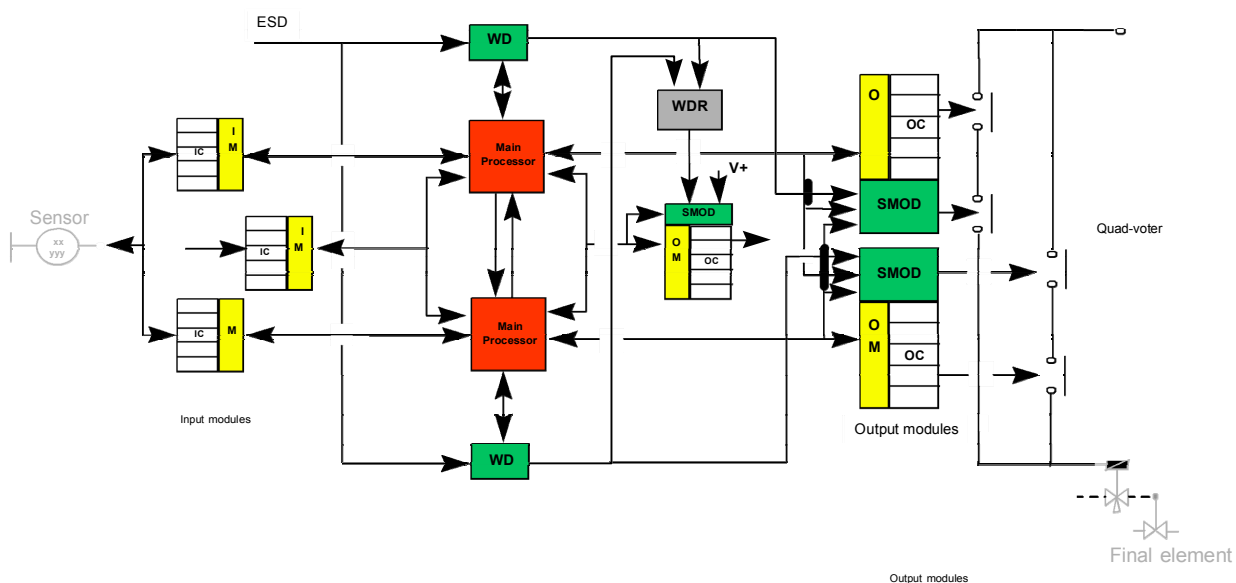


Figure 2-8 Functional diagram: redundant processor with redundant and single I/O



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Section 3 – Design phases for an E/E/PE safety-related system

3.1 Section overview

Section

This section describes the design phases for an E/E/PE safety-related system. It covers the following topics:

Subsection	Topic	See page
3.1	Section overview	23
3.2	Overall safety lifecycle.....	24
3.3	Specification of the safety class of the process	30
3.4	Specification of the instrumentation related to the safety system	31
3.5	Specification of the functionality of the safety system	34
3.6	Approval of specification	36

3.2 Overall safety lifecycle

Safety lifecycle

In order to deal in a systematic manner with all the activities necessary to achieve the required safety integrity level for the E/E/PE safety-related systems, an overall safety lifecycle is adopted as the technical framework (as defined in IEC 61508) (see Figure 3-1).

The overall safety lifecycle encompasses the following risk reduction measures:

- E/E/PE safety-related systems,
- other technology safety-related systems, and
- external risk reduction facilities.

The portion of the overall safety lifecycle dealing with E/E/PE safety-related systems is expanded and shown in Figure 3-2. The software safety lifecycle is shown in Figure 3-3. The relationship of the overall safety lifecycle to the E/E/PES and software safety lifecycles for safety-related systems is shown in Figure 3-4.

The overall, E/E/PES and software safety lifecycle figures (Figure 3-1, Figure 3-2 and Figure 3-3) are simplified views of reality and as such do not show all the iterations relating to specific phases or between phases. The iterative process, however, is an essential and vital part of development through the overall, E/E/PES and software safety lifecycles.

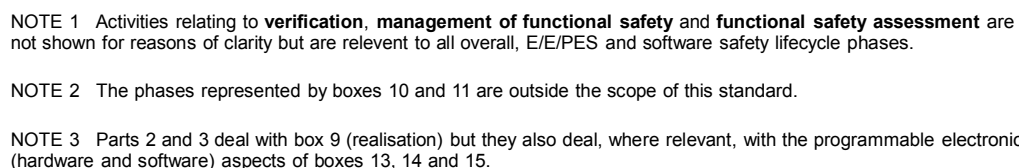


Figure 3-1 Overall safety lifecycle

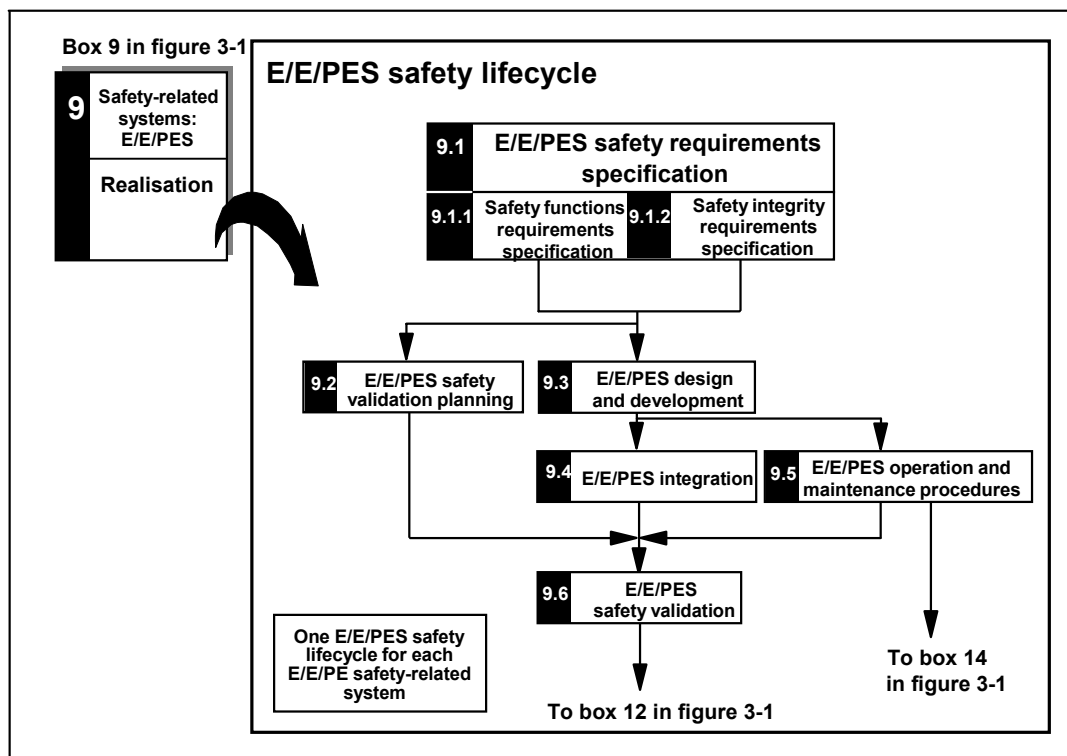


Figure 3-2 E/E/PES safety lifecycle (in realization phase)

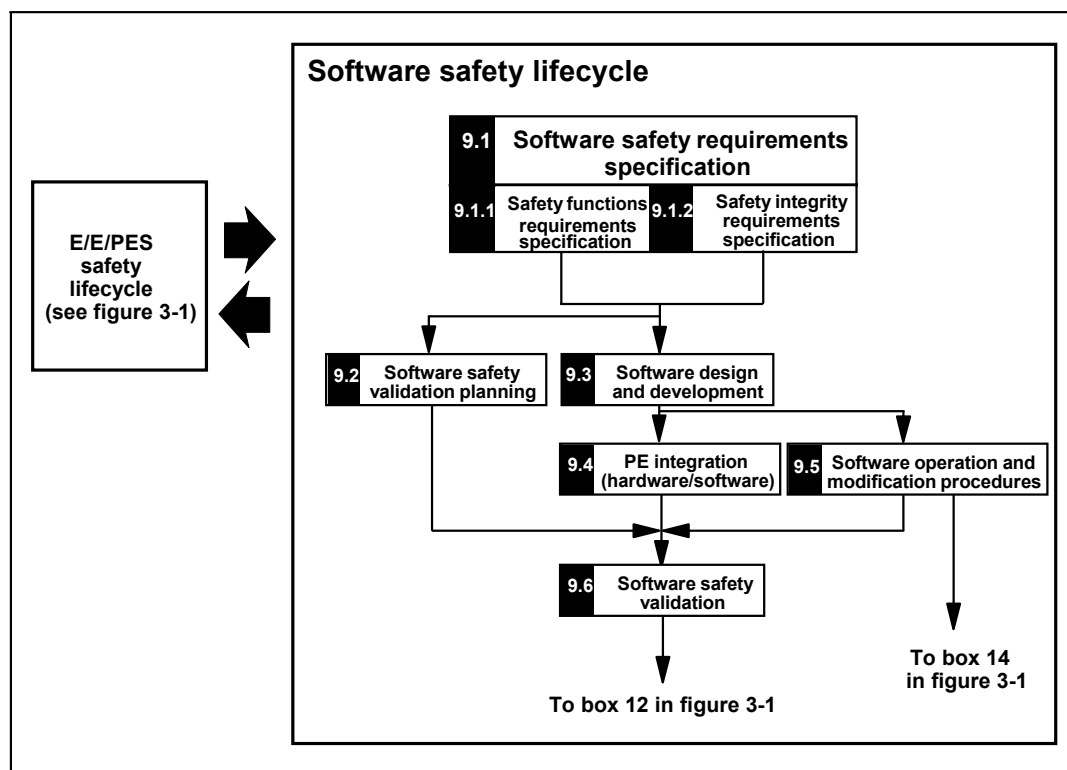


Figure 3-3 Software safety lifecycle (in realization phase)

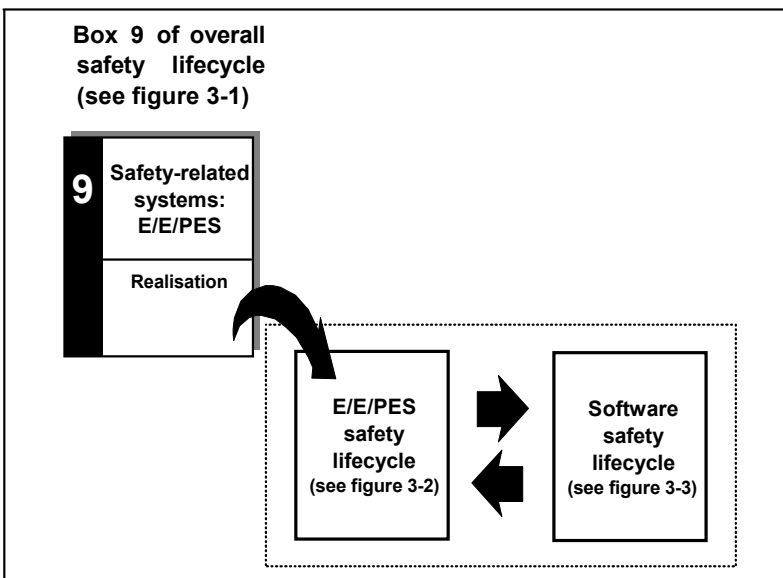


Figure 3-4 Relationship of overall safety lifecycle to E/E/PES and software safety lifecycles

Objectives

Table 3-1 indicates the objectives to be achieved for all phases of the overall safety lifecycle (Figure 3-2).

Table 3-1 Overall safety lifecycle overview

Phase	Objective	Figure 3-1 box number
Concept	To develop a level of understanding of the EUC and its environment (physical, legislative etc.) sufficient to enable the other safety lifecycle activities to be satisfactorily carried out.	1
Overall scope definition	To determine the boundary of the EUC and the EUC control system; To define the scope of the hazard and risk analysis (for example process hazards, environmental hazards, etc.).	2
Hazard and risk analysis	To identify the hazards and hazardous events of the EUC and the EUC control system (in all modes of operation), for all reasonably foreseeable circumstances including fault conditions and misuse; To identify the event sequences leading to the hazardous events identified; To determine the EUC risks associated with the hazardous events identified.	3

Table 3-1 Overall safety lifecycle overview (continued)

Title	Objective	Figure 3-1 box number
Overall safety requirements	To develop the specification for the overall safety requirements, in terms of the safety functions requirements and safety integrity requirements, for the E/E/PE safety-related systems, other technology safety-related systems and external risk reduction facilities, in order to achieve the required functional safety.	4
Safety requirements allocation	To allocate the safety functions, contained in the specification for the overall safety requirements (both the safety functions requirements and the safety integrity requirements), to the designated E/E/PE safety-related systems, other technology safety-related systems and external risk reduction facilities; To allocate a safety integrity level to each safety function.	5
Overall operation and maintenance planning	To develop a plan for operating and maintaining the E/E/PE safety-related systems, to ensure that the required functional safety is maintained during operation and maintenance.	6
Overall safety validation planning	To develop a plan to facilitate the overall safety validation of the E/E/PE safety-related systems.	7
Overall installation and commissioning planning	To develop a plan for the installation of the E/E/PE safety-related systems in a controlled manner, to ensure the required functional safety is achieved; To develop a plan for the commissioning of the E/E/PE safety-related systems in a controlled manner, to ensure the required functional safety is achieved.	8
E/E/PE safety-related systems: realization	To create E/E/PE safety-related systems conforming to the specification for the E/E/PE safety requirements (comprising the specification for the E/E/PE safety functions requirements and the specification for the E/E/PE safety integrity requirements).	9
Other technology safety-related systems: realization	To create other technology safety-related systems to meet the safety functions requirements and safety integrity requirements specified for such systems.	10
External risk reduction facilities: realization	To create external risk reduction facilities to meet the safety functions requirements and safety integrity requirements specified for such facilities.	11
Overall installation and commissioning	To install the E/E/PE safety-related systems; To commission the E/E/PE safety-related systems.	12



Table 3-1 Overall safety lifecycle overview (continued)

Title	Objective	Figure 3-1 box number
Overall safety validation	To validate that the E/E/PE safety-related systems meet the specification for the overall safety requirements in terms of the overall safety functions requirements and the overall safety integrity requirements, taking into account the safety requirements allocation for the E/E/PE safety-related systems.	13
Overall operation, maintenance and repair	To operate, maintain and repair the E/E/PE safety-related systems in order that the required functional safety is maintained.	14
Overall modification and retrofit	To ensure that the functional safety for the E/E/PE safety-related systems is appropriate, both during and after modification and retrofit activities have taken place.	15
Decommissioning or disposal	To ensure that the functional safety for the E/E/PE safety-related systems is appropriate in the circumstances during and after the process of decommissioning or disposing of the EUC.	16

Sequence of phases

The overall safety lifecycle should be used as a basis. The most important item with respect to the FSC system is the sequence of phases for the safety-related system.

The safety-related system connects to the process units, the control system and the operator interface. Consequently, the specification of the safety-related system is made late in the project. However, the first system that is required during start-up and commissioning is the safety system to ensure the safe commissioning of the total plant. The result is always a very tight schedule for the detailed design and production of the safety-related system, and this requires a system that can be designed and modified in a flexible way, and if possible is self-documenting.

The FSC safety system can be programmed during manufacturing and modified on site via the specification of the safety function (the functional logic diagrams or FLDs). The application program and updated application documentation are generated automatically and are available in a very short period of time.

Section 4 details the design phases with regard to the safety system (FSC system).

3.3 Specification of the safety class of the process

Requirement classes

Each production process must be classified with regard to safety. In Germany this classification must be done by the safety department of the company. Some applications require TÜV approval (TÜV = Technischer Überwachungsverein). The FSC system can be used in several configurations depending on the demands with respect to safety and availability. The table below shows the relation between FSC configurations and requirement classes and availability degrees, respectively.

Table 3-2 Relation between FSC configurations and requirement classes AK1-6, according to DIN V 19250

		SAFETY		
		AK 1-4	AK 1-5	AK 1-6
AVAILABILITY	NORMAL	1oo1D S + s		
	INCREASED	1oo1D S + s	R + s	R + r
	OPTIMAL	2oo2D/1oo2D R + r	R + r	

where:

- AK = Anforderungsklasse = safety requirement class
- R = Redundant processors with inter processor communication
- S = Single processor
- r = redundant I/O
- s = single I/O
- 1oo1D = 1-out-of-1 voting with diagnostics
- 1oo2D = 1-out-of-2 voting with diagnostics
- 2oo2D = 2-out-of-2 voting with diagnostics

For more information on voting refer to Section 6.



3.4 Specification of the instrumentation related to the safety system

Instrumentation related to safety system

The field instruments related to the safety system consist of valves, limit switches, high-level and low-level pressure switches, temperature switches, flow switches, manual switches, etc. Inputs and outputs used for safety applications are primarily digital. There is, however, a strong tendency towards analog I/O.

The instrumentation index generally contains:

- Tag number,
- Description,
- Make,
- Supplier, and
- Setting.



Connections to safety system

The connection to the safety system is specified in the form of a tag number with a description and termination details. The description (Service) provides additional information on the tag number and very often includes information for the signal's "health situation" (Qualification).

Configuration documents of application: DEMO_1 Date: 11-25-1997 Time: 13:39 Page: 2

Input signal specification

Type	Tag number	Service	Qualification	Location	Unit	Subunit	Sheet	Safety	Force	En.	Write	En.	SER	En.	SER	seq. no.
I	53HS-101	LAMPTEST	TEST	MCP			102	Yes	Yes	No	No	No	No	No	No	-
I	53 HS 101	LAMPTEST	"TEST"	MCP			104	Yes	Yes	No	No	No	No	No	No	-
I	91XA-651A	Door switch	Close	AH	5000	91UZ-650	0	Yes	No	No	No	No	No	No	No	-
I	ACK-PUSHBUTTON			PNL			107	Yes	Yes	No	No	No	No	No	No	-
I	ACKNOWLEDGE			DCS			106	Yes	Yes	No	No	No	No	No	No	-
I	AF_Audible			ANN			105	No	No	No	No	No	No	No	No	-
I	AF_Common_Alarm			ANN			105	No	No	No	No	No	No	No	No	-
I	ALARM-1		ALARM STATUS	DCS			107	Yes	Yes	No	No	No	No	No	No	-
I	ALARM-2		ALARM STATUS	DCS			107	Yes	Yes	No	No	No	No	No	No	-
I	AUDIBLE			ANN			107	No	No	No	No	No	No	No	No	-
I	Ack_PushButton			PNL			105	Yes	Yes	No	No	No	No	No	No	-
I	CENTR.PART-FAULT	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-
I	CLOCK-SYNC	FSC-CLOCK-SYNCHRON.	CLOCK-SYNC	SYS			0	No	No	No	No	No	No	No	No	-
I	COMMON			ANN			107	No	No	No	No	No	No	No	No	-
I	DEVICE-COM.FLT	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-
I	EARTH-LEAKAGE	EARTH LEAKAGE	PSU'S NO FAILURE	CAB			123	Yes	Yes	No	No	No	No	No	No	-
I	ENABLE	FORCE-ENABLE	ENABLE	SYS			0	Yes	No	No	No	No	No	No	No	-
I	EXT.COMMUNIC.FLT	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-
I	FIRSTUP-ALARM-1	SUBLOCATION-FSC	FIRSTUP FLAG	DCS			107	Yes	Yes	No	No	No	No	No	No	-
I	FIRSTUP-ALARM-2	SUBLOCATION-FSC	FIRSTUP FLAG	DCS			107	Yes	Yes	No	No	No	No	No	No	-
I	FIRSTUP-RESET			DCS			106	Yes	Yes	No	No	No	No	No	No	-
I	FLASHER-0.5Hz	System marker		SYS			107	No	No	No	No	No	No	No	No	-
I	FLASHER-1Hz	System marker		SYS			107	No	No	No	No	No	No	No	No	-
I	FLASHER-2Hz	System marker		SYS			105	No	No	No	No	No	No	No	No	-
I	FSC-SYSTEM-FAULT	System marker		SYS			123	Yes	No	No	No	No	No	No	No	-
I	INPUT-FAILURE	System marker		SYS			122	Yes	No	No	No	No	No	No	No	-
I	INT.COMMUNIC.FLT	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-
I	IO-COMPARE	System marker		SYS			120	Yes	No	No	No	No	No	No	No	-
I	IO-FORCED	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-
I	LAMPTEST	LAMPTEST	TEST	PNL			123	Yes	Yes	No	No	No	No	No	No	-
I	OUTPUT-FAILURE	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-
I	PSU-1	PSU-1 24VDC	NO FAILURE	CAB			123	Yes	Yes	No	No	No	No	No	No	-
I	PSU-2	PSU-2 24VDC	NO FAILURE	CAB			123	Yes	Yes	No	No	No	No	No	No	-
I	RED.INPUT-FAULT	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-
I	RESET	FSC-FAULT-RESET	RESET	SYS			121	Yes	No	No	No	No	No	No	No	-
I	RESET-ALARM	RESET ALARM	RESET	CAB			123	Yes	Yes	No	No	No	No	No	No	-
I	RESET-PUSHBUTTON			PNL			107	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR-1						109	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR-A1						111	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR-A2						111	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR-B1						112	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR-B2						112	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR-B3						112	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR-CP1						113	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR-CP2						113	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR1						110	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR2						110	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR3						110	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR_2						109	Yes	Yes	No	No	No	No	No	No	-

Figure 3-5 Specification of I/O signals for the FSC system



Process interface

The first phase of the safety system specification is the inventory of the input and output signals, i.e. the process interface.

During this specification stage, certain parameters of the I/O module must be determined by the design engineer, e.g. type of signal (digital or analog), safety relevance, fail-safe sensors, type of analog signal, scaling, etc.

System Configuration

Auto

Fail Safe Control system : Application DEMO_1 Rec. no.: 296

Variable type	: AI	Safety-related	: Yes
Tag number	: MAINLINE	Force enable	: No
Service	: NOT FAULTY	Write enable	: No
Location	:	SER enable (sequence)	: Yes 216
Unit	:	Signal type	: 4..20 mA
Subunit	:	Trm. alarm setp. low	: 3 mA
FLD(s)	: 0	Trm. alarm setp. high	: 20.93 mA
		Engineering units	: Kg
		Bottom scale	: 100 Kg
		Top scale	: 10000 Kg
		SER setpoint low	: 110 Kg
		SER setpoint high	: 9990 Kg
		Rack	: 3
		Position	: 5
		Channel	: 4
		DCS address	: 1045
		Sensor is fail-safe	: Yes

1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8

1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	2	2	5	5	1	1	1	1	9	9	9	9	9	9

Rack : 3 VBD Func.: Redundant

Press <Space> to alter, <Enter> to record status.

Start FSC Navigator System Configuration 16:02

Figure 3-6 Example of hardware specification of analog input for FSC system

The setting of the I/O parameters determine how the FSC system will treat the inputs and the outputs. The design engineer specifies the functionality required. In this way the engineer preferably delegates the safety control aspects to the main processor of the FSC system.



3.5 Specification of the functionality of the safety system

Basic function of safety system

The basic function of the safety system is to control the outputs (process) according to the predefined logic sequence based on the current status of the process received via the inputs.

The input and the output signals of a safety system are a mixture of both digital and analog signals. For digital signals, the relation between input and output can be established with logical functions including AND, OR and NOT. This is also possible with analog signals after they have been verified to be below or above a defined setpoint. In order to allow certain process conditions to occur or to continue, time functions are required within the safety system (e.g. delayed on, delayed off, pulse time). In the FSC system, the above basic functions have been extended to include a number of other functions that allow more complex functions such as counters, calculations, communication, etc.

A communication link to a supervisory control system may be required for management purposes. This is also specified in this phase of the overall design.



Relations between inputs and outputs

The second phase of the safety system specification is the detailing of the relations between inputs and outputs in order to ensure that during healthy conditions of the input signals the process stays in the predefined "operational safe status", and to ensure that the process will be directed into predefined "non-operational safe status" if an unhealthy process (input) condition occurs.

The relations are determined via functional logic diagrams (see Figure 3-7). The functional logic diagrams are created using the 'Design FLDs' option of FSC Navigator.

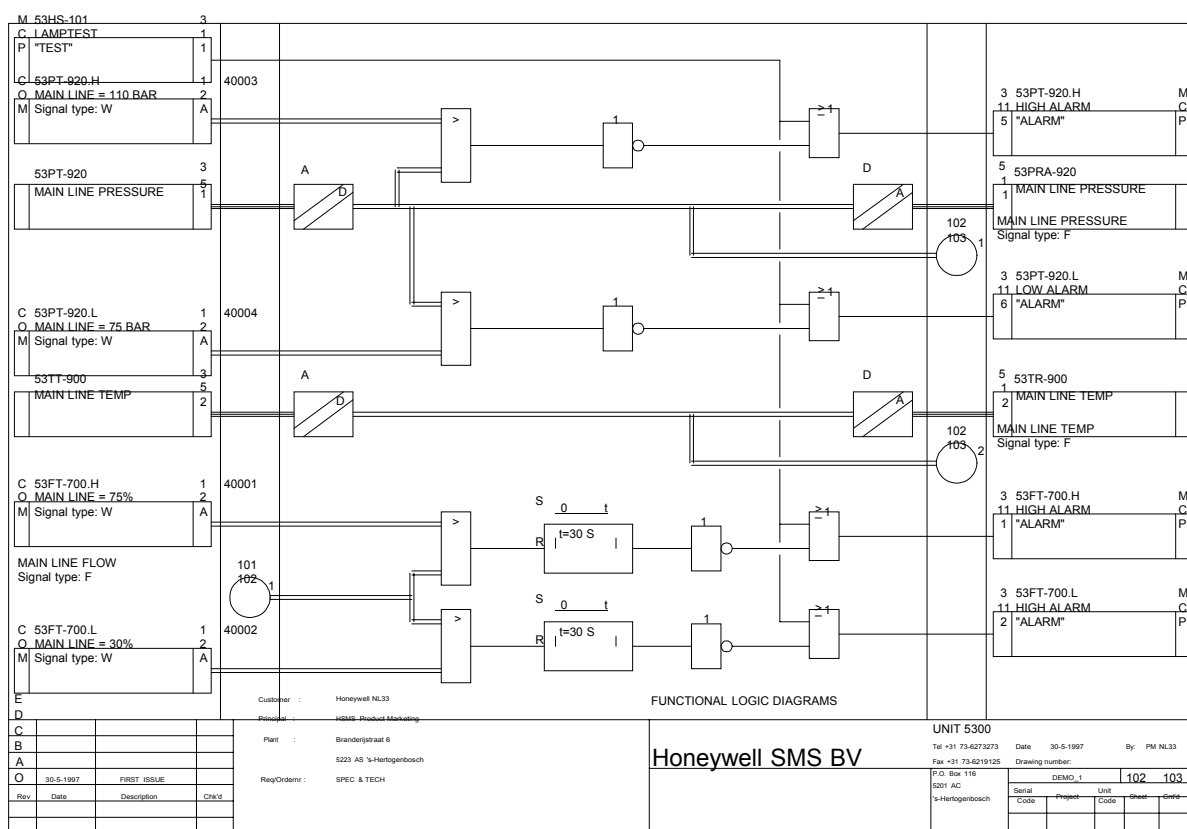


Figure 3-7 Example of functional logic diagram (FLD)

3.6 Approval of specification

Approval

The last step before acceptance of the safety system is the approval of the specifications made during the phases as described in subsections 3.3 to 3.5. The approved specification is the basis for the use of the safety system. Since the time for the specification preparation is generally too short and since the safety system influences all process units, a large number of revisions (function and termination details) to the specification may be required.

The phases as described in subsections 3.3 to 3.5 are usually performed by the customer or an engineering consultant acting on behalf of the customer. The phases that follow will normally be performed by the supplier of the safety system (e.g. Honeywell Safety Management Systems B.V. for an FSC safety system).



Section 4 – Implementation phases of FSC as a safety-related system

4.1 Overview

Section overview

This section describes the implementation phases of FSC as a safety-related system. It covers the following topics:

Subsection	Topic	See page
4.1	Overview	37
4.2	FSC project configuration.....	38
4.3	System configuration parameters	40
4.4	Specification of input and output signals.....	42
4.5	Implementation of the application software.....	43
4.6	Verification	44
4.7	Verifying an application in the FSC system.....	46



4.2 FSC project configuration

FSC Navigator

During the specification phases as described in subsections 3.3 to 3.5, the design engineer is supported by FSC Navigator (see Figure 4-1).



Figure 4-1 Main screen of FSC Navigator

FSC Navigator provides a Windows-based user interface with the FSC system. It is a powerful tool which supports the user in performing a number of design and maintenance tasks. FSC Navigator can be used to:

- configure the FSC system,
- design the application program,
- generate application documentation, and
- monitor the FSC system.

Installation database

The specification of the hardware module configuration and certain system parameters are stored in the installation database.



I/O database

The specification of the tag numbers with description, hardware configuration, etc. is stored in the input/output (I/O) database, which is created and maintained using the 'System Configuration' function of FSC Navigator. The I/O database is the basis for the design of the functionality of the safety system using functional logic diagrams (FLDs). The use of a database that contains information on the I/O signals to produce a number of different documents has the advantage that the basic information needs to be updated at one place only. Furthermore, it allows documentation to be updated in a very short period of time.

Functional logic diagrams (FLDs)

The functional logic diagrams (FLDs) define the relationship between the inputs and the outputs of the safety system (see Figure 2-14). The variable-related information entered into the I/O database is added automatically in the functional logic. FSC Navigator also checks the consistency of the information if the engineer uses tag numbers that have not been specified in the I/O database. The basic functions of FSC Navigator's project configuration features are presented in Figure 4-2.

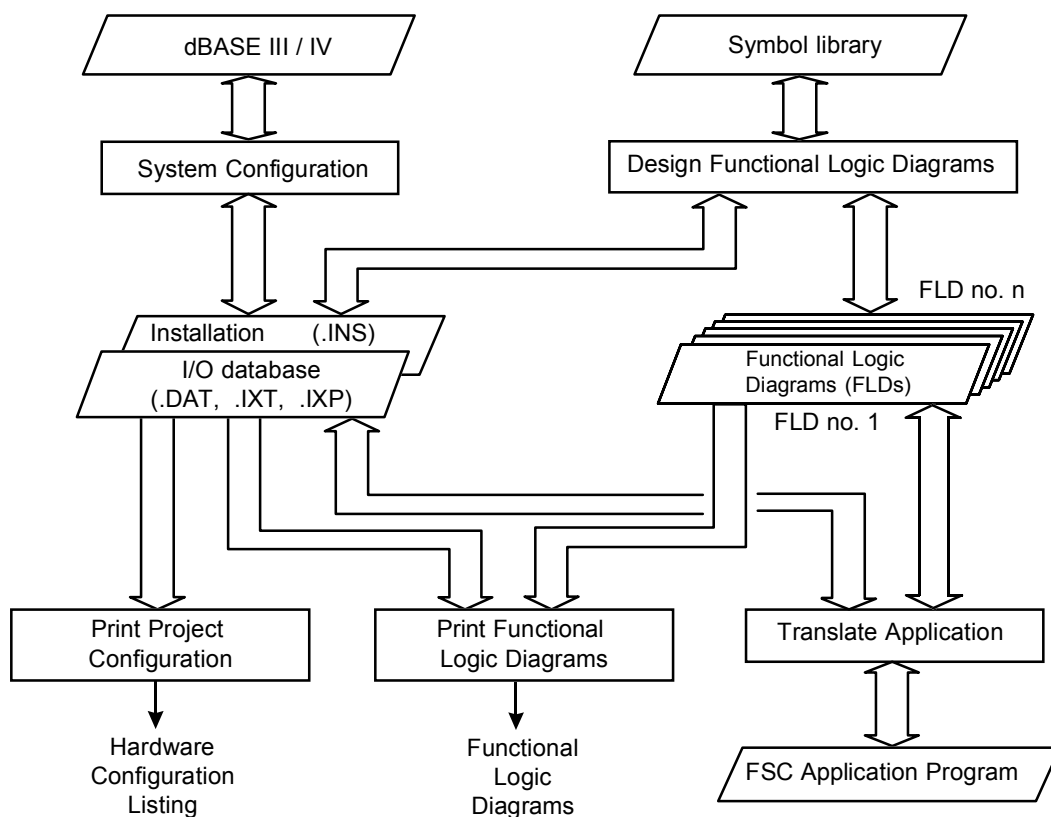


Figure 4-2 Basic functions of FSC project configuration



4.3 System configuration parameters

General

The first step in the FSC system configuration stage is the determination of the FSC system configuration parameters. The most important parameters are:

- Requirement class,
- FSC system configuration,
- Process safety time,
- Interval time between faults, and
- Power-on mode.

Each of these parameters is described in more detail below.

Requirement class according to DIN V 19250

This parameter specifies the safety requirement class for the overall system. It must be set to the requirement classification of the process parts (loops) with the highest safety demand.

Central Part configuration

One of the basic functions of the FSC system configurations is selected in accordance with the demanded safety and availability (see Table 3-2) by selecting the configuration of the Central Parts.

Process safety time

The process safety time (= fault tolerant time of the process) is the time that a fault may be present in the safety system, without possible danger for an installation or an environment. In the FSC system it specifies the period in which a self-test will be executed.

Interval time between faults

During operation, each Central Part of the FSC system performs self-tests and also tests the allocated I/O modules. If a fault is detected during self-testing, the Central Part will report the failure and take action to guarantee a safe operational result. If possible, the failure will be isolated and Central Part operation continues. If continuation of the fail-safe operation cannot be guaranteed, the Central Part shuts down. Failures of certain failure types can be isolated, but safe operation can then only be guaranteed as long as no additional faults occur, which, in correlation with the first failure, may lead to unsafe operation. Therefore, when continuing operation, there is a certain risk that such an additional correlating fault occurs. The longer the Central Part operates, the larger this risk becomes. In order to keep the risk within acceptable limits, a time



interval must be defined: the interval time between faults, which reflects the maximum period of time that the Central Part is allowed to operate after the first failure has occurred. When the interval time between faults expires, the Central Part will shut down.

The interval time between faults also defines the maximum time period allowed for a redundant system to run in single Central Part mode, in requirement classes AK5 and AK6.

The interval time between faults can be defined between 0 minutes and 22 days, or it can be completely deactivated. In the last case, organizational measures must be defined to ensure correct action on FSC system failure reports.

Power-on mode

The FSC system can operate in two basic modes:

EPROM mode

The application program is stored in EPROM using the 'Program EPROMs' option of FSC Navigator.

RAM mode

The application is stored in RAM using the load function of the 'Download Application' option of FSC Navigator. The load function can be password-protected against unauthorized use.



4.4 Specification of input and output signals

Safety	Extensive guidance in respect of safety is provided by FSC Navigator to ensure that the decisions taken by the engineer are correct. The FSC Navigator offers a number of criteria to assist in allocating the I/O signals in the safety system. For example, the system configuration function of FSC Navigator does not allow multiple allocation or connection of safety-related signals to non safety-related (untested) modules.
Input/output signals	The specification of input and output signals is partly done during the specification stage. The information entered in that stage does not contain any information on the physical allocation of the I/O signal in the safety system. The physical allocation can be described as: - the number of the rack in the cabinet(s), - the position in the rack, and - the channel number on an input or output module. This information can be sorted and presented to the user in several ways using the 'Print Project Configuration' option of FSC Navigator.
Physical allocation	The physical allocation in the FSC system can be related to a number of criteria including: - subsystems, process - units, location in the - plant, type of signal, - and personal - preference.



4.5 Implementation of the application software

Translate

The 'Translate Application' option of FSC Navigator generates the application software based on the functional logic diagrams (FLDs), the I/O database and the installation database.

Implementation

After the application software has been generated, it is transferred to EPROMs which are then placed into the FSC system.
If the FSC system is to operate in RAM mode, the application software is loaded into the RAM memory.



4.6 Verification of an application

Introduction

Throughout the design of the application, several verification steps must be accomplished to guarantee that the final application software in the FSC system meets the safety requirements of the process.

I/O signal configuration

The 'Print Project Configuration' option of FSC Navigator allows the user to create hardcopy of the I/O signal configuration as stored in the application database. The hardcopy must be reviewed to verify that the signal configuration represents the originally defined configuration.

This review may be concentrated on the safety-related configuration items, e.g. signal safety-related, force enable, hardware allocation and power-on value.

This activity covers the following aspects:

- data entry by the design engineer,
- operation of the 'System Configuration' option of FSC Navigator,
- and
- operation of the user station hardware.

Depending on local legislation, the I/O signal configuration may need to be approved by an independent certification body, e.g. TÜV.

Functional logic diagrams (FLDs)

The 'Print Functional Logic Diagrams' option of FSC Navigator allows the user to create hardcopy of the functional logic diagrams as stored in the application database. The hardcopy must be reviewed to verify that the functional logic diagrams represent the intended application program.

The activity covers the following aspects:

- data entry by the design engineer,
- operation of the 'Design FLDs' option of FSC Navigator, and
- operation of the FSC user station hardware.

Depending on local legislation, the functional logic diagrams may need to be approved by an independent certification body, e.g. TÜV.



Application software

After the application has been successfully translated and the application software has been transferred to the FSC system, the customer will verify the correct operation of the application software via a functional test which is carried out during the Factory Acceptance Test (FAT), the start-up and commissioning stage.

The customer then verifies if the original requirements have been correctly implemented in the I/O signal configuration, the system configuration and the functional logic diagrams.

The major part of this step is carried out using the 'Verify Application' option of FSC Navigator. FSC Navigator uploads the application software from the FSC system and verifies if it is "identical" to the information contained in the application database on the hard disk of the FSC user station (Figure 4-3). Subsection 4.7 describes this step in more detail.

The following aspects are covered:

- operation of the 'Translate Application' option of FSC Navigator, and
- operation of the 'Program EPROMs' option and/or the 'Download Application' option of FSC Navigator.

Finally, the assessor may carry out a sample functional test with respect to the safety-related functions in the application software.

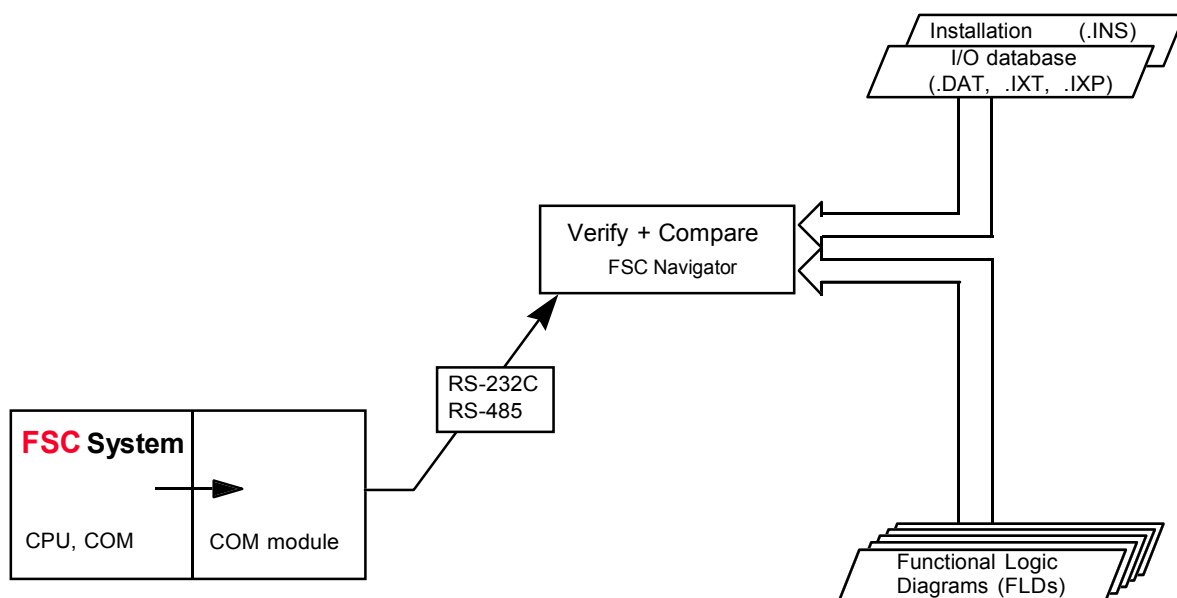


Figure 4-3 Verification of the application software

4.7 Verifying an application in the FSC system

Introduction

The 'Verify Application' option of FSC Navigator performs the verification in two main steps:

1. Verification of the FSC databases, and
2. Verification of the functional logic diagrams.

Both steps will be described briefly. For more information, refer to Section 11 of the FSC Software Manual ("Verifying an Application").

FSC database

The 'Verify Application' option of FSC Navigator compares the information in the FSC database (as stored on the FSC user station) with the application software in the FSC system. Any differences between the FSC database and the FSC application software are reported on screen and in the log file. The log file can be inspected using the 'View Log' option of FSC Navigator (see Figure 4-4)

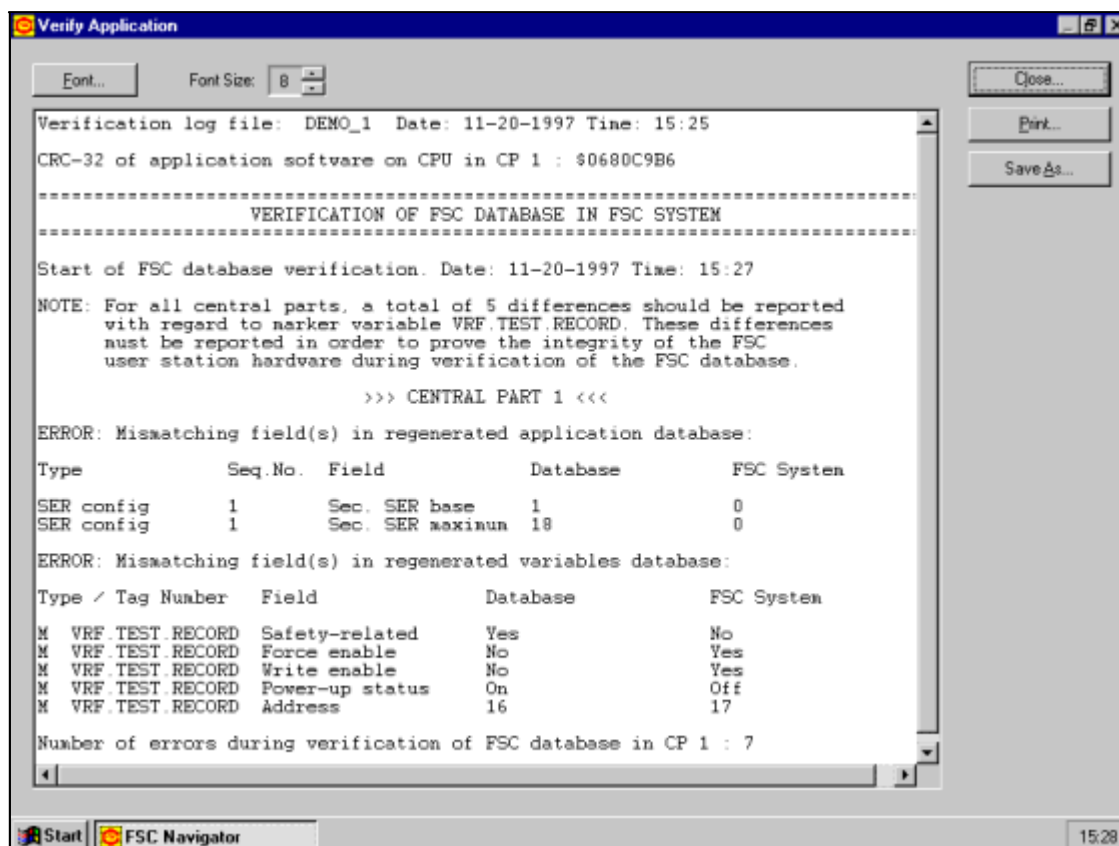


Figure 4-4 Verification log file



If any differences are detected in a field that affects related information, this field is reported. For this reason, when you decide to correct the difference and verify the application for a second time, additional differences may be reported. For example, if differences are detected in the characteristics of a specific communication channel (protocol, interface, baud rate, etc.), only the protocol is reported. Verification of the FSC database is performed once for every Central Part of the FSC configuration.

Functional logic diagrams (FLDs)

After having verified the contents of the FSC databases, FSC Navigator also verifies the functional logic diagrams (FLDs) that make up the application. Any differences found will be displayed on screen and recorded into the log file.

Test data

Due to the importance of the results of the verifications, correct execution of the 'Verify Application' option of FSC Navigator must be guaranteed.

This is realized by including test data in each application. The test data is automatically generated whenever a new application is created or when an old application is converted to a newer FSC release. When the application software is generated by the compiler, the test data is modified. During verification, these differences will then be recognized and logged. That is why the verification log file will always report a number of differences. This log file can be shown on screen or printed (see the sample report on the next page).

It must always be verified that the expected differences are actually present in the log file.

Note:

In the error report, the address field of the test variable VRF.TEST.RECORD may differ with respect to the indicated addresses contained in the database and the FSC system. The actual addresses depend on the application.



```

Verification log file: DEMO_1 Date: 11-12-1997 Time: 19:00

CRC-32 of application software on CPU in CP 1 : $05E669D6

=====
                        VERIFICATION OF FSC DATABASE IN FSC SYSTEM
=====

Start of FSC database verification: Date: 11-12-1997 Time: 19:31

NOTE: For all central parts, a total of 5 differences should be reported
with regard to marker variable VRF.TEST.RECORD. These differences
must be reported in order to prove the integrity of the FSC
user station hardware during verification of the FSC database.

>>> CENTRAL PART 1 <<<

ERROR: Mismatching field(s) in regenerated variables database:

Type / Tag number   Field                Database           FSC system
-----
M VRF.TEST.RECORD   Safety related       Yes                No
M VRF.TEST.RECORD   Force enable         No                 Yes
M VRF.TEST.RECORD   Write enable         No                 Yes
M VRF.TEST.RECORD   Power up status      On                 Off
M VRF.TEST.RECORD   Address              16                17

Number of errors during verification of FSC database in CP 1 : 5

=====
                        VERIFICATION OF FUNCTIONAL LOGICS IN FSC SYSTEM
=====

Start of functional logic diagram verification: Date: 11-12-1997 Time: 19:31

NOTE: For all central parts, a total of 4 differences should be reported
with regard to the functional logic on FLD 0. These differences
must be reported in order to prove the integrity of the FSC
user station hardware during verification of the functional logics.

>>> CENTRAL PART 1 <<<

ERROR: Regenerated symbol INVERTER not found on FLD 0
ERROR: Regenerated symbol OR GATE not found on FLD 0
ERROR: Symbol AND GATE on FLD 0 has not been regenerated.
ERROR: Symbol INVERTER on FLD 0 has not been regenerated.

Number of errors during verification of functional logics in CP 1 : 4

=====
                        TOTALS
=====

Total number of errors found during verification : 9

NOTE: All differences with regard to marker variable VRF.TEST.RECORD
and with regard to the functional logic on FLD 0 are reported
to ensure data integrity of the FSC user station.
For details refer to the FSC Safety Manual.

Verification of application completed. Date: 11-12-1997 Time: 19:31

```

Figure 4-5 Sample verification report



Section 5 – Special functions in the FSC system

5.1 Overview

Section

This section describes the special functions in the FSC system. It covers the following topics:

Subsection	Topic	See page
5.1	Overview	49
5.2	Forcing of I/O signals	50
5.3	Communication with process control systems (DCS / ICS)	53
5.4	FSC-FSC communication	55
5.5	On-line modification	59
5.6	Safety-related non-fail-safe inputs	61

Summary

The FSC system is a safety system which has a number of special functions. These functions are:

- Forcing of I/O signals (maintenance override),
- Communication with process control systems,
- Safety-related communication between FSC systems,
- On-line modification, and
- Safety-related non-fail-safe inputs.

Each of these functions is described in more detail below.

5.2 Forcing of I/O signals

General

For maintenance reasons it may be required to force an input or an output to a certain fixed state, e.g. when exchanging a defective input sensor.

This allows the sensor to be exchanged without affecting the continuity of production. While repairing the sensor, the respective input can be forced to its operational state. Forcing introduces a potentially dangerous situation as the corresponding process variable could go to the unsafe state while the force is active.

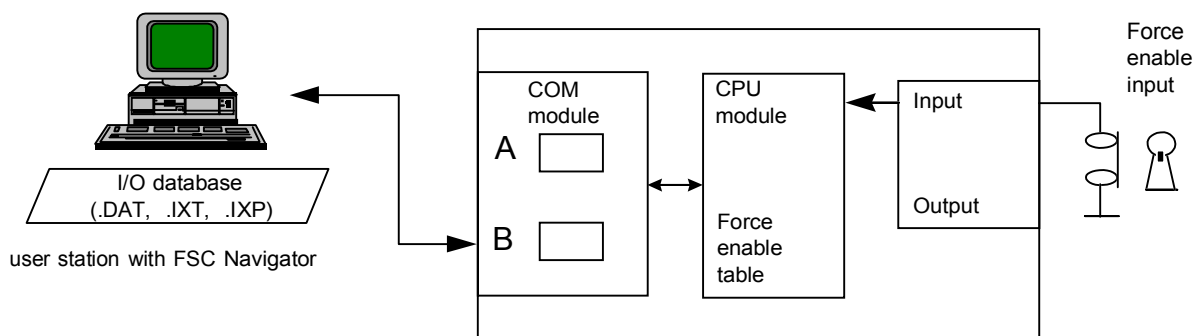


Figure 5-1 Forcing sequence

Enabling

Table 5-1 shows the procedure to include forcing in the FSC system (See also Figure 5-1):

Table 5-1 Procedure to enable the force enable flag

Step	Action
1	Define the signals that possibly require forcing during operation with regard to the safety aspects during the forced state.
2	Use the 'System Configuration' option of FSC Navigator to set the force enable flag to Yes.
3	Define the tag number and hardware allocation for the Force Enable keyswitch.
4	Translate, program EPROMs, test, etc.



Setting

I/O signals can only be forced using the Process Status Monitoring and I/O Signal Status features of FSC Navigator. Forcing is only allowed if the correct password is entered when selecting the force option.

The status of the force enable flag is also stored in the application tables in the FSC system. This has been done in such a way that a change of the force enable flag in the I/O database after translation does not allow forcing of the corresponding variable without reloading the application software.

Forces may be set high, low or on a specific value as required. Table 5-2 shows the procedure of how to use forcing.

Table 5-2 Procedure to force a variable

Step	Action
1	Activate the Force Enable keyswitch after approval by the responsible maintenance manager.
2	Use the 'Monitor System' option of FSC Navigator to select the variable that needs to be forced.
3	Select the status or value that the variable should be forced to and activate the force.

Note:

If the Force Enable key switch is deactivated, all forces are cleared.

All force actions are included in the SER report for review/historical purposes.

Checks

FSC Navigator and the FSC system carry out the following checks before the force is actually executed:

1. FSC Navigator checks if the password is activated.
2. FSC Navigator checks if the Force Enable key switch is activated.
3. FSC Navigator checks if the force enable flag in the application database is set to 'Yes'.
4. The FSC system checks if the Force Enable key switch is activated.
5. The FSC system checks if the force enable flag in the application tables is set to 'Yes'.



The FSC system continuously checks the Force Enable key switch and clears all forces immediately as soon as the Force Enable key switch is deactivated.

IO-FORCED system variable

If a force command is accepted for an input or an output, the system variable IO-FORCED is cleared, which can be used as an alarm/indication to operation.

On any subsequent force, the IO-FORCED marker will become high for one application program cycle and then become low again. When all forces are cleared, IO-FORCED becomes high again.

If one or more forces are activated, the IO-FORCED system marker is reset (see Section 6).

References

Specific TÜV requirements with the regard to forcing are described in a document by TÜV Bayern Sachsen e.V. and TÜV Rheinland entitled *Maintenance override* (available on request; please contact our Support department, tel.: +31 73-6273273, fax: +31 73-6219125). All FSC configurations meet the requirements specified in this document.



5.3 Communication with process control systems (DCS / ICS)

Exchanging process data

The FSC system can be used to exchange process data with a process control system or a man machine interface (PC).

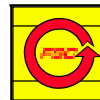
This data is represented in the functional logic diagrams (FLDs) as I/O symbols with location 'COM'. The variables with location 'COM' may only be used for *non safety-related* functions. The 'System Configuration' option of FSC Navigator sets the safety relation flag of these signals to 'No' (FALSE) and does not allow this flag to be changed. The safety relation of variables can be checked using the listing that is produced with the 'Print Project Configuration' option of FSC Navigator. Figure 5-2 below shows an example of such an input signal specification.

Configuration documents of application: DEMO_1 Date: 11-25-1997 Time: 13:39 Page: 2

Input signal specification

Type	Tag number	Service	Qualification	Location	Unit	Subunit	Sheet	Safety	Force	En.	Write	En.	SER	En.	SER	seq. no.
I	53HS-101	LAMPTEST	TEST	MCP			102	Yes	Yes	No	No	No	No	No	No	-
I	53 HS 101	LAMPTEST	"TEST"	MCP			104	Yes	Yes	No	No	No	No	No	No	-
I	91XA-651A	Door switch	Close	AH	5000	91UZ-650	0	Yes	No	No	No	No	No	No	No	-
I	ACK-PUSHBUTTON			PNL			107	Yes	Yes	No	No	No	No	No	No	-
I	ACKNOWLEDGE			DCS			106	Yes	Yes	No	No	No	No	No	No	-
I	AF Audible			ANN			105	No	No	No	No	No	No	No	No	-
I	AF_Common_Alarm			ANN			105	No	No	No	No	No	No	No	No	-
I	ALARM-1		ALARM STATUS	DCS			107	Yes	Yes	No	No	No	No	No	No	-
I	ALARM-2		ALARM STATUS	DCS			107	Yes	Yes	No	No	No	No	No	No	-
I	AUDIBLE			ANN			107	No	No	No	No	No	No	No	No	-
I	Ack_PushButton			PNL			105	Yes	Yes	No	No	No	No	No	No	-
I	CENTR.PART-FAULT	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-
I	CLOCK-SYNC	FSC-CLOCK-SYNCHRON.	CLOCK-SYNC	SYS			0	No	No	No	No	No	No	No	No	-
I	COMMON			ANN			107	No	No	No	No	No	No	No	No	-
I	DEVICE-COM.FLT	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-
I	EARTH-LEAKAGE	EARTH LEAKAGE PSU'S	NO FAILURE	CAB			123	Yes	Yes	No	No	No	No	No	No	-
I	ENABLE	FORCE-ENABLE	ENABLE	SYS			0	Yes	No	No	No	No	No	No	No	-
I	EXT.COMMUNIC.FLT	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-
I	FIRSTUP-ALARM-1	SUBLOCATION-FSC	FIRSTUP FLAG	DCS			107	Yes	Yes	No	No	No	No	No	No	-
I	FIRSTUP-ALARM-2	SUBLOCATION-FSC	FIRSTUP FLAG	DCS			107	Yes	Yes	No	No	No	No	No	No	-
I	FIRSTUP-RESET			DCS			106	Yes	Yes	No	No	No	No	No	No	-
I	FLASHER-0.5Hz	System marker		SYS			107	No	No	No	No	No	No	No	No	-
I	FLASHER-1Hz	System marker		SYS			107	No	No	No	No	No	No	No	No	-
I	FLASHER-2Hz	System marker		SYS			105	No	No	No	No	No	No	No	No	-
I	FSC-SYSTEM-FAULT	System marker		SYS			123	Yes	No	No	No	No	No	No	No	-
I	INPUT-FAILURE	System marker		SYS			122	Yes	No	No	No	No	No	No	No	-
I	INT.COMMUNIC.FLT	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-
I	IO-COMPARE	System marker		SYS			120	Yes	No	No	No	No	No	No	No	-
I	IO-FORCED	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-
I	LAMPTEST	LAMPTEST	TEST	PNL			123	Yes	Yes	No	No	No	No	No	No	-
I	OUTPUT-FAILURE	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-
I	PSU-1	PSU-1 24VDC	NO FAILURE	CAB			123	Yes	Yes	No	No	No	No	No	No	-
I	PSU-2	PSU-2 24VDC	NO FAILURE	CAB			123	Yes	Yes	No	No	No	No	No	No	-
I	RED.INPUT-FAULT	System marker		SYS			0	Yes	No	No	No	No	No	No	No	-
I	RESET	FSC-FAULT-RESET	RESET	SYS			121	Yes	No	No	No	No	No	No	No	-
I	RESET-ALARM	RESET ALARM	RESET	CAB			123	Yes	Yes	No	No	No	No	No	No	-
I	RESET-PUSHBUTTON			PNL			107	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR-1						109	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR-A1						111	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR-A2						111	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR-B1						112	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR-B2						112	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR-B3						112	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR-CP1						113	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR-CP2						113	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR1						110	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR2						110	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR3						110	Yes	Yes	No	No	No	No	No	No	-
I	SENSOR_2						109	Yes	Yes	No	No	No	No	No	No	-

Figure 5-2 Example of a printout of engineering documents



Protocols

The following communication protocols are used for communication with process control systems and computer equipment running visualization programs:

- TPS network protocol,
- Modbus RTU and Modbus H&B protocol,
- RKE3964R protocol, and
- FSC-DS protocol.

For details on these communication protocols refer to Appendix F of the FSC Software Manual ("Communication Manual").



5.4 FSC-FSC communication

Networks

FSC systems may be interconnected to form a safety-related communication network (see Figure 5-3).

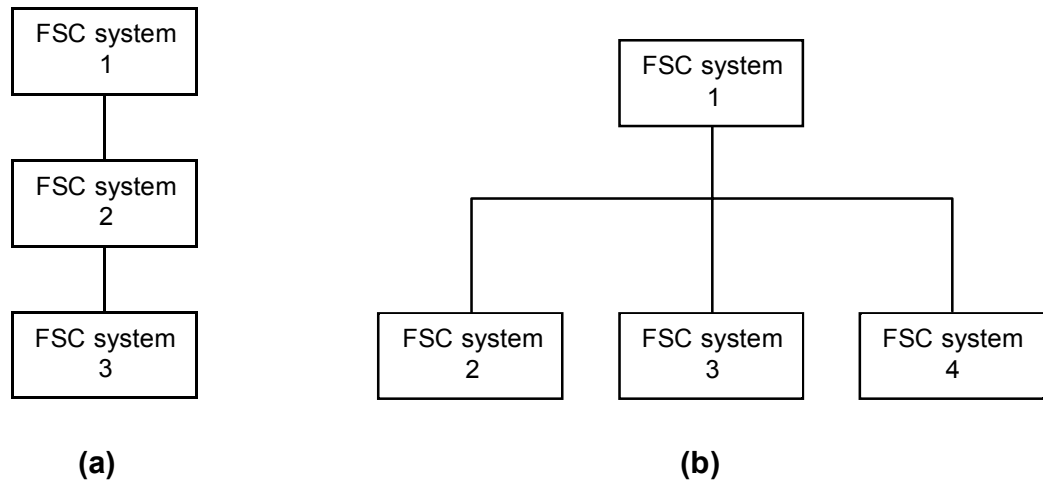


Figure 5-3 Examples of FSC communication networks

FSC communication networks can be used to allow multiple FSC systems to exchange data in order to perform a joint task. Another possibility is gathering of sequence-of-event (SOE) data of multiple FSC systems at a single point in the network.

Master/slave

Within the network, systems may be connected in pairs (point-to-point) (see Figure 5-3a), or multiple systems may be connected to the same link (multidrop) (see Figure 5-3b).

For every communication link, one FSC system operates as a master and the other systems operate as a slave. Every program cycle the master sends data to the slave and initiates a request for data from the slave. The slave sends data after receipt of the data request from the master. Data integrity is ensured by using the same protocol and surveillance mechanisms as used for communication between Central Parts in redundant FSC configurations.

More than one slave may be connected to one master. One slave may have multiple master (see Figure 5-4).

All systems within the network must have a unique system number.

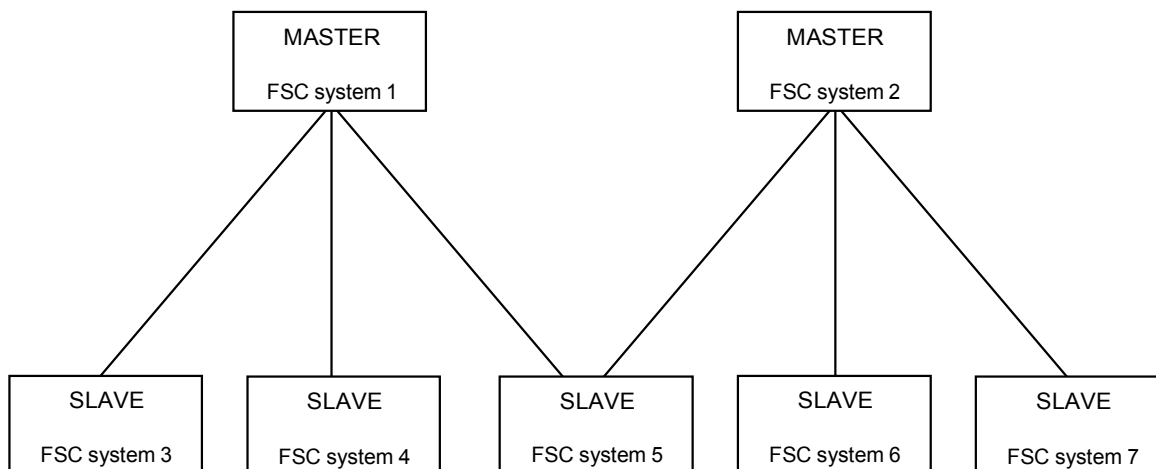


Figure 5-4 FSC master/slave interconnection

Data that is used for communication between FSC systems is represented in the function logic diagrams as I/O symbols with the location 'FSC'. Variables with location 'FSC' can be of type I, O (markers), BI or BO (registers), and may be configured for both safety-related and non safety-related functions.

Redundant communication

For redundant systems, redundant FSC links must be used (see Figure 5-5). This results in a single-fault-tolerant communication network.

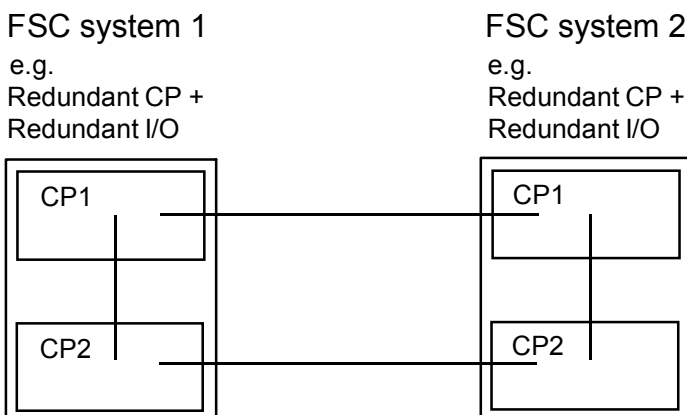


Figure 5-5 Redundant FSC communication link



Response time

The response time depends on the application program cycle time of the systems and the type of the communication link.

Point-to-point

The response time is the sum of the application program cycle times of the master and slave system. The result will always be less than 1 second. This is represented in the following formula:

$$T_{\text{resp}} = T_{\text{am}} + T_{\text{as}}$$

Where: T_{am} = Master application program cycle time.

T_{as} = Slave application program cycle time.

Multidrop

The maximum response time is the sum of the application program cycle times of the master and the slave system plus the total communication time needed to serve all systems connected to the multidrop network. This is represented in the following formula:

$$T_{\text{resp}} = T_{\text{am}} + T_{\text{as}} + \sum_{S=1}^{63} (F_1 + 2 T_r) + (F_2 + 8 T_r) (M_{\text{bs}} + R_{\text{bs}}) + F_3 (M_{\text{cs}} + R_{\text{cs}})$$

Where:

T_{am} = Master application program cycle time.

T_{as} = Slave application program cycle time.

T_r = Transmission delay in the physical communication network (0 for direct cable connections < 1 km).

F_1, F_2, F_3 = Performance factors (in ms), depending on the baud rate (see table below)

Table 5-3 Performance factors

Baud rate	Performance factors		
50 K:	$F_1 = 19$	$F_2 = 19$	$F_3 = 6.5$
125 K:	$F_1 = 11.5$	$F_2 = 11.5$	$F_3 = 2.5$
1 M:	$F_1 = 7$	$F_2 = 13$	$F_3 = 0$
2 M:	$F_1 = 6.5$	$F_2 = 10$	$F_3 = 0$

Notes:

- 1) With both redundant links operational, a typical value of F_1 , F_2 and F_3 is half the maximum value.
- 2) T_r , F_1 , F_2 and F_3 are 0 if the system number is not used as a system number for a slave system.

M_{bs}, R_{bs} = The number of data blocks to be sent.

$M_{bs} (R_{bs})$ is the number of 256-byte blocks configured for transfer of Marker (Register) data from the slave system to the master system. If the number of bytes is not an exact multiple of 256 bytes, an extra block must be allocated, for example:

1. A slave sends 48 bytes of marker data and 400 bytes of register data to the master system.

For this slave system $M_{bs} = 1$ and $R_{bs} = 2$.

2. A slave sends 256 bytes of marker data to the master system. No register data is sent.

For this slave $M_{bs} = 1$ and $R_{bs} = 0$.

M_{cs}, R_{cs} = The number of data bytes to be sent.

$M_{cs} (R_{cs})$ is the number of 16-byte blocks configured for transfer of Marker (Register) data from the slave system to the master system. If the number of bytes is not an exact multiple of 16 bytes, an extra block must be allocated.

Timeout time

All systems within the network monitor the operation of the communication link by means of timeouts.

The timeout depends on the system function and the type of the communication link (see Table 5-4).

Table 5-4 FSC-FSC communication timeout

Link type	System	Timeout
Point to point	Master	Response of the slave is expected within the same application program cycle.
	Slave	1 second
Multidrop	Master	Configured communication timeout (refer to Section 4 of the FSC Software Manual).
	Slave	2x configured communication timeout time (refer to Section 4 of the FSC Software Manual).

Note:

If communication fails via all links, the safety-related variables I and BI of location 'FSC' that are allocated to the system connected to the link are set to 0. The non safety-related variables are frozen at their last received state.



5.5 On-line modification

Introduction

On-line modification (OLM) is an FSC system option which allows you to modify the application software, system software and the FSC hardware configuration of redundant systems while the system remains operational.

During on-line modification, the changes are upgraded in one Central Part at a time. Meanwhile, the other Central Part can continue safeguarding the process.

Compatibility check

During the upgrade, the FSC system performs a compatibility check across the application-related data, in order to guarantee a safe changeover from the old software to the new software. The system reports the FLD numbers of the functional logic diagrams that have changed (see Figure 5-6). This allows easy verification of the implemented modifications.

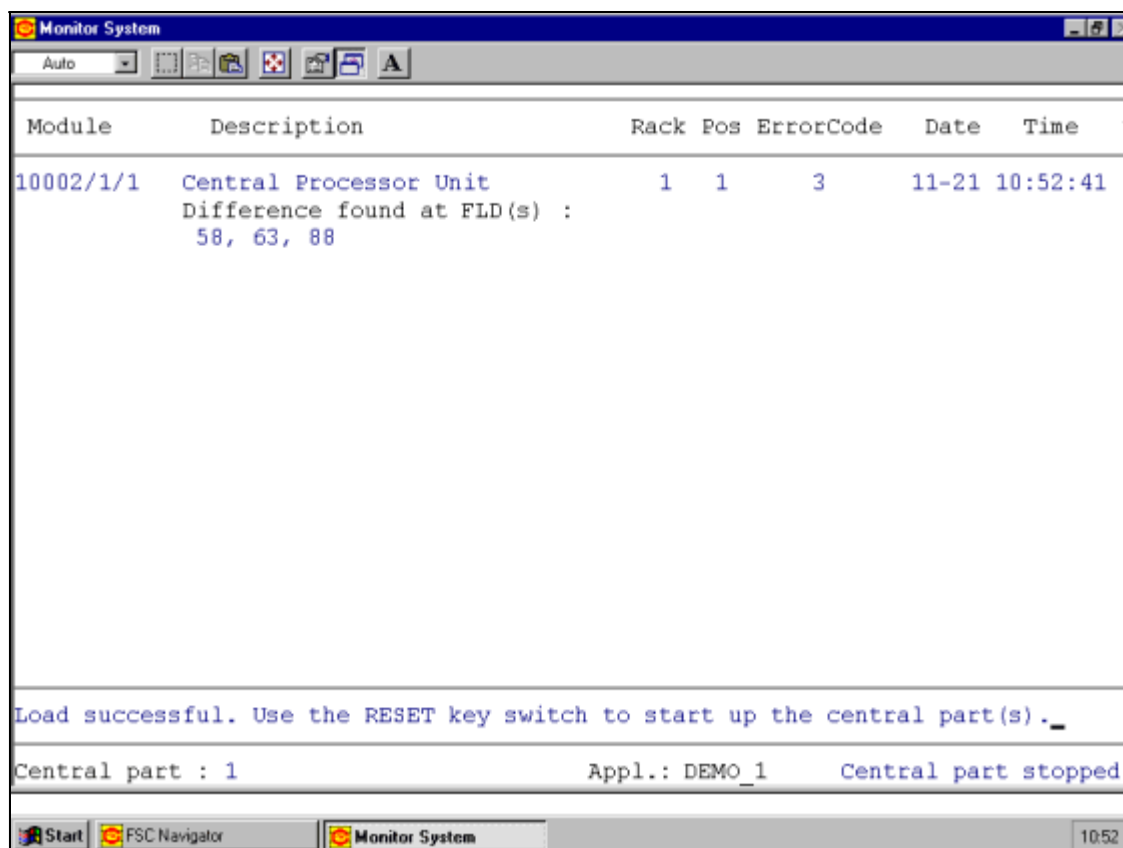


Figure 5-6 Sheet differences



Using the on-line modification option of the FSC system, changes in the functional logic diagrams (FLDs), the FSC system configuration and the system software can be implemented in the system without the need for a plant shutdown.

For details on on-line modification, refer to Appendix D of the FSC Software Manual ("On-Line Modification").

When modifications in the application are implemented, only a functional logic test of the modified functions is required by, for example, TÜV, when the final verification of the implemented changes is obtained via the sheet difference report of the FSC system and the 'Verify Application' option of FSC Navigator.

Note:

If a function block is changed, a difference will be reported for all functional logic diagrams that use this function block.

During on-line modification, the 'Verify Application' option of FSC Navigator may be used to log all revision information. For more information, refer to Section 11 of the FSC Software Manual ("Verifying an Application").

FSC networks

If a system has been integrated into an FSC communication network, it performs a compatibility check for all connected systems.

If inconsistencies are detected or if the check for a specific system cannot be completed for any other reason, an error message is generated in the extended diagnostics. In case of such an error, no data will be exchanged with the system after start-up. The communication can only be re-established after successful completion of the compatibility check by any of the systems that communicate with each other, initiated via a CPU reset.



5.6 Safety-related non-fail-safe inputs

Introduction

Safety-related inputs require the use of fail-safe input module (e.g. 10101/2/1 for digital inputs and 10105/2/1 for analog inputs). In addition, it is also required that fail-safe input devices are used (e.g. sensors, switches and transmitters). If the input device is not fail-safe, then redundant sensors (transmitters) and redundant inputs are required.

Depending on the number of sensors and the FSC configuration applied, the system offers a variety of "sensor redundancy configurations".

Figure 5-7 shows an example of redundancy type 2o2, which can be used for VBD functions with redundant I/O.

System Configuration

Auto

Fail Safe Control system : Application DEMO_1 Rec. no.: 72

Variable type	: I	Safety-related	: Yes
Tag number	: 91XA-651A	Force enable	: No
Service	: Door switch	Write enable	: No
Qualification	: Close	SER enable	: Yes
Location	: AH	Rack	: 4
Unit	: 5000	Position	: 1
Subunit	: 91UZ-650	Channel	: 4
FLD(s)	: 0	DCS address	: 123
		Sensor is fail-safe	: No
		Sensor is redundant	: Redund. 2O2
		Max. discrepancy time	: 10 s
		Maximum on-time	: 28800 s

1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	
1	1	1	1					1	1	1	1	1	1	1			1	1
0	0	0	0					0	0	0	0	0	0	0			0	0
1	1	1	1					2	2	2	2	2	2	2			2	2
0	0	0	0					0	0	0	0	0	0	0			0	0
1	1	5	5					1	1	9	9	1	1	1	1		1	1

Rack : 4 VBD Func.: Redundant

Press <Space> to alter, <Enter> to record status.

Start FSC Navigator System Configuration 14:11

Figure 5-7 Configuration of a redundant input

Digital inputs

To check the safety capability of the sensors, they must switch within a certain time interval specified in the configured maximum on time, which can be set in the range of 1 second to 2047 minutes.

If the maximum on-time is exceeded, the resulting sensor status is executed as 'unhealthy'. To detect if all inputs execute the switch-defined function, an extra timer is added: the maximum discrepancy timer. If the maximum on timer or the maximum discrepancy timer expires, a redundant input fault (system alarm marker) and a sensor fault alarm are generated.

Note:

The maximum on time may also be deactivated. In that case organizational procedures must exist that ensure periodical testing of the sensors.

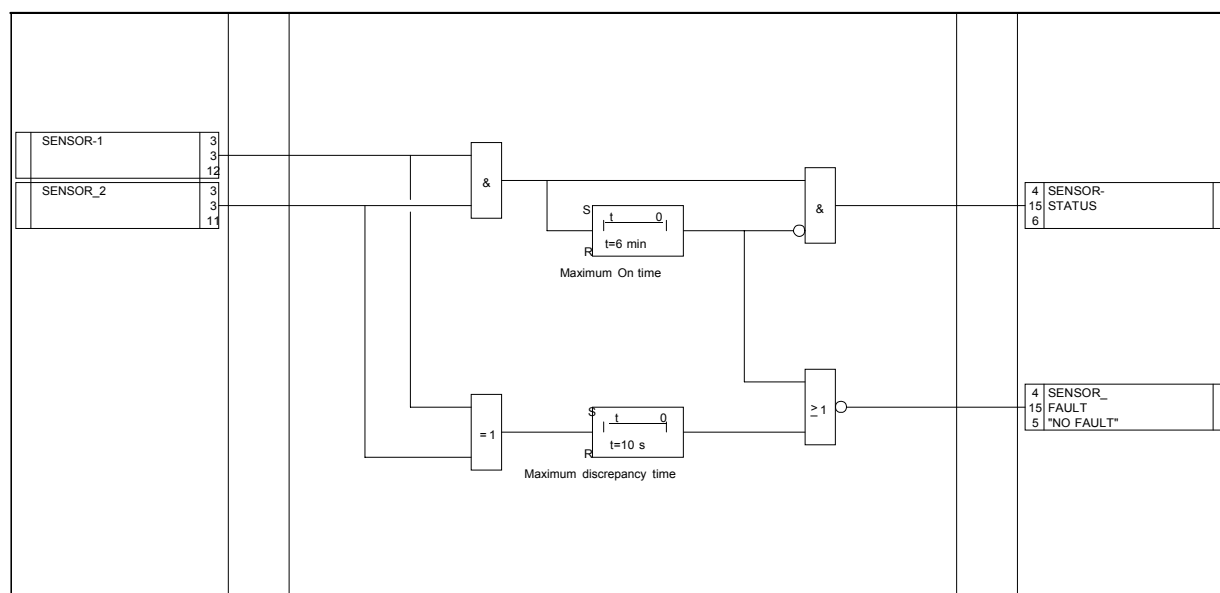


Figure 5-8 Example of functionality of a redundant digital input function



Analog inputs

For analog inputs, the system monitors if the difference between the transmitter values does not exceed a predefined value. The maximum allowable difference is specified in the maximum discrepancy value. If the difference between the transmitter values exceeds the maximum value, a redundant input fault (system alarm marker) and transmitter fault alarm are generated.

The safety-related redundant input configurations are described in detail in Appendix C of the FSC Software Manual ("Safety-related inputs with non-fail-safe sensors").



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Section 6 – FSC system fault detection and response

6.1 Section overview

Section overview

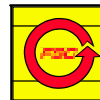
This section describes how the FSC detects system faults and how it responds to them. It covers the following topics:

Subsection	Topic	See page
6.1	Section overview	65
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6.3	FSC diagnostic inputs	69
6.4	FSC system alarm markers	70
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Introduction

Progressive test software and the use of dedicated hardware allow the FSC system to detect a number of faults in the field instrumentation and all predefined faults according to the FMEA model applied within the FSC system itself, and to provide adequate diagnostics on any detected fault. As a result, the system is able to respond as a fail-safe system in accordance with its specifications as projected during the safety specification stage.

Apart from safety, the FSC system fault detection and response strategy also provides optimum availability. As the system is able to locate faults accurately, the faulty part can be isolated from the process to obtain a safe process state while minimizing the effect on the remaining process parts.



Detected faults are reported via extended diagnostics of the FSC system, via channel-specific diagnostic markers and via system alarm markers. The diagnostic and alarm markers can be used in the application software, e.g. to generate an operator alarm or to be passed to a control system for further processing.

This section describes the behavior of the FSC system in case of faults and how alarms can be used within the application.



6.2 Voting

Voting

The FSC system is available in single and redundant mode, both for Central Part and I/O, in several combinations. For details on the various FSC configurations refer to Section 2. If the Central Part and I/O are operating in single mode, it is obvious what will happen in case a fault is detected: the Central Part or I/O will go to the safe (i.e. non-operational) state. For redundant Central Parts and/or I/O, this is less obvious, and users may want to define the system response in case a fault is detected in one part of the redundant components. This is the reason that voting has been incorporated into the system, which allows the users to optimize the system response to his safety needs.

Single components

For all single components in the FSC system, two voting schemes are available depending on the hardware that is being used. The table below lists the various options.

Table 6-1 Voting schemes for single FSC components

Voting scheme	Used for hardware modules...	Fault results in...
1oo1D	diagnostics capabilities (e.g. 10101/./. digital input modules)	switch-off
1oo1	without diagnostic capabilities (e.g. 10104/./. digital input modules)	incorrect operation or switch-off

The default voting scheme for single Central Parts is 1oo1D.

Redundant components

Redundant components have more voting schemes to choose from, depending on the hardware that is being used and on the primary action in case a fault is detected: switch-off or continue. Table 6-2 and Table 6-3 below list the various options.

Table 6-2 Voting schemes for redundant components

	Hardware	
Primary action at fault	Fail-safe	Non fail-safe
Safety (switch-off)	1oo2D	1oo2
Availability (continue)	2oo2D	2oo2

The default voting scheme for redundant Central Parts is 1oo2D.

Table 6-3 Explanation of redundancy voting schemes

Voting scheme	Used for hardware modules...	Primary action directed at...	Response to faults
1oo2	without diagnostics capabilities (e.g. 10104//. digital input modules)	safety (switch-off)	The first fault may result in switch-off as the faulty module may overrule the correct one.
2oo2	without diagnostics capabilities (e.g. 10104//. digital input modules)	availability (continue)	The first fault may result in incorrect operation as the faulty module may overrule the correct one.
1oo2D	with diagnostics capabilities (e.g. 10101//. digital input modules)	safety (switch-off)	For detected faults, operation continues as desired. A fault that cannot be detected by the diagnostics (probability = $1 - \text{diagnostic coverage}$) may result in switch-off as the faulty module may overrule the correct one.
2oo2D	with diagnostics capabilities (e.g. 10101//. digital input modules)	availability (continue)	For detected faults, operation continues as desired. A fault that cannot be detected by the diagnostics (probability = $1 - \text{diagnostic coverage}$) may result in incorrect operation as the faulty module may overrule the correct one.



6.3 FSC diagnostic inputs

General

Apart from the alarm markers, a variety of diagnostic inputs are available to indicate the diagnostic status of a specific I/O channel allocated to an FSC fail-safe I/O module.

Diagnostic inputs

The inputs can be used in the functional logic diagrams. Table 6-4 shows the available diagnostic inputs and the I/O modules for which they exist.

Table 6-4 Diagnostic inputs

Type	I/O module
I/O type I	10101/1/1, 10101/1/2, 10101/1/3, 10101/2/1, 10101/2/2, 10101/2/3
I/O type O	10201/1/1, 10201/2/1, 10202/1/1, 10203/1/2, 10212/1/1, 10213/1/1, 10213/1/2, 10213/1/3, 10213/2/1, 10213/2/2, 10213/2/3, 10214/1/2, 10215/1/1, 10215/2/1, 10216/1/1, 10216/2/1, 10216/2/3
I/O type AI	10102/1/1, 10102/1/2, 10102/2/1, 10105/2/1
I/O type AO	10205/1/1, 10205/2/1
Sens AI	10102/1/1, 10102/1/2, 10102/2/1, 10105/2/1 transmitter status
LoopO	10214/1/2, 10216/1/1, 10216/2/1, 10216/2/3 loop status
WD ../../..	10201/1/1, 10201/2/1, 10202/1/1, 10203/1/2, 10212/1/1, 10213/1/1, 10213/1/2, 10213/1/3, 10213/2/1, 10213/2/2, 10213/2/3, 10214/1/2, 10215/1/1, 10215/2/1, 10216/1/1, 10216/2/1, 10216/2/3

If the channel status is healthy, its diagnostic input is high. If a fault is detected for the channel, the diagnostic input becomes low. The status of the diagnostic inputs does not depend on the safety relation of the channel.

Note:

The application diagnostic input for redundant analog inputs is calculated on the basis of the transmitter value after the values of the transmitters of each Central Part have been synchronized. For all other signal types, the application diagnostic inputs are an 'AND' function of the diagnostic inputs per Central Part.



6.4 FSC system alarm markers

Function of alarm markers

The FSC system uses a number of alarm markers to indicate the occurrence of abnormal situations. The following alarm markers are used:

Table 6-5 FSC alarm markers

Alarm marker	Description
CENTR.PART-FAULT	Fault detected within a Central Part.
DEVICE-COM.FLT	Communication with a connected device (e.g. a DCS) is faulty.
EXT.COMMUNIC.FLT	Communication with a connected FSC system is faulty.
FSC-SYSTEM-FAULT	Overall alarm marker, any fault exists.
INPUT-FAILURE	Fault detected for an input channel or input module.
INT.COMMUNIC.FLT	Communication between Central Parts faulty.
IO-COMPARE	I/O value discrepancy between Central Parts.
OUTPUT-FAILURE	Fault detected for an output channel or output module.
RED.INPUT-FAULT	A sensor of a safety-related input with non-fail-safe sensors is faulty.
TEMP.PRE-ALARM	The temperature within the FSC system exceeds the pre-alarm setting. (For details refer to the data sheet of the 10006/. diagnostic and battery module).
TRANSMIT.-FAULT	An analog transmitter gives a value outside its specified range.
IO-FORCED	One or more variables are forced (see subsection 5.2).

The normal state of the markers, if no fault is present, is '1'. If the first fault occurs, the associated alarm marker changes to '0'. Any subsequent fault will cause the alarm marker to be pulsed to '1' for one application program cycle (see Figure 6-1).

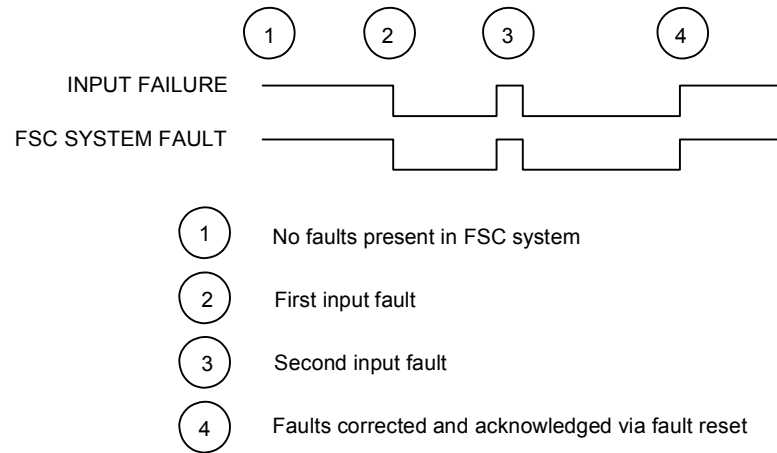


Figure 6-1 Input failure alarm marker function

The alarm markers are available in the application program, e.g. to generate an alarm.



6.4.1 Input fault detection

Input fault detection	Input fault detection applies to hardware inputs that are allocated to fail-safe, tested input modules. The tests include detection of faults affecting: - a single input channel, - a group of input channels at the same input module, and - all channels of an input module.
Possible faults	Possible faults are: - inability to represent both the '0' and the '1' state, and - correlation between inputs.
Tested modules	Input fault detection applies to hardware inputs allocated to the following fail-safe input modules: 10101/1/1, 10101/1/2, 10101/1/3, 10101/2/1, 10101/2/2, 10101/2/3, 10102/1/1, 10102/1/2, 10102/2/1, and 10105/2/1. Hardware inputs can be configured to be safety-related or not.
Safety-related inputs	If a fault affects an input configured for a safety-related signal connected to a tested input module, the faulty input is isolated from the application. For digital inputs, a '0' value is applied to the application, regardless of the value present at the input channel. For analog inputs, the application value is clamped to the configured bottom scale.
Non safety-related inputs	If a fault affects an input configured for a non safety-related signal connected to a tested input module, the fault is only alarmed. The input value is applied to the application program as read from the input channel.
Fault alarm	Occurrence of an input fault is indicated in the INPUT-FAILURE alarm marker and the associated diagnostic input(s).



6.4.2 Transmitter fault detection

Transmitter fault detection

A transmitter fault is detected if the value obtained from a transmitter, via an analog input, is outside its configured range.

If an underrange fault is detected, the application value is clamped to the configured bottom scale. If an overrange is detected, it is clamped to max. 6.25 V, 12.5 V or 25 mA, depending on the selected range.

Tested modules

Transmitter fault detection applies to inputs allocated to the following fail-safe analog input modules:

10102/1/1, 10102/1/2, 10102/2/1, and
10105/2/1

Fault alarm

Occurrence of a transmitter fault is indicated in the TRANSMIT.-FAULT alarm marker and the associated sensor diagnostic input.



6.4.3 Redundant input fault detection

Redundant input fault detection

Redundant input fault detection applies to fail-safe inputs with redundant non-fail-safe sensors.

Digital inputs

For digital inputs, a fault is detected if:

the input value is 'ON' for a longer time period than specified in the maximum on timer, or

the input values of the redundant sensors differ for a longer time period than specified in the maximum discrepancy time.

If a fault is detected, a '0' value is applied to the application.

Analog inputs

For analog inputs, a fault is detected if the transmitter values differ more than the specified maximum discrepancy value. If a fault is detected, the configured bottom scale is applied to the application.

Fault alarm

Occurrence of a redundant input fault is indicated in the RED.INPUT-FAULT alarm marker.



6.4.4 Output fault detection

Output fault detection

Output fault detection applies to hardware outputs that are allocated to tested output modules. The tests include detection of faults affecting:

- a single output channel,
- a group of output channels at the same output module, and
- all channels of an output module.

Possible faults

Possible faults are:

- inability to represent the '0' state,
- inability to represent the '1' state (for digital outputs with loop monitoring),
- inability to represent the correct value, bottom value, top value and variations of the current value (for analog outputs),
- output short circuit,
- correlation between outputs,
- arc-suppressing diode faulty (for digital outputs),
- open circuit in the output loop (for outputs with loop monitoring, i.e. 10205/1/1, 10205/2/1, 10214/1/2, 10216/1/1, 10216/2/1, 10216/2/3), and
- external power supply voltage below the minimum operating voltage.

Tested modules

Output fault detection applies to the following fail-safe output modules:

Module	Group specification
10201/1/1 and 10201/2/1:	Group 1: channels 1 to 4 Group 2: channels 5 to 8
10203/1/2 (<i>see note below</i>):	Group 1: channels 1 to 4
10205/1/1 and 10205/2/1:	Each channel is a separate group.
10212/1/1	Group 1: channels 1 to 4 Group 2: channels 5 to 8 (non saf.-rel.)
10213/1/1 and 10213/2/1:	Group 1: channels 1 to 4
10213/1/2 and 10213/2/2:	Group 1: channels 1 to 4
10213/1/3 and 10213/2/3:	Group 1: channels 1 to 4



10214/1/2:	Group 1: channels 1 to 3
10215/1/1 and 10215/2/1:	Group 1: channels 1 and 2 Group 2: channels 3 and 4
10216/1/1 and 10216/2/1:	Group 1: channels 1 to 4
10216/2/3:	Group 1: channels 1 to 4

Note:

The channels of the 10203/1/2 module are single fault tolerant. In case of a fault within a channel, full output control is still guaranteed. Therefore, any first channel fault is only reported. No additional corrective actions will be taken.

Hardware outputs can be configured to be safety-related or not.

Safety-related outputs

If a fault affects an output configured for a safety-related signal, the faulty output is forced to the safe state (i.e. '0'). The '0' value is applied to the process, regardless of the value calculated by the application program. Depending on the predefined effects of the fault, a single channel, a group of channels or all channels of an entire module are forced to '0'.

If a short-circuit is detected for one output channel, that channel is forced to '0'. If a short-circuit is detected for two or more channels within a single output group, all channels of the entire group are forced to '0'.

If any other fault is detected for an output channel, the entire group is regarded faulty and all channels of the group are forced to '0'.

If an entire group of safety-related output channels is regarded faulty, the second fault timer is started.

If all groups at the same output module are faulty, the entire module is regarded faulty.

If an entire safety-related output module is regarded faulty, the Central Part that controls the affected output module will trip. If the module is located in a single I/O section, the entire FSC system will trip.

Non-safety-related outputs

If a fault affects an output configured for a non safety-related signal, the fault is only reported. The output value is applied to the process as calculated by the application program as a result of the faulty module.

**External power failure**

External power failure is an exceptional fault, which does not cause a trip of the Central Part that controls the output module, even if safety-related output signals are allocated to the module.

Fault alarm

Occurrence of an output fault is indicated in the OUTPUT-FAILURE alarm marker and the associated output diagnostic input(s) and/or diagnostic loop-monitoring input.



6.4.5 I/O compare error detection

I/O compare error detection

The FSC system includes two high-level safety check functions which are active in redundant FSC configurations:

1. Input compare, and
2. Output compare.

Compare errors occur when a different status for inputs or outputs between the Central Parts is detected which cannot unambiguously be allocated to faults in the field or within the FSC system hardware. Because of the high level of self-testing by the FSC system, compare errors will be very rare.

If the FSC system is used for surveillance of processes which are classified in requirement class 5 (AK5) and which must meet the requirements of DIN V VDE 0801 A1 in its full extent, the IO-COMPARE alarm marker should be used to initiate a system shutdown if an I/O compare error is detected in the outputs (see programming example in Figure 7-1). The final decision whether automatic shutdown must be programmed lies with the approval authority (e.g. TÜV) during the acceptance of the plant.

Input and output compare faults are discussed in more detail below.

Tested modules

Input compare error detection applies to safety-related hardware inputs.

Output compare error detection applies to all digital hardware outputs and to communication outputs (O, BO) with location 'FSC'.

Fault alarm

Occurrence of an **input compare error** is indicated in the IO-COMPARE alarm marker. As the fault applies to inputs, the INPUT-FAILURE alarm marker is also asserted.

Occurrence of an **output compare error** is indicated in the IO-COMPARE alarm marker. If the error concerns an output with location 'FSC', the EXT.COMMUNIC.FLT alarm marker is also asserted because communication will halt on the affected communication link.



Input compare

In redundant FSC configurations, with dual Central Parts, the process inputs are scanned every application program cycle by both Central Parts.

Each Central Part executes the application program independently of the other Central Part. For proper operation of the system, both Central Parts must have an identical application status at all time. It is therefore essential that they use identical values for the process inputs.

There is no problem if the process inputs are stable. However, if an input value changes, both Central Parts could read a different value. In such cases, an identical input value in the Central Parts is obtained via input synchronization.

Differences in the input status read should be momentary. Persisting differences could be the result of hardware faults. In that case, the faulty input channel is reported in the diagnostics, and both Central Parts use the process value read from the healthy input channel. A persisting difference in status of an input while no faults are detected at the accessory hardware channels leads to an input compare error. Different synchronization algorithms are used for digital and analog inputs.

Digital input synchronization

A digital input compare error is detected if the inputs of both Central Parts are stable but different (e.g. CP1 continuously '0', CP2 continuously '1'), for the duration of the configured Process Safety Time (PST).

The input compare error detection algorithm puts the following demands on the dynamic nature of the digital process inputs:

1. If an input changes of state, it must become stable again within the configured Process Safety Time.
2. The frequency of continuously changing inputs must be less than $1/PST$.

The synchronization algorithm for digital inputs (I and BI) depends on the voting scheme that has been configured for the affected module. Table 6-6 below specifies the system response to a digital input compare error.

For details on the available voting schemes for the FSC input modules refer to Section 4 of the FSC Software Manual ("System Configuration"). For details on voting refer to subsection 6.2.



Table 6-6 System response in case of digital hardware input compare error

IF INPUT COMPARE ERROR AND...			THEN...					
			System markers			Applied state		
AK class	Voting	Safety-related	IO-COMPARE	FSC-SYSTEM-FAULT	INPUT-FAILURE	Digital Input	Channel diagnostic input	System shutdown
1-6	1oo2D 1oo1D	Yes	0	0	0	0	0	No
1-6	1oo2D 1oo1D	No	0	0	0	0	0	No
1-6	2oo2D	Yes	0	0	1	1	0	No
1-6	2oo2D	No	0	0	1	1	0	No

0 = false, low, de-energized
1 = true, high, energized

Notes:

- 1) 1oo1D voting is treated as 1oo2D as the voting of redundant Central Parts is 1oo2D by default.
- 2) 2oo2D voting for inputs that must satisfy safety requirement class higher than AK4 are not allowed. FSC Navigator does NOT check for this.

Analog input synchronization

For analog inputs, the synchronized value is the mean value of the input values. An input compare error is detected if the input values differ more than 2% of the full scale for the duration of the configured process safety time.

The input compare error detection algorithm puts the following demands on the dynamic nature of the analog process inputs:

1. For inputs located at modules within a redundant I/O section (10102/1/2, 10102/2/1 and 10105/2/1), the slope steepness must be less than 125 mA/s.
2. For inputs located at modules within a single I/O section (10102/./ and 10105/2/1), the slope steepness must be less than 20 mA/s.

Note:

Analog input compare errors may, for example, occur when calibrating smart transmitters using hand-held terminals. Refer to the project maintenance manual for details on to calibrate smart transmitters that are connected to FSC analog inputs.



The synchronization algorithm for analog inputs (AI) depends on the voting scheme that has been configured for the affected module. Table 6-7 below specifies the system response to an analog input compare error.

Table 6-7 System response in case of analog input compare error

IF INPUT COMPARE ERROR AND...			THEN...					
			System markers			Applied state		
AK class	Voting	Safety-related	IO-COMPARE	FSC-SYSTEM-FAULT	INPUT-FAILURE	Analog output	Channel diagnostic input	System shutdown
1-6	1oo2D 1oo1D	Yes	0	0	0	bottom scale	0	No
1-6	1oo2D 1oo1D	No	0	0	0	last healthy value	0	No
1-6	2oo2D	Yes	0	0	1	last healthy value	0	No
1-6	2oo2D	No	0	0	1	last healthy value	0	No

0 = false, low, de-energized

1 = true, high, energized

Notes:

- 1) 1oo1D voting is treated as 1oo2D as the voting of redundant Central Parts is 1oo2D by default.
 - 2) 2oo2D voting for inputs that must satisfy safety requirement class higher than AK4 are not allowed. FSC Navigator does NOT check for this.
-

Output compare

As a result of the synchronization algorithms within the FSC system, both Central Parts will continuously have an identical application status, which results in identical process outputs.

An output compare error is detected if there is a difference between the Central Parts with regard to the status of digital outputs (O, BO) or communication outputs (O, BO) with location 'FSC'.

The synchronization algorithm for digital outputs (O, BO) depends on the voting scheme that has been configured for the affected module. Table 6-8 below specifies the system response to a digital output compare error.

Note:

Table 6-8 does not apply for outputs with location 'FSC'. If an output compare error is detected for outputs with location 'FSC', communication with the system that the outputs are allocated to is halted.

Table 6-8 System response in case of digital output compare error

IF OUTPUT COMPARE ERROR AND...			THEN...				
			System markers		Applied state		
AK class	Voting	Safety-related	IO-COMPARE	FSC-SYSTEM-FAULT	Digital output	Channel diagnostic input	System shutdown
1-5	1oo2D 1oo1D	Yes	0	0	0	0	No
1-5	1oo2D 1oo1D	No	0	0	1	0	No
1-5	2oo2D	Yes	0	0	1	0	No
1-5	2oo2D	No	0	0	1	0	No
6	1oo2D 1oo1D	Yes	0	0	0	0	Yes
6	1oo2D 1oo1D	No	0	0	0	0	Yes
6	2oo2D	Yes	0	0	0	0	Yes
6	2oo2D	No	0	0	0	0	Yes

0 = false, low, de-energized

1 = true, high, energized



6.4.6 Central Part fault detection

Central Part fault detection

Central Part fault detection applies to Central Part modules, horizontal bus driver modules (HBD) and system internal buses. If an error is detected, the faulty part will be isolated, which may result in the Central Part trip. Exceptions are faults detected at non-safety-related HBD modules (10100/1/1, 10100/2/1) and some faults on the Diagnostic and Battery Module (10006/./.), e.g. if the battery fuse is defective.

Tested modules

Central Part fault detection applies to the following FSC modules:

- 10001/./1, 10002/1/., 10003/1/1, 10004/./., 10005/1/1, 10006/./., 10007/1/1, 10008/./.,
- 10100/1/1, 10100/2/1,
- System bus, and
- V-bus, H-bus.

Fault alarm

Occurrence of a Central Part fault is indicated in the CENTR.PART-FAULT alarm marker.



6.4.7 Internal communication error

Internal communication error

An internal communication error is detected if communication between the Central Parts in a redundant FSC configuration fails. One of the Central Parts will trip.

In fully redundant configurations (without single I/O sections), Central Part 2 will trip. In systems with a single I/O section, one of the Central Parts will trip, depending on the internal status of the system.

An internal communication error is always reported by the running Central Part.



6.4.8 FSC-FSC communication fault detection

FSC-FSC communication fault detection

For communication with a connected FSC system, a fault is detected if communication with the connected FSC system fails. If the systems are interconnected via redundant communication links, fault detection applies to each link separately resulting in single fault tolerance overall.

Inputs and outputs allocated for communication with a connected FSC system (location 'FSC') can be configured to be safety-related or not.

If all links to a connected system are faulty, the safety-related inputs that are received from the connected system are forced to the safe state (i.e. '0'). The non safety-related inputs are frozen to the state that was last received from the connected system. The outputs are not affected.

Fault alarm

Occurrence of an FSC-FSC communication fault is indicated in the EXT.COMMUNIC.FLT alarm marker



6.4.9 Device communication fault detection

Device communication fault detection

The FSC system monitors for several device types if the communication link with the device is operating correctly.

Distributed control system

For distributed control systems (DCS) that communicate with the FSC system via the Modbus or RKE3964R protocol, continuous communication is expected. If no communication is established within a predefined timeout period, the link to the device is regarded faulty. If the device is connected to the FSC system via a redundant communication link, the fault detection applies to each link separately resulting in single-fault-tolerant communication.

The timeout period for the Modbus protocol can be configured using the 'System Configuration' option of FSC Navigator. The timeout period for the RKE3964R protocol is fixed at 3 seconds.

Inputs and outputs that are allocated to the distributed control system (location 'COM') are always non-safety-related.

If all links to the DCS are faulty, the inputs remain frozen at the state that was last received from the DCS. The outputs are not affected.

SOE collecting devices

A communication fault for SOE collecting devices is detected if the device is off-line for more than 1 minute.

Fault alarm

Occurrence of a device communication fault is indicated in the DEVICE-COM.FLT alarm marker.



6.4.10 Temperature alarm

Temperature alarm

During configuration of the FSC system, the user may define the temperature range within which the FSC system must operate. Temperature prealarm values can also be configured.

If the temperature of the running system exceeds the alarm settings, a fault will be reported. If the temperature exceeds the configured operating boundaries, the Central Part will shut down.

Tested modules

Temperature alarms apply to the operational temperature within the Central Part as measured at the Diagnostic and Battery module (10006/./.).

Fault alarm

If the temperature exceeds the alarm settings, this is indicated in the TEMP.PRE-ALARM alarm marker.



6.5 Calculation errors

General

Calculation errors result from the application program and occur if:

- the calculated value for an analog value is outside the specified range of the analog output,
- the square root of a negative number is taken,
- a divide-by-zero occurs,
- an overflow of the result of addition, subtraction, multiplication and division functions occurs,
- a timer is loaded with a value > 2047 , or
- a counter is loaded with a value > 8191 .

Calculation errors reflect incorrect design of the application program for the intended function. Once a calculation error occurs for a specific process variable, the result of successive calculations based on this variable cannot be ensured and escalation of the anomaly needs to be prohibited. The FSC system will therefore trip if a calculation error occurs.

Guidelines on how to avoid calculation errors in the FSC application program are presented below.

Preventing calculation errors

Calculation errors can be prevented in a number of ways:

- prevention from occurrence through overall process design,
- inclusion of FSC diagnostic data,
- validation of signals when entering the Functional Logic Diagrams (FLDs), and
- exception handling during the actual calculation.

Prevention by design

In line with good software engineering practice, as promoted by IEC 61508, calculation errors should be avoided by design. This means that an application should be designed in such a way that the operands of a symbol in the FLDs can never get an invalid value. The design approach starts with the assurance that input values as obtained from the process remain within a deterministic range, and subsequently ensuring that the derived values are valid for successive operations.



Sometimes, however, it cannot be guaranteed that an input value remains within a deterministic area which is valid for all functions. For example, a signal derived from a reverse-acting, non-linear 4-20 mA transmitter which has been configured for a zero top scale in the application domain could become negative if the transmitter fails and delivers a signal beyond 20 mA. If the signal is then linearized through a square-root function, a system trip will occur (square root of negative number).

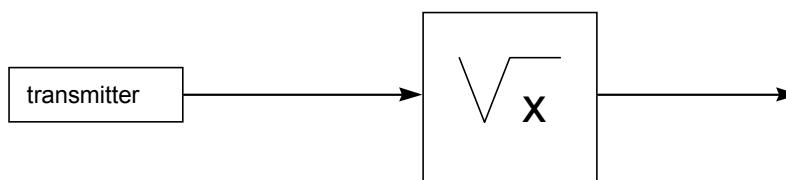


Figure 6-2 Intended square-root function

Preventive measures

If a valid input value cannot be guaranteed, preventive measures must be built into the design. A comparison function can be used as an indicator that the transmitter value has left its normal operational band and that the calculation should not be done. The alarm signal is used to implement corrective action and to indicate the exception to the operator (see Figure 6-3).

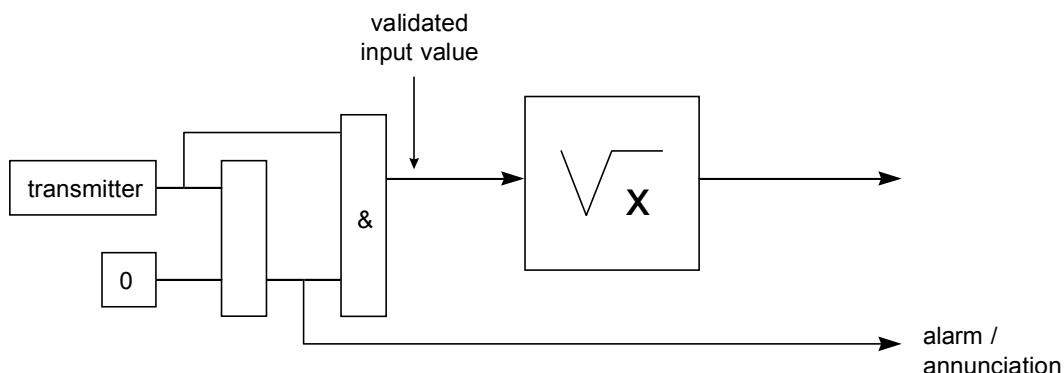


Figure 6-3 Square-root function with validated input value

If diagnostics are not available (e.g. for 0-20 mA transmitters), it is necessary to implement range checking in the application program itself. The result of the boundary check is again used for implementation of corrective actions.

An important advantage of input validation is that it can be implemented on input values for which a valid range cannot be guaranteed. Furthermore, the deviating input can be exactly identified. This allows the implementation of effective correction strategies which only apply to the affected part of the process.

Common function block

A last option is to create a common function block, e.g. square root, which is used for all such calculations. The function block validates the operand(s) and only performs the intended function if the operands are valid. Otherwise a predefined value is returned. An additional function block output should be provided which indicates if the calculation result is valid or not. This output signal can then again be used for implementation of corrective actions in the application program (see Figure 6-4).

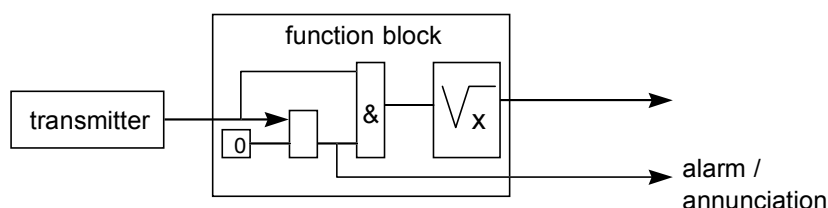


Figure 6-4 Square-root function with validity check in function block



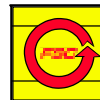
Section 7 – Using the FSC alarm markers and diagnostic inputs

7.1 Section overview

Section overview

This section describes how FSC alarm markers and diagnostic inputs are used. It covers the following topics:

Subsection	Topic	See page
7.1	Section overview	91
7.2	Applications of alarm markers and diagnostic inputs.....	92
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7.4	Unit shutdown.....	94
7.5	Diagnostic status exchange with DCS	99



7.2 Applications of alarm markers and diagnostic inputs

Applications

FSC alarm markers and diagnostic inputs can be used within the functional logic diagrams (FLDs) to respond to abnormalities or to initiate an alarm. This is illustrated in three examples below.

Shutdown at assertion of FSC alarm markers

This example shows how to program a shutdown in case of assertion of FSC alarm markers. This kind of programming could be used if the system is intended to run in AK5 without operator surveillance. (See subsection 7.3.)

Unit shutdown

This example shows how diagnostic inputs of type I/O-TYPE O can be used to realize independent safeguarding of process units including only unit shutdown in case of defects. (See subsection 7.4.)

Diagnostic status exchange with DCS

This example discusses the functional logic which can be used to report the status of alarm markers and diagnostic inputs to a distributed control system (DCS). (See subsection 7.5.)



7.3 Shutdown at assertion of FSC alarm markers

If it is not sufficient to initiate an alarm in case the FSC system detects a fault, and direct system response is required, the FSC alarm markers can be used to shut down the system via the application program.

Figure 7-1 shows an example of how to shut down the system in case of an I/O compare error. An additional manual shutdown hardware input is provided which the operator can use to initiate a shutdown by hand.

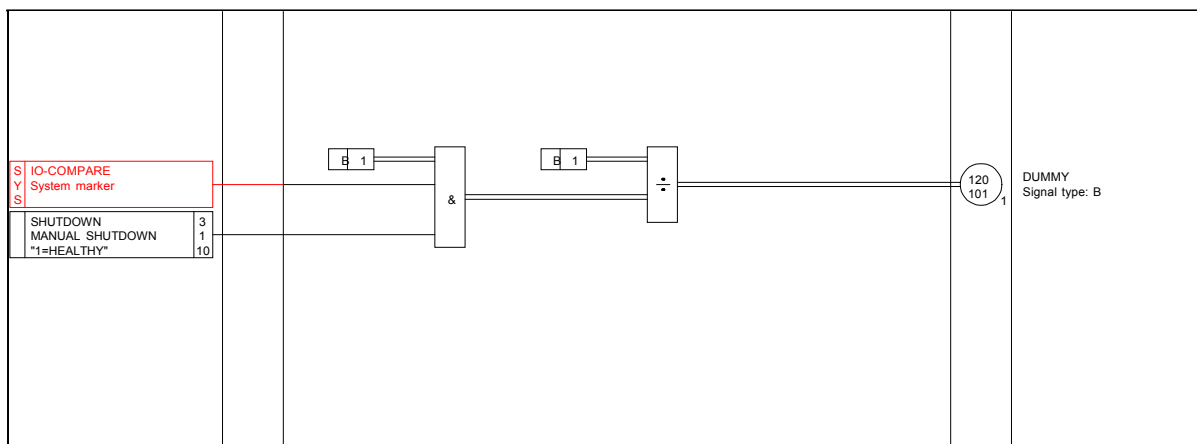


Figure 7-1 Diagram to shut down system in case of output compare error

If an I/O compare error is detected or a manual shutdown is initiated, a divide-by-zero is initiated and the FSC system will shut down. Other alarm markers can be used in a similar way.

Note:

A manual shutdown can also be realized via the ESD input of the watchdog module (10005/1/1). This module enables the use of a tested solid-state hardwired connection, which allows the secondary means of de-energization of all outputs to be activated. This unique feature allows an ESD pushbutton chain to be connected to the FSC system which can then be used to initiate an emergency shutdown (ESD), fully independently of the central processor.

7.4 Unit shutdown

Process units

If a process can be divided into independent process units, the overall process availability can be increased by separate shutdown of the units within the FSC system. Thus, in case a fault is detected within the hardware of a process unit, only the affected unit needs to be shut down, while the remaining parts of the process are not affected.

Configuration of unit shutdown

This subsection discusses the configuration, application programming and wiring required to achieve shutdown per process unit.

Figure 7-2 shows a standard wiring diagram to realize unit shutdown for three separate process units.

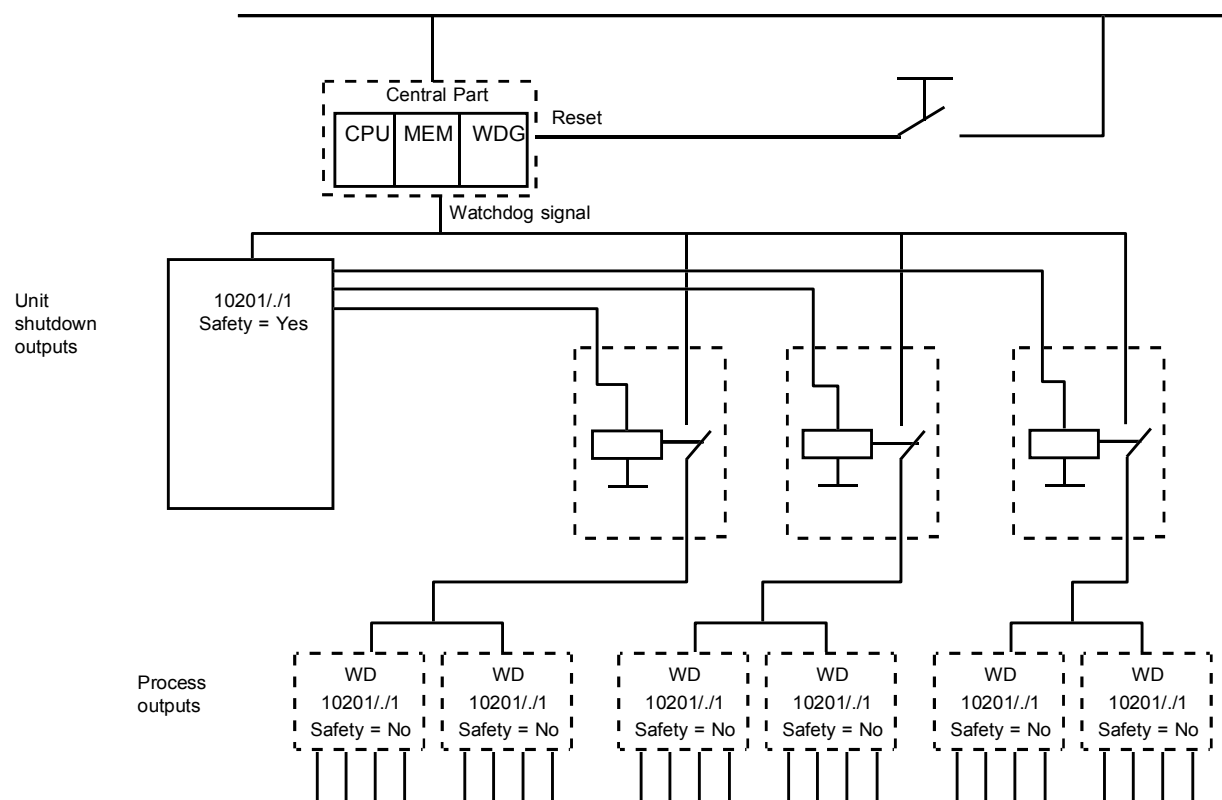


Figure 7-2 Wiring diagram for unit shutdown

For each unit, a relay is used to connect the watchdog input signal of the unit output to the output of the FSC watchdog module (10005/1/1). This relay is controlled via an output of the FSC system: the unit shutdown output. In normal operation, all relays are activated. If a



fault is detected within a process unit, the corresponding relay is deactivated, which results in a shutdown of the unit.

The unit relays must meet the requirements of DIN VDE 0116, part 8.7.4.5 and 8.7.4.6 of October 1989, i.e.:

- a) Mechanical reliability $> 3 \cdot 10^6$ switches.
- b) Contacts protected (e.g. fuses, series resistors, etc.) at 0.6 nominal contact current.
- c) Electrical reliability $> 2.5 \cdot 10^5$ switches.

Unit shutdown outputs

The unit shutdown outputs must be safety-related (e.g. allocated to a 10201/.1 or 10216/.1 module). This will guarantee that the FSC system will direct the process to its safe state if a fault occurs which affects this output.

The power-up status of the output must be on, to allow correct start-up of the FSC system with activated unit relays (see Figure 7-3).

For optimum availability it is recommended that the unit shutdown outputs are allocated to redundant output modules.

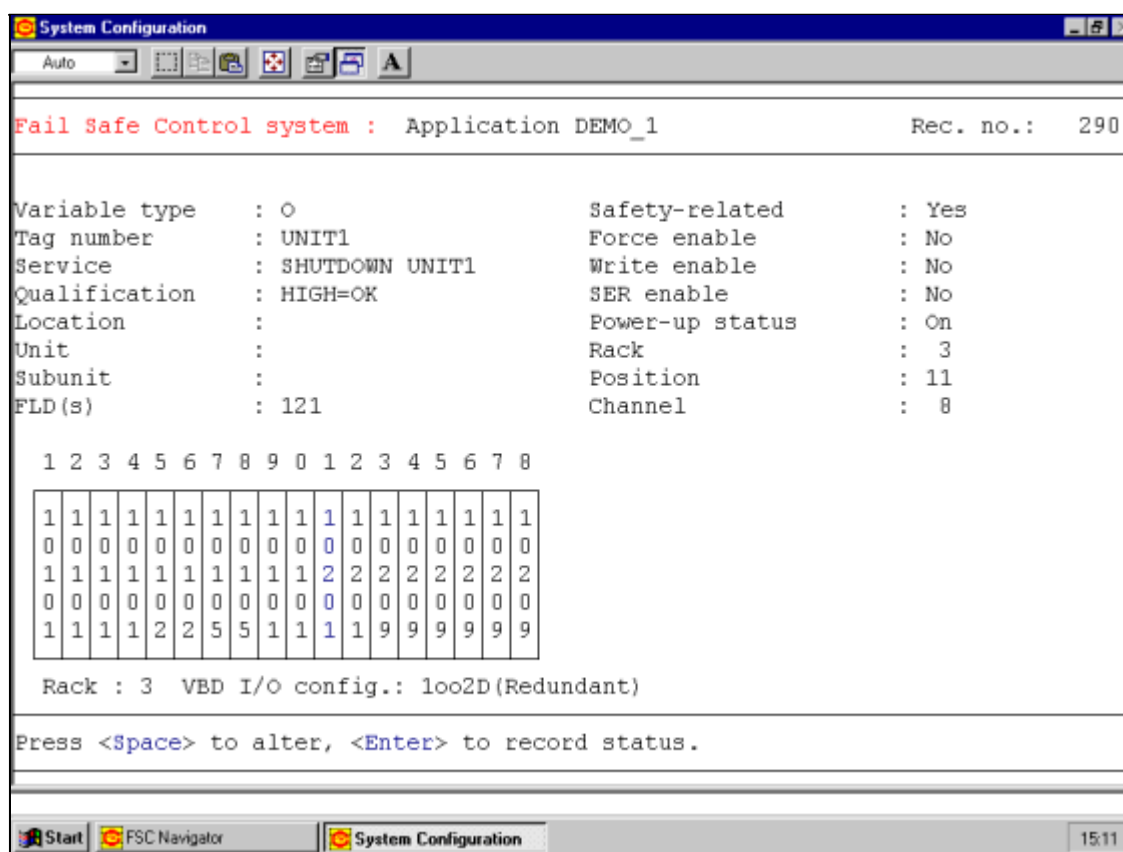


Figure 7-3 Configuration of the unit shutdown output



Process outputs (safety-related)

The process outputs must be allocated to an FSC fail-safe output module:

10201/1/1	Fail-safe digital output module (24 Vdc, 0.55 A, 8 channels)
10201/2/1	Fail-safe digital output module (24 Vdc, 0.55 A, 8 channels)
10203/1/2	Fail-safe output module with double switch-off (24 Vdc, 0.9 A, 4 channels)
10205/1/1	Fail-safe analog output module (0(4)-20 mA, 2 channels)
10205/2/1	Fail-safe analog output module (0(4)-20 mA, 2 channels)
10212/1/1	Digital output module (24 Vdc, 0.9 A, 16 channels)
10213/1/1	Fail-safe digital output module (110 Vdc, 0.32 A, 4 channels)
10213/2/1	Fail-safe digital output module (110 Vdc, 0.32 A, 4 channels)
10213/1/2	Fail-safe digital output module (60 Vdc, 0.67 A, 4 channels)
10213/2/2	Fail-safe digital output module (60 Vdc, 0.67 A, 4 channels)
10213/1/3	Fail-safe digital output module (48 Vdc, 0.75 A, 4 channels)
10213/2/3	Fail-safe digital output module (48 Vdc, 0.75 A, 4 channels)
10214/1/2	Fail-safe digital output module (220 Vdc, 0.25 A, 3 channels)
10215/1/1	Fail-safe digital output module (24 Vdc, 2 A, 4 channels)
10215/2/1	Fail-safe digital output module (24 Vdc, 2 A, 4 channels)
10216/1/1	Fail-safe loop-monitored digital output module (24 Vdc, 1 A, 4 channels)
10216/2/1	Fail-safe loop-monitored digital output module (24 Vdc, 1 A, 4 channels)
10216/2/3	Fail-safe loop-monitored digital output module (48 Vdc, 0.5 A, 4 channels)



The safety relation for the outputs must be 'No' (see Figure 7-4). This will suppress the automatic response of the FSC system if faults occur at safety-related output modules, which allows programming of the response via the application.

System Configuration

Auto

Fail Safe Control system : Application DEMO_1 Rec. no.: 310

Variable type	: O	Safety-related	: No
Tag number	: EARTH-LEAKAGE	Force enable	: No
Service	: EARTH LEAKAGE PSU'S	Write enable	: No
Qualification	: FAILURE	SER enable	: No
Location	: PNL	Power-up status	: Off
Unit	:	Rack	: 3
Subunit	:	Position	: 13
FLD(s)	: 123	Channel	: 2

1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	2	2	5	5	1	1	1	1	9	9	9	9	9	9

Rack : 3 VBD I/O config.: 1oo2D(Redundant)

Press <Space> to alter, <Enter> to record status.

Start FSC Navigator System Configuration 15:14

Figure 7-4 Configuration of the process outputs

Application programming

To realize the unit shutdown in the functional logic diagrams, all diagnostic inputs ('SYS' internal markers related to output modules available in the database) of one process unit are connected to an AND gate.

The output signal of the AND gate is connected to the unit shutdown output (see Figure 7-5).

As long as all the diagnostic inputs are healthy, the diagnostic inputs will be high, the unit shutdown output will be high and the unit relay is activated (relay contact closed).

If one diagnostic input of an output channel within the unit becomes 'not healthy', the corresponding unit shutdown output becomes low and the unit relay is deactivated (relay contact open).

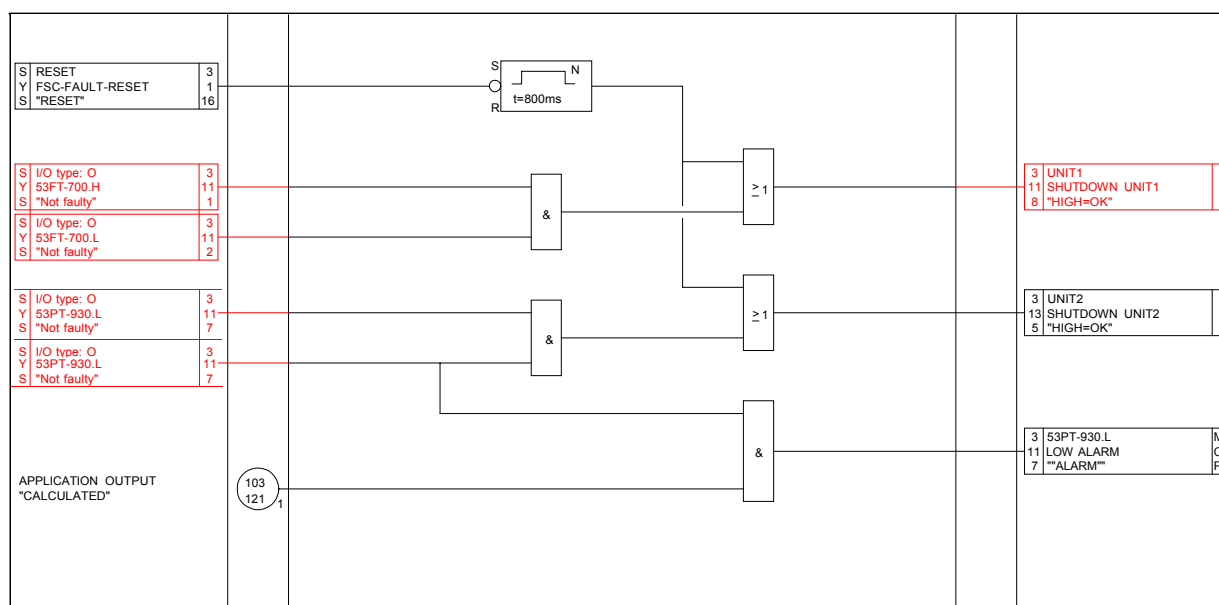


Figure 7-5 Functional logic diagram of unit shutdown

In order to realize a switch-off of a defective output channel in accordance with the normal FSC response for safety-related signals, the calculated application output should be applied to the output channel via an AND gate with the channel diagnostic input.

The FSC-FAULT-RESET alarm marker is connected to all unit shutdown outputs via an OR gate. After an error is detected and repaired in one unit, that unit may be restarted using the FSC-FAULT-RESET alarm marker.

The minimum and maximum time the unit output is enabled by the FSC-FAULT-RESET is limited to ensure that the FSC-FAULT-RESET is detected by the output. The pulse length may not exceed the process safety time (timer typically set at 800 ms).



7.5 Diagnostic status exchange with DCS

Distributed control systems (DCS)

FSC alarm markers and the diagnostic inputs can be transferred to distributed control systems (DCSs), e.g. to generate an operator alarm or to initiate corrective action within the DCS.

Figure 7-6 shows the functional logic diagram to report the occurrence of an input fault (INPUT-FAILURE alarm marker) and the use of a diagnostic input (I/O type AI) to report the status of an analog input channel to a DCS system.

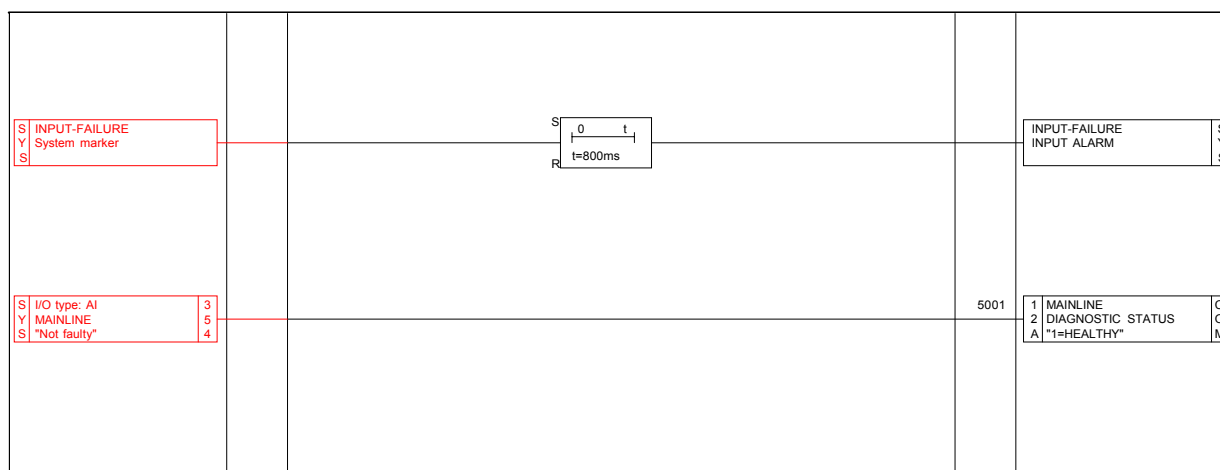


Figure 7-6 FSC system information to DCS

The status of both variables is transferred to the DCS via outputs with location 'COM', which are allocated to the communication channel that the DCS is connected to.

Behavior of alarm markers

The behavior of the alarm markers is quasi-static. Normally, if no fault is present, the value of the markers is high. If a fault is detected, the corresponding alarm marker will become low. On subsequent faults the alarm marker will become high during one application program cycle of the FSC system (e.g. 300 ms) and then low again (see subsection 6.2).

If the scan cycle of the DCS is larger than the FSC application program cycle, it is possible that any subsequent faults are not detected by the DCS. The FSC alarm marker is therefore connected to the output of the DCS via a delayed off timer. Thus, a pulse on the alarm



marker is extended to the configured timer value. To ensure detection by the DCS, the timer value must be larger than the DCS scan time.

Behavior of diagnostic inputs

The behavior of the diagnostic inputs is static. Normally, an I/O channel is healthy and the value of the corresponding diagnostic input is high. If the I/O channel becomes faulty, the diagnostic input will be low. It remains low until the fault is repaired and a fault reset has been given. The diagnostic input can therefore be connected directly to the output to the DCS.



Section 8 – Wiring and 1oo2D output voting in AK5 and AK6 applications

Using standard wiring

The FSC configuration with redundant Central Parts and redundant I/O is a versatile configuration which may be used in applications of requirement class AK1 up to AK6. In applications up to AK4, standard redundant I/O wiring is used. In applications of requirement class AK5, standard wiring can be used if the process runs under continuous operator surveillance, i.e. if the operator:

is able to monitor the process, and

is able to respond to achieve the safe process state within acceptable time.

For this purpose a pushbutton can be provided which the operator can use to shut down the FSC system connected to the ESD input of the watchdog module (10005/1/1).

Using special wiring

If the system is intended for safeguarding a non-surveiled process, DIN V VDE 0801 A1 requires that each Central Part by itself is able to shut down the process, independent of the status of the other Central Part. This requires specific wiring of the outputs of the FSC system.

Furthermore, all AK6 applications require independent Central Part shutdown capability.

Single Central Part operation

Single Central Part operation in AK5 and AK6 is only allowed for a limited time. AK6 applications also require additional measures with regard to the field instrumentation for the signals that are AK6, e.g. diverse and/or redundant sensors.

Example

This section provides an example of how the outputs of an FSC configuration with redundant Central Parts and redundant I/O can be wired for non-surveiled applications in AK5 and for all applications in AK6.

Figure 8-1 shows the wiring principle. The figure shows cross-wiring of an output channel which each Central Part can use to de-energize the output channels of the other Central Part via the 24 Vdc emergency

shutdown input of the watchdog module (10005/1/1). The 24 Vdc ESD input is switched via a normally closed relay contact. The relay must meet the requirements of DIN VDE 0116 part 8.7 of October 1989 (see subsection 7.4).

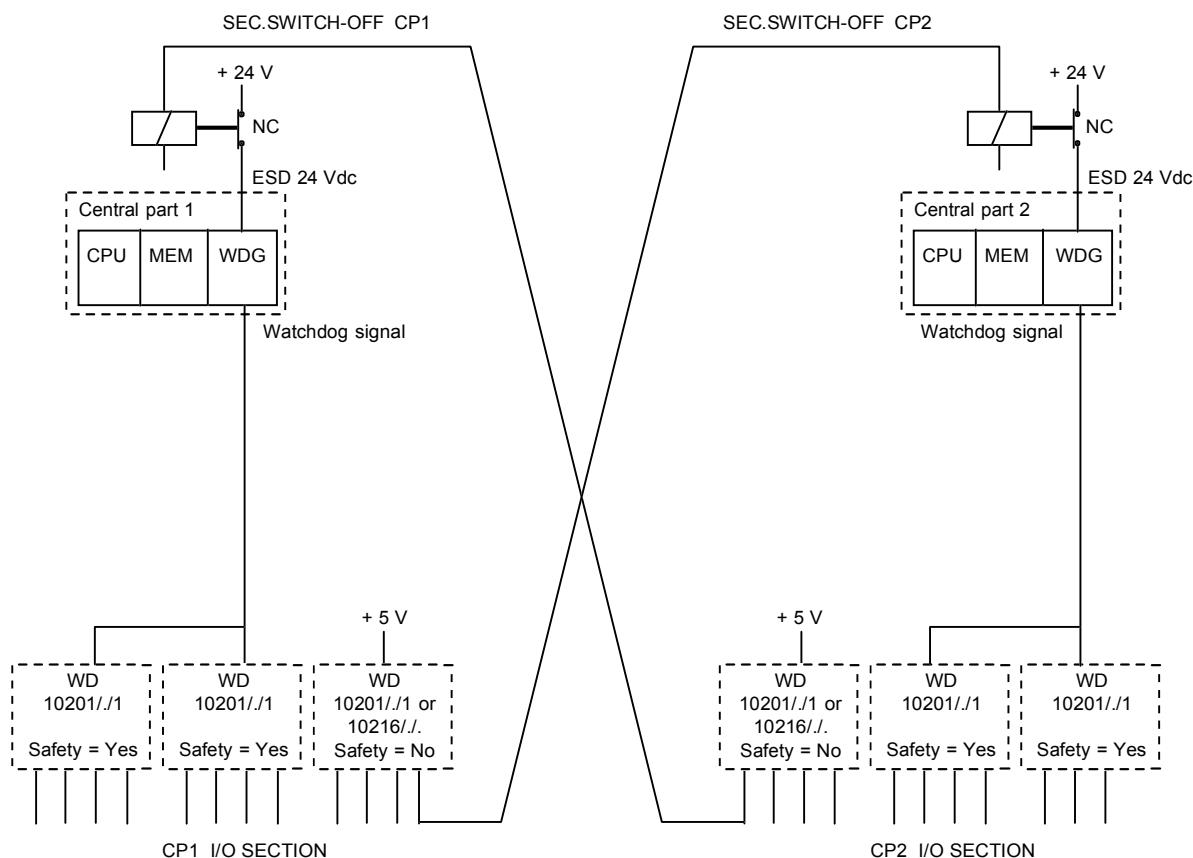


Figure 8-1 Redundant I/O wiring in AK6 and non-surveilled AK5 applications

Secondary switch-off

The output which is used to realize the ESD function is a dedicated system output, the 'secondary switch-off' (tag number: SEC.SWITCH-OFF). The name 'secondary switch-off' refers to the capability to switch off the outputs of the other Central Part via the secondary means of de-energization. During normal operation, the SEC.SWITCH-OFF output is low and the relay contact is closed. If a condition occurs which, for example, requires Central Part 2 to deactivate the outputs of Central Part 1, the SEC.SWITCH-OFF output is set to high, the relay contact is opened, and an emergency shutdown is effected on the watchdog module of Central Part 1. The outputs of Central Part 1 are de-energized via the



watchdog output signal. Similarly, Central Part 1 is able to de-energize the outputs of Central Part 2.

The SEC.SWITCH-OFF output may be used in the application program to initiate a shutdown at a user-specified condition.

The SEC.SWITCH-OFF output is allocated to a channel of a fail-safe output module (10201/./1 or 10216/./.) in the I/O section of the Central Part. A fail-safe output module is used to benefit from the FSC self-tests which will provide diagnostic information if faults are detected at the module. During the test, the switch-on capability of the output is also verified. Application of the 10216/./ has the additional advantage that the total output loop, including the relay coil, is covered.

The Central Part must be able to activate the SEC.SWITCH-OFF output, not only when running, but also while in shutdown. To enable activation of the output while in shutdown, the safety relation of the output module must be configured at 'No' and the watchdog input signal of this module must be connected to +5 V.

The remaining channels of the output module may be used to drive non-safety-related process output signals. Contrary to normal redundant I/O wiring, the outputs controlling the relays may not be wired in parallel.



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Section 9 – Fire and gas application example

Application example This section describes an application program for a Fire & Gas (F&G) application which is designed according to the requirements of EN-54 part 2, with the OVERRIDE and TEST options installed. The FSC system does not support alphanumeric displays, so this option of EN-54 part 2 is not shown here.

The figures in this section are identified by a descriptive text and the functional logic diagram (FLD) number which is used in the sheet references. Where applicable, references to the EN-54 part 2 standard are shown in *italics* in square brackets.

The status of the installation which is monitored and the status of the FSC system must be uniquely displayed [*EN-54 part 2, 2.1.3*]. Within the complete example this is accomplished by the use of hardwired digital I/O signals which can drive LEDs or lamps. Another option is to have the display on a remote location, and communicate the status via the FSC-FSC communication link [*EN-54 part 2, 2.2.13, 2.3.10, 2.4.1.2*]. For details on configuring the FSC-FSC communication refer to Section 4 of the FSC Software Manual ("System Configuration"). Failure of the communication link must be alarmed [*EN-54 part 2, 2.3.2.4, 2.3.2.6, 2.3.2.11*].

Please note that the sheet references in the functional logic diagrams must point to a higher FLD number, which means that they are used in the same application program cycle in order to get the best possible response time. This response time for automatic fire detectors resulting in the required outputs is 1 second [*EN-54 part 2, 2.2.8*].

Functional logic diagrams (FLDs)

The system alarm FLD (see Figure 9-1) covers the status indication for the redundant power supplies (PSU 1 and 2) [*EN-54 part 2, 2.3.2.5*], the indication for an earth leakage alarm [*EN-54 part 2, 2.3.2.7*] and the common failure alarm which is set in case of a failure of any component in the Fire & Gas detection system, including failures in the F&G detectors.

The failures in the F&G detectors are handled on other FLDs, in this example in the FLD for each input loop as shown in Figure 9-2 [*EN-54 part 2, 2.3.1*]. Function Block (FB) 905 handles the latching function for the alarm status, the alarm reset function and the lamp test function.

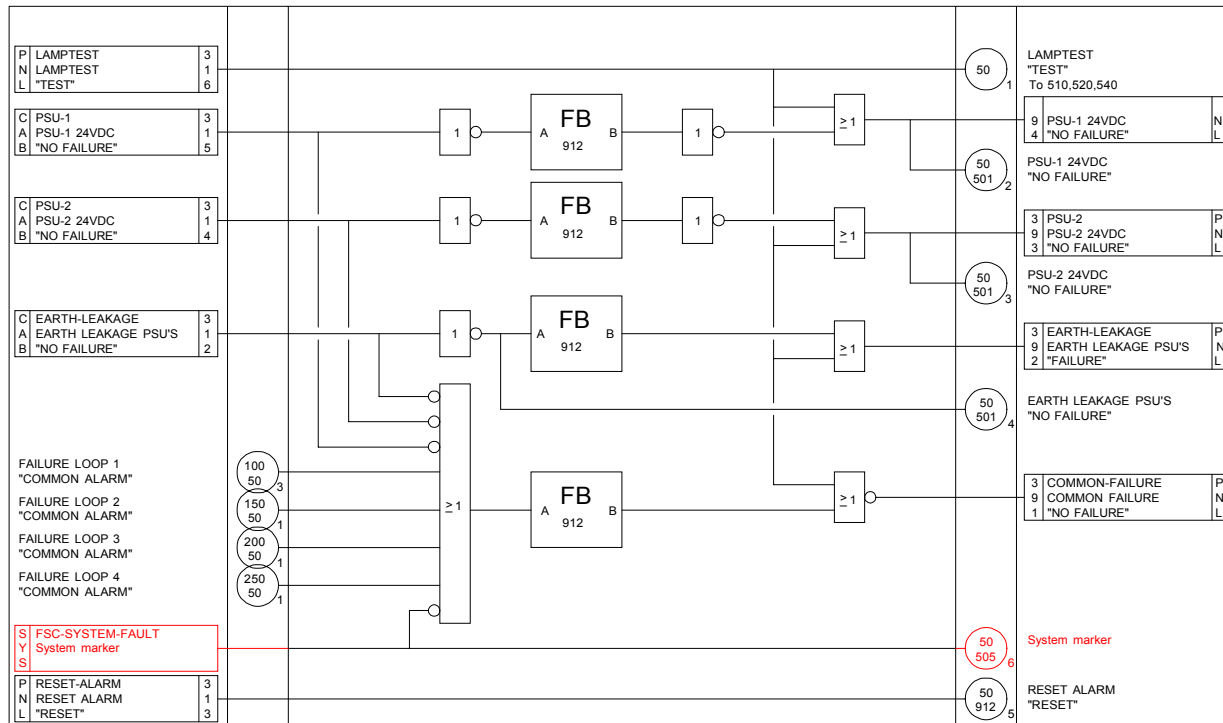


Figure 9-1 System alarm (FLD 50)

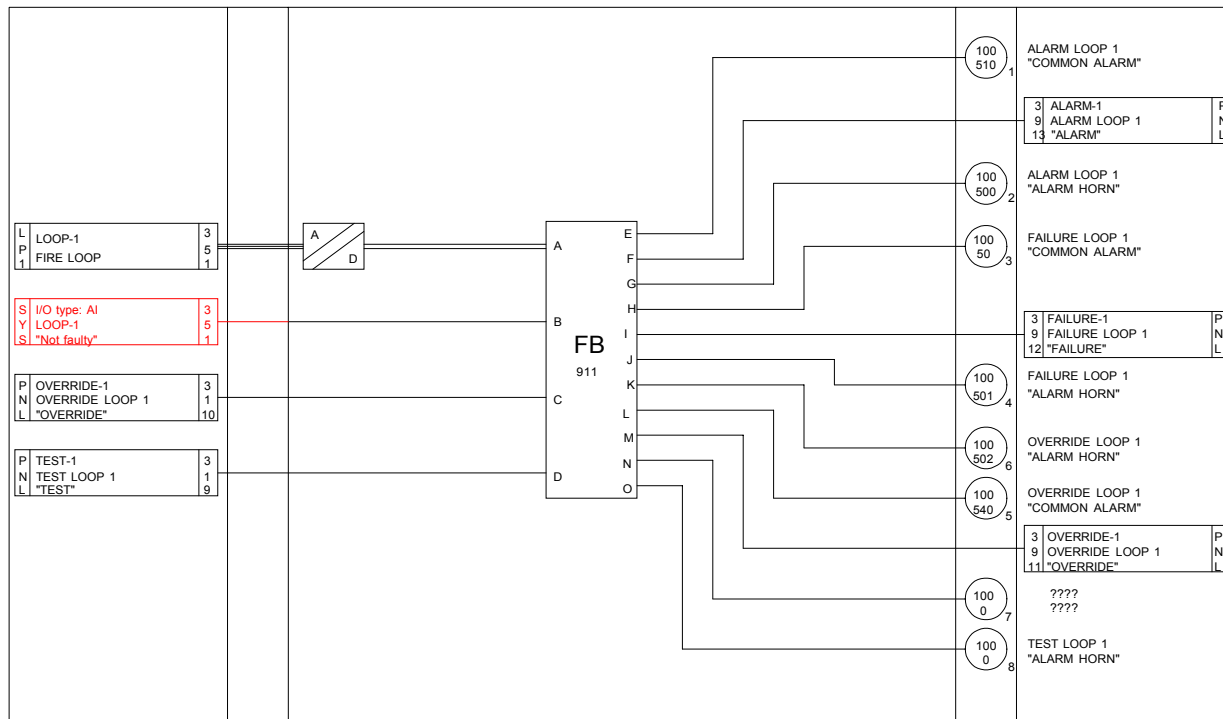


Figure 9-2 Input loop 1 (FLD 100)



Input loops

The example presented here has four input loops which could come from Fire & Gas detectors (the other FLD numbers are 150, 200, 250 but they are not shown here as they are identical to FLD 100). The Fire & Gas detectors are connected using analog input modules. The output of the detectors can be a digital contact with loop-monitoring or an analog signal. The function block 911 (FB-911) handles all functions that can be executed on an input loop [EN-54 part 2, 2.1.5]. These functions are:

Setting of alarm levels (in this example they are identical for all loops. In general, these settings are set per input loop, which means that the alarm levels detection part of the FB must to be transferred to the FLD of the input loop) [EN-54 part 2, 2.2.1.2].

Loop status (open loop, short-circuit) as determined via the system software of the FSC system [EN-54 part 2, 2.3.2.3, 2.3.2.8, 2.3.2.11].

Override for the input loop [EN-54 part 2, 2.4.3].

Test function for the input loop [EN-54 part 2, 2.5.2].

Loop status

The loop status (operational status, failure status, override status and test status) is indicated on panel indications with an indication per status [EN-54 part 2, 2.1.3]. All states are also transferred to other FLDs via sheet transfers to generate the common status indication and to drive the audible indications (horn) [EN-54 part 2, 2.2.12].

Failure indication and override indication

In this example the failure indication and the override indication is done using separate digital outputs. It is possible to use the same digital output per channel but with different common outputs in order to distinguish uniquely between failure and override [EN-54 part 2, 2.4.4].

Test function

The test function is implemented per input loop. The test function on one input loop may not override or prohibit detection of a fire or gas alarm on another input loop which is not in test or override [EN-54 part 2, 2.5.1].

Monitoring for alarm status

The input loops are monitored for an alarm status. If an alarm status occurs, an audible alarm (horn) must also be activated [EN-54 part 2, 2.2.1.1, 2.2.1.2]. The example FLD in Figure 9-3 creates a common signal of the alarm status in order to activate the horn. The cycle pulse logic for each loop combined in the NOR gate is required to activate the horn for every subsequent alarm in the same alarm group. For each alarm in an alarm group, an entry to the top OR gate is required as well as a cycle pulse and entry to the bottom NOR gate. If more than one alarm group is used in one Fire & Gas detection system, logic as shown in the diagram below is required for each alarm group.

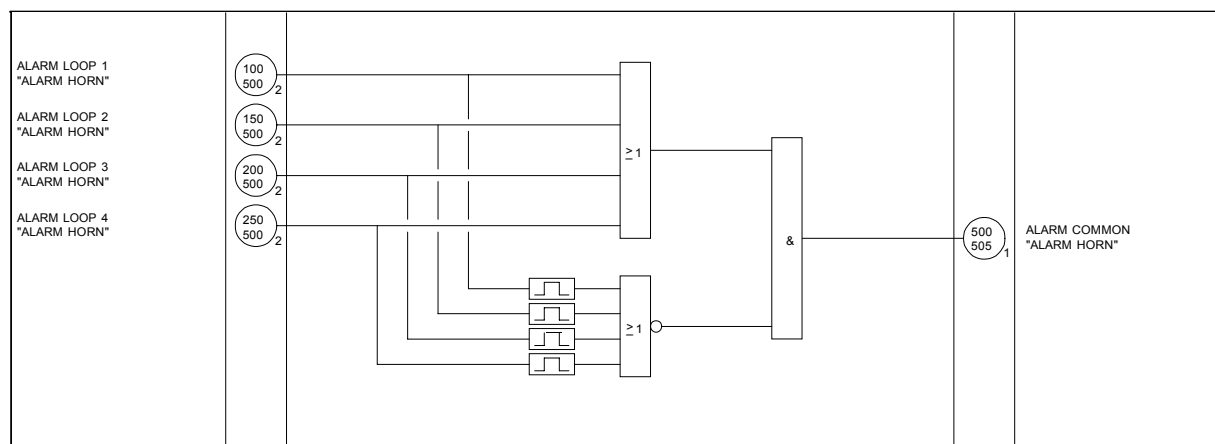


Figure 9-3 Control of the alarm horn (FLD 500)

Monitoring for failure status

All components of the Fire & Gas system, including the input loops and output loops, are monitored for a failure status. If a failure occurs, an audible alarm (horn) must also be activated which has a different frequency than the Fire & Gas audible alarm. The example FLD in Figure 9-4 creates a common signal of the failure status in order to activate the failure horn. The cycle pulse logic for each loop combined in the NOR gate is required to activate the horn for every subsequent failure in a failure group [EN-54 part 2, 2.3.9]. An entry to the top OR gate is required for each failure in a failure group, as well as a cycle pulse and entry to the bottom NOR gate. Failures which must be covered are power supply failures and earth leakage failures. Depending on the application, other internal failures of the FSC system can also be covered by the common failure alarm. If more than one failure group is used in one Fire & Gas detection system, logic as shown in the diagram below is required for each failure group.

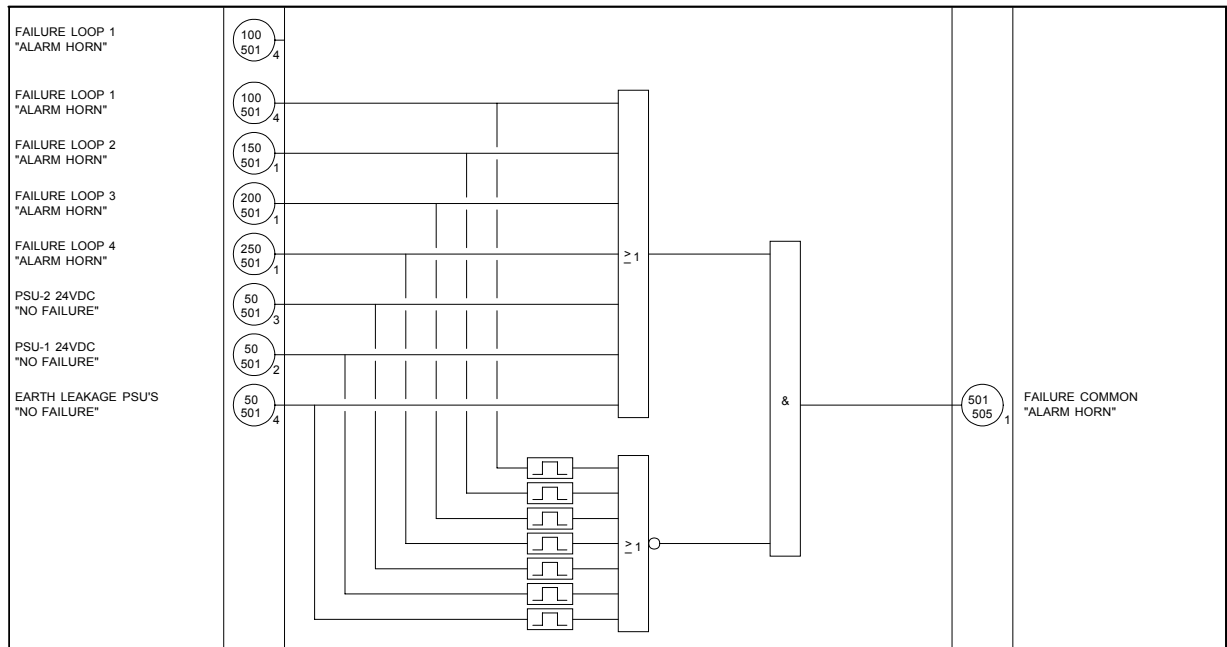


Figure 9-4 Control of the failure alarm horn (FLD 501)

Override function

Input sensors can go faulty during operation. To allow exchanging of a faulty input sensor without a constant Fire or Gas alarm, it is necessary to have an override function. The override function is also visually indicated on the operator panel. Although not required by the EN-54 part 2 standard, it is possible to generate an override audible alarm as indicated in the FLD shown in Figure 9-5. The cycle pulse logic for each loop combined in the NOR gate is required to activate the horn for every subsequent override in the same alarm group. An entry to the top OR gate is required for each override in an alarm group, as well as a cycle pulse and entry to the bottom NOR gate. If more than one alarm group is used in one Fire & Gas detection system, logic as shown in the diagram below is required for each alarm group.

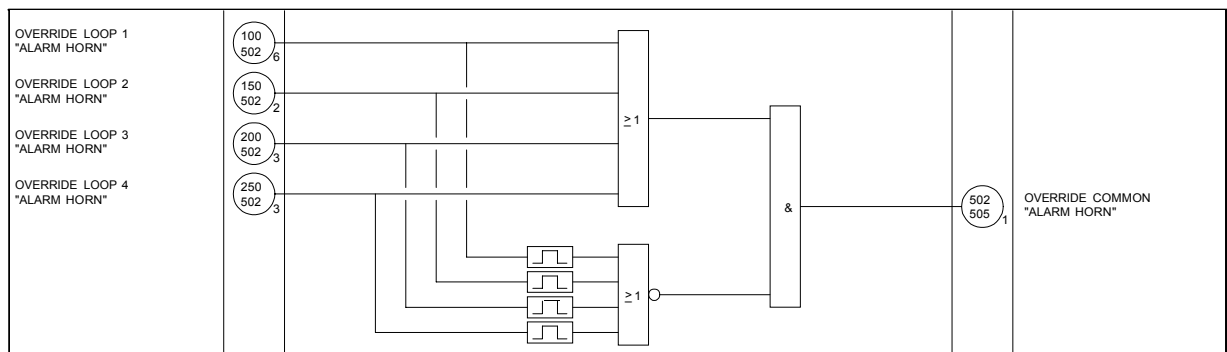


Figure 9-5 Control of the override alarm horn (FLD 502)

Simulation

Fire & Gas sensors can go faulty during normal operation. In order to test the functionality of the sensors, a test function must be implemented which overrides the audible alarms. A simulation of fire or gas at the input sensor will generate the alarm indication but will block the audible indication. The test function is also visually indicated on the operator panel. Although not required by the EN-54 part 2 standard, it is possible to generate a test audible alarm as indicated in the FLD shown in Figure 9-6. The cycle pulse logic for each loop combined in the NOR gate is required to activate the horn for every subsequent test operation in the same alarm group. An entry to the top OR gate is required for each test in an alarm group, as well as a cycle pulse and entry to the bottom NOR gate.

If more than one alarm group is used in one Fire & Gas detection system, logic as shown in the diagram below is required for each alarm group [EN-54 part 2, 2.5.2].

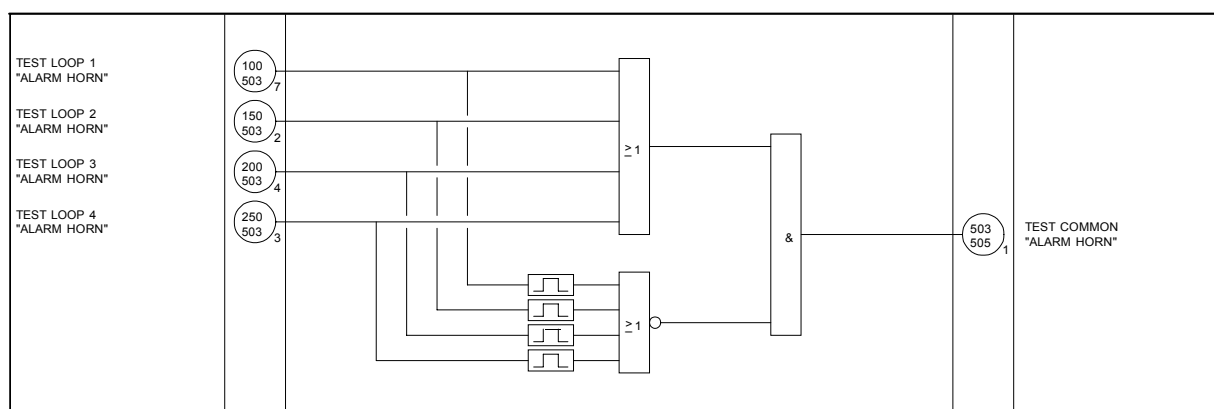


Figure 9-6 Control of the test alarm horn (FLD 503)

Cycle pulse

The signals controlling the horn are used to set the horn flip-flop via a cycle pulse [EN-54 part 2, 2.2.1.1 (alarm), 2.3.2.1 (failure)] (see Figure 9-7). The horn flip-flops can be reset via a horn reset digital input signal [EN-54 part 2, 2.3.8]. If multiple alarm groups are used in a Fire & Gas detection system, these can be combined via an OR gate between the cycle pulse and the flip-flop. A cycle pulse must be used for each individual alarm group.

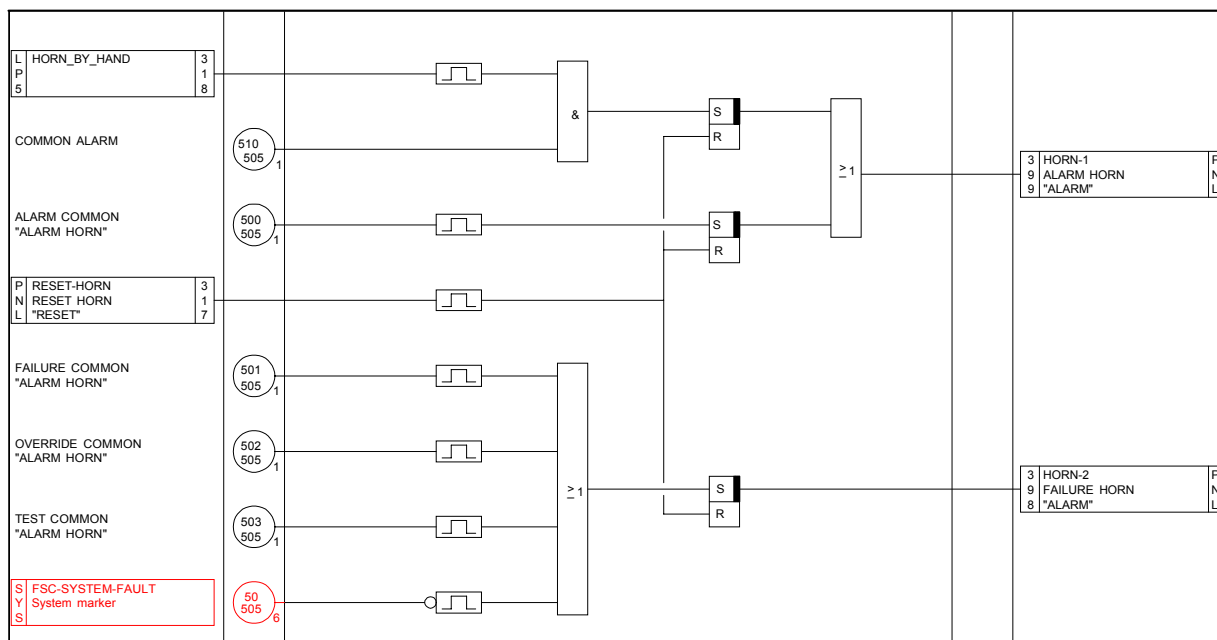


Figure 9-7 Control and acknowledge of the alarm horns (FLD 505)

Common alarm

The alarm indications for Fire or Gas alarm must be combined into a common alarm according to the EN-54 part 2, 2.2.1.2, 2.2.1.3, 2.2.19. This combination is shown in Figure 9-8 as a number of signals combined in an OR gate. The common alarm indication is combined with the lamp test function in order to test this visual indication too. The combination of Fire and Gas alarms into a common alarm must be done for each individual alarm group.

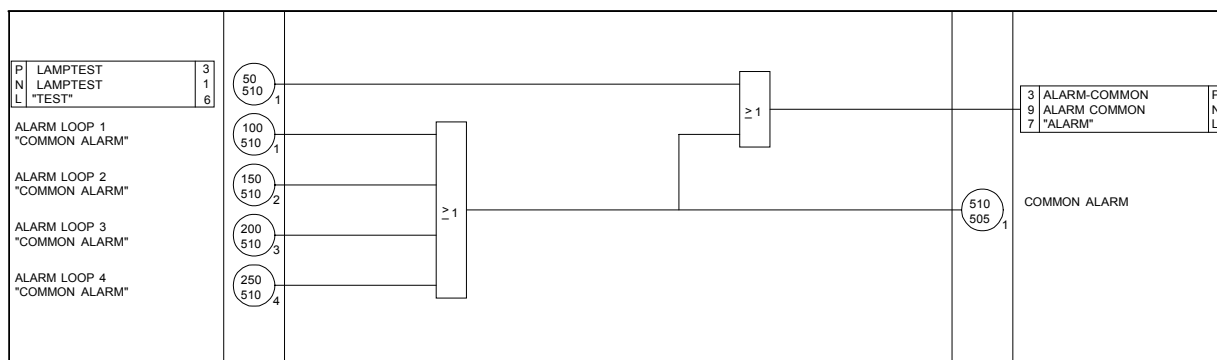


Figure 9-8 Control of the common alarm indication (FLD 510)

Common test indication

The indications that tests are executed for Fire or Gas detectors must be combined into a common test indication according to EN-54 part 2, 2.5.2. This combination is shown in Figure 9-9 as a number of signals combined in an OR gate. The common test indication is combined with the lamp test function in order to test also this visual indication.

The combination of Fire and Gas detector test indications into a common test indication must be done for each individual alarm group.

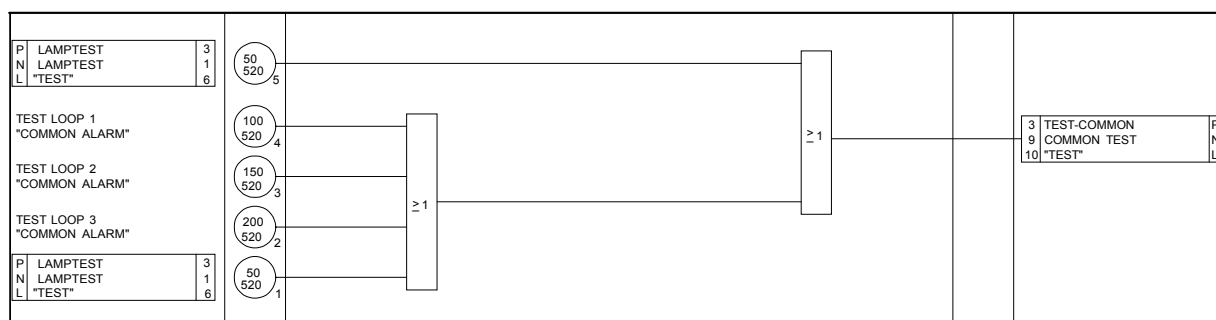


Figure 9-9 Control of the common test indication (FLD 520)

Common failure indication

The indications that failures have been detected in Fire or Gas detectors must be combined into a common failure indication according to EN-54 part 2, 2.3.1, 2.3.2.2. This combination is shown in Figure 9-10 as a number of signals combined in an OR gate. The common failure indication is combined with the lamp test function in order to test also this visual indication.

The combination of Fire and Gas detector failure indications into a common failure indication must be done for each individual alarm group.

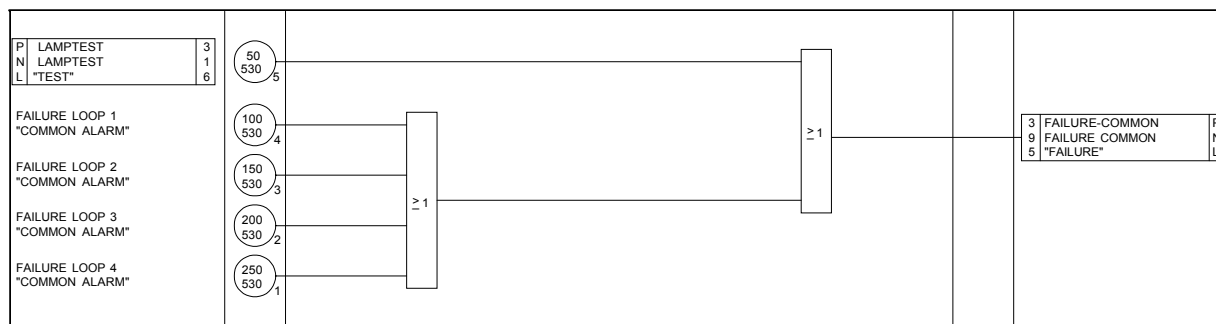


Figure 9-10 Control of the common failure alarm indication (FLD 530)



Common override indication

The indications that overrides have been made active for Fire or Gas detectors must be combined into a common override indication according to EN-54 part 2, 2.4.3.1.

This combination is shown in Figure 9-11 as a number of signals combined in an OR gate. The common override indication is combined with the lamp test function in order to test also this visual indication. The combination of Fire and Gas override indications into a common override indication must be done for each individual alarm group [EN-54 part 2, 2.4.3.2]. The display of the common override signal can be done remotely using the FSC-FSC communication [EN-54 part 2, 2.4.3.3] or via hardwired outputs using a digital output with loop-monitoring [EN-54 part 2, 2.4.4.4].

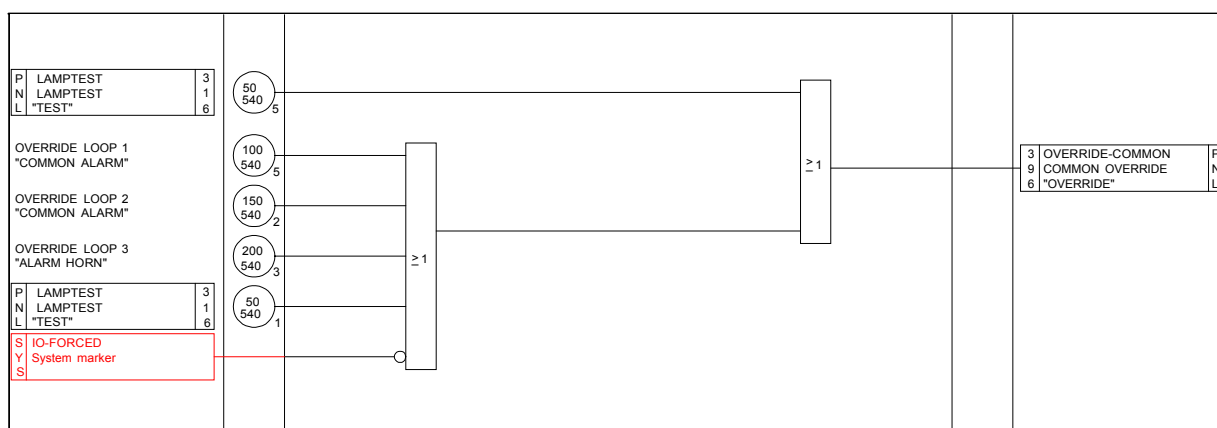


Figure 9-11 Control of the common override indication (FLD 540)

Alarm sequence function block

The alarm sequence function block handles the control of all visual and audible indications associated with an input loop [EN-54 part 2, 2.2.1.1, 2.2.1.2, 2.3.1]. For the example application, all alarm settings are identical so the determination of the alarm levels is included in this function block, but they may differ depending on the fire & gas detector (see Figure 9-12).

If the alarm levels are not the same for all input loops, the alarm detection should be included on the FLDs where this function block is called.

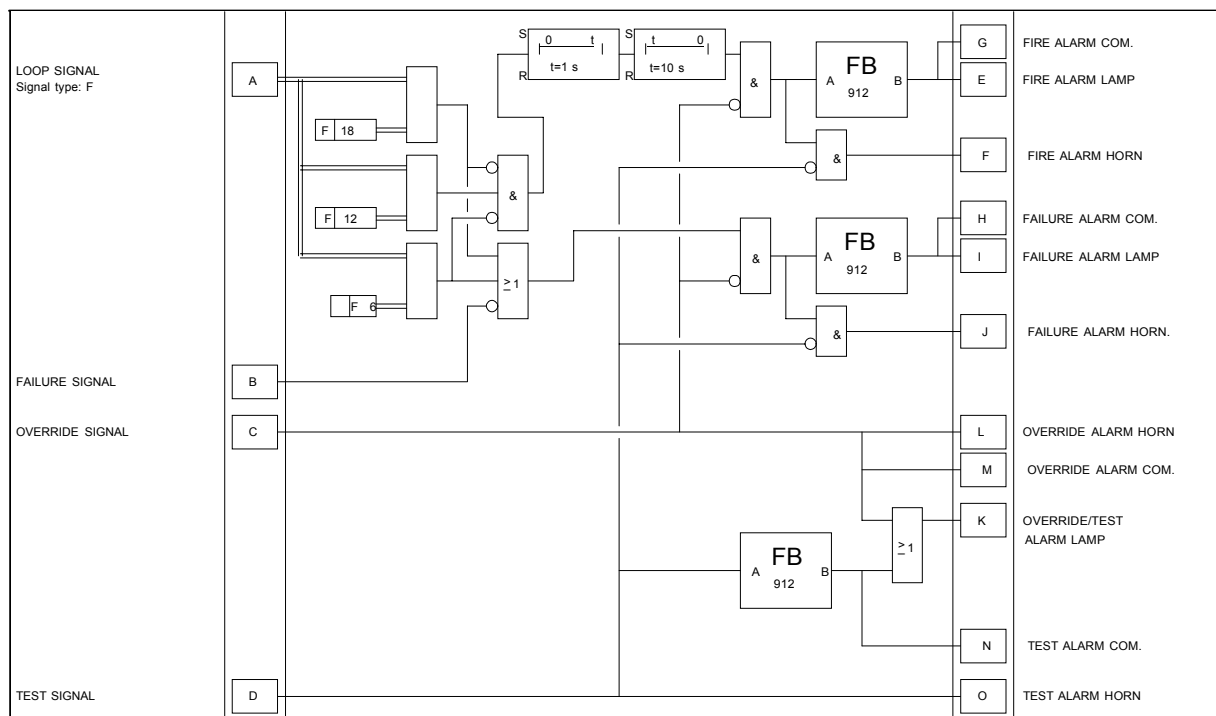


Figure 9-12 Alarm sequence function block (FLD FB-900)

The control of the indication is described via Function Block 905 (see Figure 9-13). This function handles the control of the indications and the control of the horn in case of the test function (alarms are passed but the horn is suppressed) and the override function (alarms and horn are suppressed).

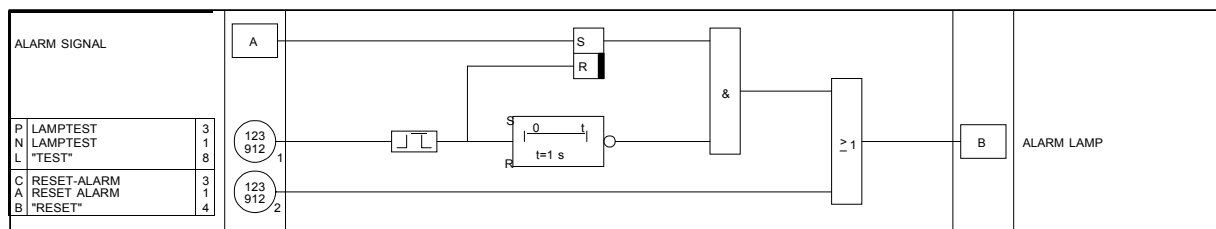


Figure 9-13 Alarm latching, alarm reset and lamp test function block (FLD 905)

Function Block 905 (FB-905) controls the indication status of lamps. It contains a latching function for each status that needs to be indicated until a manually initiated reset (key switch) occurs [EN-54 part 2, 2.2.10, 2.3.6]. If the indication status is still active, it will return to the On status after a defined period. (EN-54 part 2, 2.2.10 defines < 20 seconds; the time in the diagram above is 1 second.)



Section 10 – Special requirements for TÜV-approved applications

Requirements for TÜV approval

The FSC system can be used for those processes that require TÜV approval. The requirements for the safety applications are the following:

1. The maximum application program cycle time is half the process safety time. For example, the process safety time of a burner control system is 1 second in accordance with TRD-411 for boilers > 30 kW (July 1985) Table 1, TRD-412 (July 1985) Table 1 and DIN 4788 (June 1977) Part 2 Chapter 3.2.3.2 1. This implies that the application program cycle time must be 0.5 second or less. The application program cycle time is calculated by the compiler. It is listed in the log file (.LOG) produced by the compiler, and also shown on screen during translation. The application program execution time is limited to 0.5 seconds by hardware on the watchdog module, which means that the FSC system can be used without checking of the execution time for those applications that have a process safety time of 1 second or more.
2. If the FSC system detects a fault in its safety-related output hardware it is possible to de-energize part of the process instead of de-energizing all outputs. The de-energization of process parts or all outputs is fully implemented in the system software and cannot be influenced by the user (see also item 3). The de-energization depends on the output module type:

10201/1/1	Fail-safe digital output module
10201/2/1	(24 Vdc, 0.55 A, 8 channels) De-energization per group of output channels: Group 1: outputs 1, 2, 3, 4. Group 2: outputs 5, 6, 7, 8.
10203/1/2	Fail-safe output module with double switch-off (24 Vdc, 0.9 A, 4 channels) De-energization can be effected per channel or for the entire group i.e. outputs 1, 2, 3, 4, after the detection of a second fault.
10205/1/1	Fail-safe analog output module
10205/2/1	(0(4)-20 mA, 2 channels) De-energization per channel.



10212/1/1	Digital output module (24 Vdc, 0.9 A, 16 channels) De-energization of group 1: outputs 1, 2, 3, 4 (these are the 4 fail-safe outputs).
10213/1/1	Fail-safe digital output module
10213/2/1	(110 Vdc, 0.32 A, 4 channels) De-energization of group 1: outputs 1, 2, 3, 4.
10213/1/2	Fail-safe digital output module
10213/2/2	(60 Vdc, 0.67 A, 4 channels)
10213/1/3	Fail-safe digital output module
10213/2/3	(48 Vdc, 0.75 A, 4 channels) De-energization of group 1: outputs 1, 2, 3, 4.
10214/1/2	Fail-safe digital output module (220 Vdc, 0.25 A, 3 channels) De-energization of group 1: outputs 1, 2, 3.
10215/1/1	Fail-safe digital output module
10215/2/1	(24 Vdc, 2 A, 4 channels) De-energization of group 2: outputs 3, 4.
10216/1/1	Fail-safe loop-monitored digital output module
10216/2/1	(24 Vdc, 1 A, 4 channels) De-energization of group 1: outputs 1 to 4.
10216/2/3	Fail-safe loop-monitored digital output module (48 Vdc, 0.5 A, 4 channels) De-energization of group 1: outputs 1 to 4.

If a complete safety-related module is detected faulty, all outputs connected to the Central Part that controls the output module are de-energized via the watchdog module (10005/1/1) of the Central Part. If the output is located in a non-redundant I/O section, all outputs of the FSC system are de-energized. De-energization is only effected if safety-related outputs are configured to the faulty module.

3. If the FSC system detects a fault in its safety-related output hardware (see item 2 above), a timer is started. When this timer expires, all outputs are de-energized via the watchdog module (10005/1/1). This timer can be set to the following values:
 - Not used. The timer is not started so an output fault may be present in the system without further action.
 - 0 minutes. This results in immediate de-energization of all outputs in case of an output fault.
 - 1 minute to 22 days. This represents the interval time between the fault occurring and automatic system shutdown.



The "interval time between faults" can be set using the 'System Configuration' option of FSC Navigator (Install \ Configuration).

4. If the FSC system detects a fault in its safety-related input hardware, the faulty input is set to low (off) for digital inputs and to bottom scale for the analog inputs. This represents the safe status for both digital and analog inputs.
5. The watchdog module (10005/1/1) contains an emergency shutdown (ESD) input. For normal operation, the ESD input must be 24 Vdc. If the input is forced to 0 V, a Central Part shutdown and de-energization of the outputs are initiated, independent of the CPU.
6. For further details on I/O wiring details, termination of I/O signals and power supply distribution refer to the FSC Hardware Manual
7. The setting of the watchdog and the safety time (the time in which all I/O tests are executed once) and the time between faults can be checked using the 'Monitor System' option of FSC Navigator (FSC system \ Sys info \ Parameters) (see Figure 10-1).

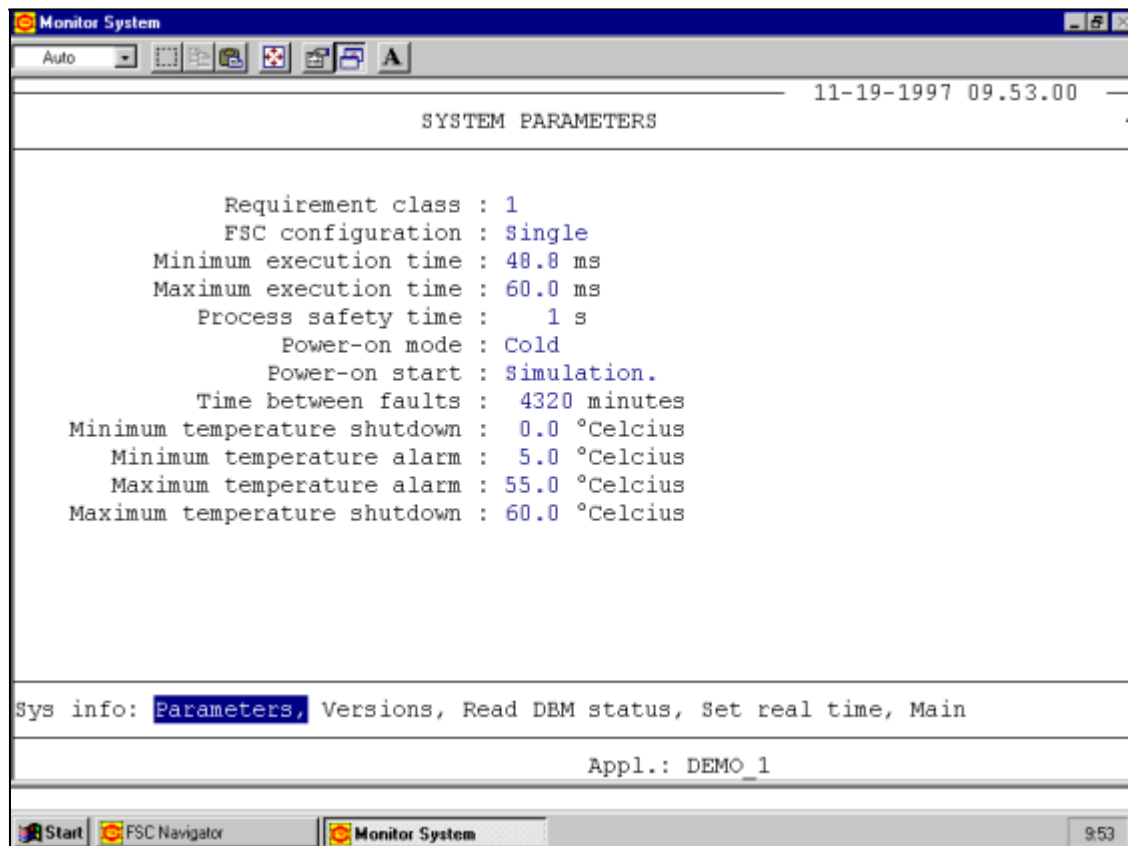


Figure 10-1 System parameters



8. The 24 Vdc to 5 Vdc DC/DC converter (PSU: 10300/1/1) has limited capacity. Larger FSC systems may require the use of more than one power supply unit (PSU). In that case, each additional PSU requires a watchdog repeater module (10302/1/1 or 10302/2/1) to monitor the 5 Vdc of the PSU which controls the WD input of all fail-safe output modules connected to that PSU.
9. The M24-20 HE and M24-12 HE power supply units provide 24 Vdc as output voltage. If these power supply units are used, a watchdog repeater module must be placed to monitor the 24 Vdc voltage. This watchdog repeater may also be used to monitor the 5 Vdc of a second PSU (see item 8).

Note:

The 1200 S 24 P067 power supply does not require a watchdog repeater module.

10. The value of the voltage monitor analog input channels of the 10105/2/1 modules must be checked in the application software for the correct operating range for the transmitters connected to that analog input module.
11. To reduce the influence of disturbances on the power supply lines, all major metal parts (cabinet side walls, doors, 19-inch racks, horizontal bus rack and flaps, swing frames, etc.) must be grounded properly.
12. All power supply inputs (except 110/230 Vac) require a power supply filter to be fitted immediately after the power supply input terminals.
13. Grounding of the power supplies of the FSC system is only permitted for the 0 Vdc. Grounding of the +24 Vdc / +48 Vdc / +60 Vdc / +110 Vdc / +220 Vdc is NOT allowed as an earth fault will result in an unsafe situation.
14. To maintain the separation between the external power supply (24 Vdc) and the internal power supply (5 Vdc), the wiring of these voltage levels must be physically separated. This can be obtained by using separate ducts and a separate power supply distribution.
15. Do not use radio transmitting equipment within a radius of 1 m (3 ft) of the system cabinet when the doors are opened.
16. For details on power supply distribution and watchdog wiring (especially FSC configurations with redundant Central Parts and both redundant and single I/O) refer to the FSC Hardware Manual.



17. Safety-related inputs require the use of fail-safe input modules (10101/1/1, 10101/1/2, 10101/1/3, 10101/2/1, 10101/2/2, 10101/2/3, 10102/1/1, 10102/1/2, 10102/2/1 or 10105/2/1) and fail-safe input sensors (transmitters). If the input sensors (transmitters) are not fail-safe, redundant sensors (transmitters) must be used. Refer to appendix C of the FSC Software Manual ("Safety-related inputs with non-fail-safe sensors") for further details.
18. If non-fail-safe sensors/transmitters are used to realize safety-related inputs (see Appendix C of the FSC Software Manual), a maximum on time and a maximum discrepancy time must be configured. The maximum on time specifies the time that a signal can remain high before the system will regard the input as faulty. The maximum discrepancy time specifies the maximum time that redundant inputs may have different values before the system regards the input as faulty. Both the maximum on time and maximum discrepancy time should be configured according to the dynamic behavior of the input signal.
19. If non-fail-safe transmitters are used to realize safety-related analog inputs (see Appendix C of the FSC Software Manual), a maximum discrepancy value must be configured. The value specifies the tolerable difference between the value of the transmitters before the system will regard the input as faulty.
20. If the FSC system runs without operator surveillance, one of the following measures shall be taken:
 - Inspection of the FSC system status if the FSC system application is fault free, at least once per 72 hours.
 - Alarm indication of the FSC system (e.g. via DCS) if a fault is detected and subsequent inspection of the FSC system status within 72 hours after generation of the fault report.
21. The operating conditions of the FSC system shall not exceed the following ranges:
 - Operating temperature: 0 to 60°C (32 to 140°F)
 - Relative humidity: 95%, non-condensing
 - Vibration: 2.5 G (10-55-10 Hz)
 - Shock: 15 G (11 ms, 3 axes)

The operating temperature is measured on the diagnostic and battery module (DBM) in the Central Part rack. This location has a higher temperature than outside the cabinet, which results in a lower ambient temperature for the cabinet. Depending on the internal dissipation in the cabinet and the ventilation provided, a temperature difference of 20°C (39°F) is possible, which results in

a maximum ambient temperature of 40°C (104°F). To minimize the temperature difference, forced ventilation with one or more fans can be applied. By using the temperature pre-alarm system variable, an alarm can be given if the internal temperature rises too high. For further details on the DBM refer to Section 4 of the FSC Software Manual ("System Configuration").

22. The storage conditions of the FSC hardware modules shall not exceed the following ranges:
Storage temperature: −25 to +80°C (−13 to 176°F)

F&G applications

Fire and Gas (F&G) applications have the following additional requirements:

1. Each visual indication (alarm, override or test, failure) shall have its own dedicated digital output. This digital output may be a hardware output or a communication output, e.g. to a DCS system. Override and test status may be combined in one visual indication. No support for alphanumeric displays is available.
2. Redundant power supplies must be connected to the FSC system in such a way that the redundant power supplies do not fail at the same time, e.g. by using diverse primary power sources (e.g. 220 Vac mains and a 24 Vdc from a battery backup). Detection of power supply failure (e.g. via a voltage-monitoring module) shall be part of the system design.

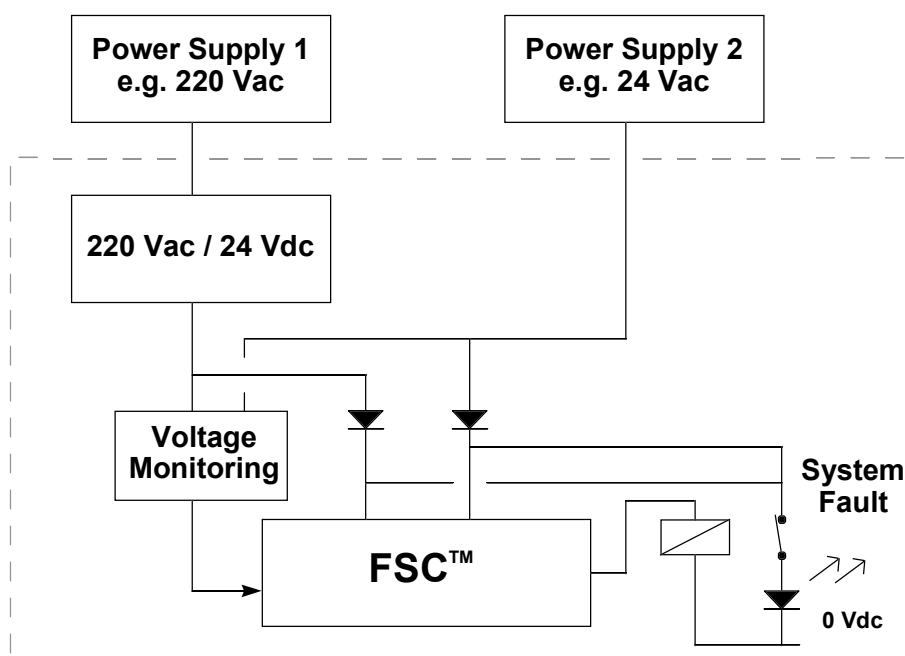


Figure 10-2 Power supply



3. Any faults in the Fire & Gas detection system shall be indicated visually. This indication shall also be active if the Fire & Gas detection system has been switched off. This can be realized as shown in Figure 10-2 above, using a normally de-energized relay, or via a visual indication on a DCS display which is activated if the communication to the Fire & Gas detection system fails. The protected side of the fuses are connected to the voltage-monitoring device in order to detect blown fuses.
4. The field instruments, including panel instruments such as (key) switches, which are used in conjunction with the FSC system, must meet the requirements of the applicable parts of the EN-54 standard. Visual and audible indications shall be as per paragraph 3.2 of EN-54 part 2.
5. Field inputs must have loop-monitoring (short-circuiting and open loop). Input module types that can be used are: 10102/1/1, 10102/1/2, 10102/2/1 and 10105/2/1.
Field outputs must have loop-monitoring (short-circuiting and open loop). Output module types that can be used are: 10216/1/1, 10216/2/1, 10216/2/3 and 10214/1/2.
6. The FSC system performs loop testing of output channels allocated to 10216/1/1, 10216/2/1 or 10216/2/3 modules in groups of five modules per user-defined Process Safety Time. The test interval for each module shall not exceed 100 seconds.
The number of 10216/1/1, 10216/2/1 and 10216/2/3 modules in an FSC configuration for Fire & Gas applications, in a non-redundant I/O section, shall therefore not exceed the number $(5 \times 100 \text{ seconds})$ divided by the Process Safety Time. The number of 10216/1/1, 10216/2/1 and 10216/2/3 modules in redundant I/O sections shall not exceed the number $(5 \times 100 \text{ seconds})$ divided by the $2 \times$ Process Safety Time.
7. The Fire & Gas detection system shall have earth leakage monitoring/detection facilities.
8. Remote display of alarms, failures etc. may only be executed via interconnection of FSC systems using the FSC-FSC communication option or via hardwired outputs with loop-monitoring via the 10216/1/1, 10216/2/1, 10216/2/3 and 10214/1/2 digital output modules. Communication and loop monitoring failures must be alarmed.



9. The FSC system is only the basis for an EN-54 compliant application. The responsibility for a full EN-54 compliant application lies with the person(s) responsible for configuring and application programming of the FSC system. The requirements of EN-54 which must be covered in the application program can be found in the attached example, which references the requirements that must be fulfilled in the application program.
 10. For details on the mechanical construction requirements (cabinet, indications, horns) refer to EN-54 part 2 paragraph 3.2.
-



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Title of Document: **Fail Safe Control
Safety Manual 500 Rev. 01**

Issue Date: **01/98**

Document Number: **PM.MAN.8047**

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